

Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

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1 Orientation and rigour boundary

The north star, and its status

The target of this project is the classical (stochastic) stability statement **op-classical**: there are universal constants $\eta_0, C > 0$, independent of the dimension, such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta \leq \eta_0$ admits a stochastic idempotent E (**def-stochastic**) with

$$\|Q - E\|_{\infty \rightarrow \infty} \leq C\sqrt{\eta}, \quad (1)$$

the exponent $\frac{1}{2}$ being sharp.

This is OPEN. Nothing in this document proves it, and nothing in this document should be read as evidence that it is close to proved. What follows is a rigorous *partial* record: the components of the currently live route that have cleared this repository's top in-repo rigour rung.

Two pictures, one bridge

Two equivalent formulations are in play throughout. In the *stochastic picture* the object is a genuinely positive row-stochastic Q that is only approximately idempotent, with defect η measured in the $\infty \rightarrow \infty$ map norm (**def-almost-idempotent**). In the *signed picture* the object is an exact signed affine retraction $P \dashv P\mathbf{1} = \mathbf{1}$ and $P^2 = P$ *exactly* (**def-signed-idempotent**) — whose failure of positivity is quarantined in the single scalar δ , the negative mass (**def-negative-mass**). The two are interchangeable up to universal constants, and the licence to cross between them is **lem-classical-equiv** (§2); a bound must always name the picture it lives in and cite the bridge when it crosses.

What this report contains

Four blocks, in dependency order.

1. **The bridge** (§2). **lem-classical-equiv**, the signed $\leftrightarrow$ stochastic equivalence, with explicit constants in both directions.
2. **Positive-retract hardening** (§3). **lem-prh** converts a *positive approximate retract* — a pair of positive unital maps A, M between finite commutative C^* -algebras whose composite MA is ε -close to the identity (**def-positive-approximate-retract**) — into an exact stochastic idempotent at distance $2\sqrt{2\varepsilon}$. It is the step that turns approximate retraction data into the exact object (1) asks for, at the sharp square-root rate.
3. **The COMP tier** (§§4–17, interleaved). This tier builds, over an *extended* ε - C^* -algebra (**def-extended-epsilon-cstar-algebra**), the approximate corner calculus of δ -projections: the compressed corners $\mathcal{S}_{P,Q}$, the compression maps $\text{Co}_{P,Q}$ and the compressed product (**def-compressed-corner**). Its content is that all of this is *matrix-uniform*: every estimate holds at every amplification level n with one constant. The keystones are the amplification identity $1_{M_n} \otimes \text{Co}_{P,Q} = \text{Co}_{I_n \otimes P, I_n \otimes Q}$, the uniform rectangular-product estimate, and the statement that a nonvanishing corner $\mathcal{S}_P$ is itself an *extended* $O(e)$ - C^* -algebra — the object the next tier needs.
4. **The H-CB tier** (§§11–29, interleaved). Over a closed H-CB datum (**def-hcb-datum**) this tier establishes the amplified estimates for the map $\text{Ha}_{P,R}^Q$ of **def-ha-map**: the column-Hilbert comparison, the variational identity that defines it, its column action, exact adjointness, its

multiplicative defect, the diagonal unit and upper-norm estimates, and — under explicit level-one hypotheses — the diagonal lower modulus and the diagonal inverse bound, all uniform in n .

5. **The H-CB parent contract and EXT tier** (§30 and §§31–36). This tier develops corner arithmetic through a one-dimensional index, the dimension selection $(r, 1)$ of an EXT-CB datum, the four-corner merge, and the EXT parent that adds one dimension to a level-one datum.

The honesty boundary

Three rules govern what may appear here.

Only af-validated results are stated as lemmas. Every numbered lemma below is a registry result whose adversarial proof tree has root node **validated** with clean taint, produced by fresh **codex** provers and *separate* fresh per-node verifiers; the orchestrator never judged a proof. Each shard records the node count and names its workspace. Nothing else in the registry — the inherited **proved-mod-audit** transcriptions, the **conjecture** rows, the **numerical** evidence, the **heuristic** arguments — is restated here as a theorem; those keep their tags in **argument/** and are surveyed, without promotion, in §37.

Every lemma is exactly as strong as its registry contract. Each statement is a typeset transcription of the one-line **contract** of its **argument/** shard, reproduced byte-verbatim beneath it. Where a contract is deliberately conditional or one-sided — the two-sided compressed-unit bound only for *nonvanishing* projections, the level-one hypotheses of the inverse tier — that shape is load-bearing and is preserved. In particular a level-one conclusion is never glossed as a matrix-uniform one.

No definition is restated. Terms resolve to their canonical **definitions/** shards, cited by **def-**identifier. Where a proof tree consumed a byte-matched auxiliary registered inside its own **af** workspace — the ε - C^* norm axioms, the compressed-product display, the one-dimensional nonvanishing bridge — it is named as such, and identified as a workspace external rather than a registry definition.

What is deliberately absent

No claim, direct or implied, that **op-classical** follows. The H-CB tier feeds the parent contract **conj-hcb** (§30), and that in turn feeds the EXT parent **conj-extcb** (§36); both are **proved** with **af: validated**. What is open beyond them is the assembly, whose registry carrier **lem-thmainext-conditional** is **proved-mod-audit** with **af: none**, together with the quarantined assembly interface. The route is a route, not a proof. Also absent: any numerical statement (those live in **runs/**, and are never rigorous), and the earlier signed-geometry material, which is off the live route and inventoried in §37.

2 The signed-stochastic bridge

Lemma 1 (`lem-classical-equiv`). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants. There is a universal $C > 0$ such that:*

1. *(stochastic $\rightarrow$ signed) if Q is row-stochastic with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta$, then $P = \theta(2Q - I)$ is a signed affine retraction (**def-signed-idempotent**) satisfying $\|P - Q\|_{\infty \rightarrow \infty} \leq C\eta$ and negative mass $\delta \leq C\eta$;*
2. *(signed $\rightarrow$ stochastic) conversely, row-normalising the positive parts p_i^+ of a signed idempotent P yields a row-stochastic Q with $\|P - Q\|_{\infty \rightarrow \infty} \leq 2\delta$ and $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq 6\delta + 4\delta^2$.*

Transcription note. The contract writes the map norm as $\|\cdot\|$ without a subscript; per `CONVENTIONS.md` (b) it is the $\infty \rightarrow \infty$ map norm $\|\cdot\|_{\infty \rightarrow \infty}$. The operator θ is carried verbatim from the contract, which does not expand it; the validated tree instantiates it as the functional calculus $\theta(X) = \frac{1}{2}(I + \text{sgn } X)$ of **def-theta-idempotent-approximation**.

Registry contract (`verbatim`).

The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants: Q row-stochastic with $\|Q^2 - Q\| \leq \eta$ gives $P = \theta(2Q - I)$ signed affine retraction with $\|P - Q\| \leq C\eta$ and neg mass $\delta \leq C\eta$, and conversely row-normalising p_i^+ gives Q with $\|P - Q\| \leq 2\delta$, $\|Q^2 - Q\| \leq 6\delta + 4\delta^2$.

Proof (prose account of the af-validated tree). *Forward direction.* Put $X = 2Q - I$. Row-stochasticity gives $X\mathbf{1} = \mathbf{1}$ and $\|X\|_{\infty \rightarrow \infty} \leq 3$, while the algebraic identity

$$X^2 - I = 4(Q^2 - Q) \tag{2}$$

converts the almost-idempotence hypothesis into $\|X^2 - I\|_{\infty \rightarrow \infty} \leq 4\eta$. Write $Y = X^2 - I$. Two properties of Y drive everything: $Y\mathbf{1} = 0$ (because $X\mathbf{1} = \mathbf{1}$ forces $X^2\mathbf{1} = \mathbf{1}$), and $\|Y\|_{\infty \rightarrow \infty} \leq 4\eta \leq \frac{1}{2}$ once $\eta \leq \frac{1}{8}$.

The smallness of Y puts the inverse square root inside its convergence radius: the binomial series

$$B = (I + Y)^{-1/2} = \sum_{m \geq 0} \binom{-\frac{1}{2}}{m} Y^m$$

converges absolutely in the map norm. Being a norm limit of polynomials in $Y = X^2 - I$, the operator B commutes with X ; since $Y\mathbf{1} = 0$ kills every positive power, $B\mathbf{1} = \mathbf{1}$; and the absolute coefficient sum gives the quantitative bound $\|B - I\|_{\infty \rightarrow \infty} \leq (1 - t)^{-1/2} - 1 \leq 2t$ at $t = \|Y\|_{\infty \rightarrow \infty} \leq \frac{1}{2}$, hence $\|B - I\|_{\infty \rightarrow \infty} \leq 8\eta$. The scalar identity $((1 + z)^{-1/2})^2(1 + z) = 1$ transfers to the absolutely convergent series in Y , so $B^2X^2 = I$; commutation then makes $S = XB$ an exact involution,

$$S^2 = XBXB = X^2B^2 = I, \quad S\mathbf{1} = XB\mathbf{1} = X\mathbf{1} = \mathbf{1},$$

with $S - X = X(B - I)$ and therefore $\|S - X\|_{\infty \rightarrow \infty} \leq 3 \cdot 8\eta = 24\eta$.

Setting $P = \theta(X) = \frac{1}{2}(I + S)$ now gives $P\mathbf{1} = \mathbf{1}$ and $P^2 = \frac{1}{4}(I + 2S + S^2) = \frac{1}{2}(I + S) = P$, so P is an exact signed idempotent in the sense of **def-signed-idempotent**. Since $Q = \frac{1}{2}(I + X)$,

the difference is $P - Q = \frac{1}{2}(S - X)$ and hence $\|P - Q\|_{\infty \rightarrow \infty} \leq 12\eta$. Finally the negative mass: row-stochastic Q is entrywise nonnegative (**def-stochastic**), so for every entry $\max\{-P_{ij}, 0\} \leq |P_{ij} - Q_{ij}|$ — trivially where $P_{ij} \geq 0$, and because $-P_{ij} \leq Q_{ij} - P_{ij}$ where $P_{ij} < 0$. Summing over j and maximising over i gives, with δ as in **def-negative-mass**,

$$\delta(P) \leq \|P - Q\|_{\infty \rightarrow \infty} \leq 12\eta. \quad (3)$$

So $C = 12$ serves. The export records an instructive scope correction here: the negative-mass step was first blocked because **def-negative-mass** — although permitted by the registry shard — had not been registered inside the workspace, and the tree carried an explicit gap certificate rather than the estimate; registering the definition discharged it.

Converse direction. Let P be an exact signed idempotent with negative mass δ . Split each row into disjointly supported nonnegative parts, $p_i = p_i^+ - p_i^-$, and set $a_i = \sum_k (P_{ik})_- \leq \delta$. Total mass one (**def-signed-idempotent**) gives $\sum_k p_{ik}^+ = 1 + a_i$, so

$$q_i = \frac{p_i^+}{1 + a_i}$$

is a probability vector and the matrix Q with rows q_i is row-stochastic. The correction $e_i = q_i - p_i$ is

$$e_i = p_i^- - \frac{a_i}{1 + a_i} p_i^+,$$

a difference of two disjointly supported vectors of ℓ^1 -mass a_i each; hence $\|e_i\|_1 = 2a_i \leq 2\delta$ and $\|P - Q\|_{\infty \rightarrow \infty} \leq 2\delta$.

For the idempotence defect, write $D = Q - P$. The row-geometry clause of **def-signed-idempotent** gives $\|P\|_{\infty \rightarrow \infty} \leq 1 + 2\delta$, hence $\|P - I\|_{\infty \rightarrow \infty} \leq 2 + 2\delta$. Using $P^2 = P$ exactly, each row satisfies the exact decomposition

$$q_i Q - q_i = e_i(P - I) + q_i D. \quad (4)$$

Since q_i is a probability vector, $\|q_i D\|_1 \leq \|D\|_{\infty \rightarrow \infty} \leq 2\delta$, while $\|e_i(P - I)\|_1 \leq 2\delta(2 + 2\delta)$. Adding, $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq 6\delta + 4\delta^2$, exactly as contracted.

Status. proved; **af:** validated (T0). Workspace **proofs/lem-classical-equiv**: a 29-node tree (21 live nodes validated, 8 superseded nodes archived), root **validated**, taint clean.

Role in the chain. This is the licence to move between the two pictures of §1, and the only such licence in the registry. It has no dependencies. Everything stated in the signed picture reaches the stochastic formulation of **op-classical** only through it, and only up to the universal constants above; **op-classical** itself remains open.

3 Positive-retract hardening

Lemma 2 (`lem-prh`). *Let $k, n \geq 1$ and let*

$$A : \ell^\infty(k) \longrightarrow \ell^\infty(n), \quad M : \ell^\infty(n) \longrightarrow \ell^\infty(k)$$

be positive unital maps — equivalently, maps whose matrices have probability-vector rows. If

$$\|MA - I_k\|_{\infty \rightarrow \infty} \leq \varepsilon, \quad 0 \leq \varepsilon < \frac{1}{2},$$

*then there is a stochastic idempotent $E : \ell^\infty(n) \rightarrow \ell^\infty(n)$ (**def-stochastic**) with*

$$\|AM - E\|_{\infty \rightarrow \infty} \leq 2\sqrt{2\varepsilon}. \quad (5)$$

Transcription note. The pair (A, M) is the datum of **def-positive-approximate-retract**, with retract defect $\varepsilon_{\text{ret}} = \|MA - I_k\|_{\infty \rightarrow \infty}$. Following the validated tree, A is presented by rows $a_i \in \Delta_k$ indexed by $i \in \{1, \dots, n\}$ and M by rows μ_s indexed by $s \in \{1, \dots, k\}$, so that the s -th row of MA is the probability vector $\sum_i \mu_s(i) a_i$.

Registry contract (`verbatim`).

Positive-retract hardening (PRH): let $k, n \geq 1$ and let $A : \ell^\infty(k) \rightarrow \ell^\infty(n)$ and $M : \ell^\infty(n) \rightarrow \ell^\infty(k)$ be positive unital maps (equivalently, have probability-vector rows); if $\|MA - I_k\|_{\infty \rightarrow \infty} \leq \varepsilon$ with $0 \leq \varepsilon < 1/2$, then there is a stochastic idempotent $E : \ell^\infty(n) \rightarrow \ell^\infty(n)$ with $\|AM - E\|_{\infty \rightarrow \infty} \leq 2\sqrt{2\varepsilon}$.

Proof (prose account of the af-validated tree). The tree splits on the defect. If $\varepsilon = 0$ the hypothesis reads $MA = I_k$ exactly, and then $E := AM$ is already row-stochastic with $E^2 = A(MA)M = AM = E$, so (5) holds with error 0. The substance is the case $0 < \varepsilon < \frac{1}{2}$.

The scale and the cores. Fix

$$\lambda = \sqrt{\varepsilon/2}, \quad C_s = \{i : a_{is} > 1 - \lambda\}, \quad \beta_s = \mu_s(C_s^c).$$

The single quantitative input is a coordinate defect identity. The s -th row of MA is the probability vector $\sum_i \mu_s(i) a_i$, and for probability vectors the ℓ^1 distance to the basis vector e_s is governed by one coordinate:

$$\|(MA)_{s,*} - e_s\|_1 = 2(1 - (MA)_{ss}) = 2 \sum_i \mu_s(i) (1 - a_{is}) =: 2D_s, \quad (6)$$

so the norm hypothesis is exactly $D_s \leq \varepsilon/2$ for every s . Two consequences follow. First, the cores are pairwise disjoint: if $i \in C_s \cap C_t$ with $s \neq t$ then $a_{is} + a_{it} > 2(1 - \lambda) > 1$ (using $\lambda < \frac{1}{2}$, which is where $\varepsilon < \frac{1}{2}$ enters), contradicting that a_i is a probability vector. Second, off the core one has $1 - a_{is} \geq \lambda$, so dropping the on-core terms of the nonnegative sum D_s gives $\lambda\beta_s \leq D_s \leq \varepsilon/2$, whence

$$\beta_s \leq \frac{\varepsilon}{2\lambda} = \lambda < \frac{1}{2}. \quad (7)$$

The tree records a correction at this point: an earlier node asserted the strict inequality $\beta_s < \lambda$, which the threshold $a_{is} \leq 1 - \lambda$ off the core does not give; the weak form (7) is what holds and all that is used.

The exact construction. Condition each decoder row on its core and replace each encoder row inside a core by a basis vector:

$$\nu_s(i) = \frac{\mu_s(i) \mathbf{1}_{C_s}(i)}{1 - \beta_s}, \quad \hat{A} \text{ has row } e_s \text{ on } C_s \text{ and row } a_i \text{ off } \bigcup_s C_s.$$

Both are well defined: $\beta_s < 1$ makes ν_s a probability vector, and disjointness of the cores makes $\hat{A}$ unambiguous. The construction is engineered so that the retraction becomes *exact*: the s -th row of $N\hat{A}$ is $\sum_{i \in C_s} \mu_s(i) e_s / (1 - \beta_s) = e_s$, i.e.

$$N\hat{A} = I_k, \quad \text{hence} \quad E := \hat{A}N \text{ satisfies } E^2 = \hat{A}(N\hat{A})N = \hat{A}N = E. \quad (8)$$

As a product of positive unital maps, E is row-stochastic, so it is a stochastic idempotent in the sense of **def-stochastic**.

The two error terms. Write $AM - E = A(M - N) + (A - \hat{A})N$ and bound the summands separately.

For the conditioning term, ν_s is μ_s conditioned on C_s , so the two rows differ by mass β_s removed from C_s^c and mass β_s redistributed on C_s ; hence $\|\mu_s - \nu_s\|_1 = 2\beta_s \leq \varepsilon/\lambda$ by (7) and $\|M - N\|_{\infty \rightarrow \infty} \leq \varepsilon/\lambda$. Because A has probability rows it is an $\infty \rightarrow \infty$ contraction — concretely, $\sum_j |(AB)_{ij}| \leq \sum_s a_{is} \sum_j |B_{sj}| \leq \|B\|_{\infty \rightarrow \infty}$ — so $\|A(M - N)\|_{\infty \rightarrow \infty} \leq \varepsilon/\lambda$.

For the core-replacement term, the rows of $(A - \hat{A})N$ vanish outside $\bigcup_s C_s$. For $i \in C_s$ the corresponding row is $\sum_t a_{it} \nu_t - \nu_s$, and since every ν_t is a probability vector its ℓ^1 norm is at most $(1 - a_{is}) + \sum_{t \neq s} a_{it} = 2(1 - a_{is}) < 2\lambda$. Hence $\|(A - \hat{A})N\|_{\infty \rightarrow \infty} \leq 2\lambda$.

Optimisation. Adding the two bounds,

$$\|AM - E\|_{\infty \rightarrow \infty} \leq \frac{\varepsilon}{\lambda} + 2\lambda,$$

and the choice $\lambda = \sqrt{\varepsilon/2}$ balances the two terms at the common value $\sqrt{2\varepsilon}$, giving (5). The square-root rate is intrinsic to this balance: conditioning costs ε/λ and core replacement costs 2λ , and no choice of threshold does better than their geometric mean.

Status. proved; af: validated (T0). Workspace **proofs/lem-prh**: 14 nodes, all validated, root validated, taint clean.

Role in the chain. **lem-prh** has no registry dependencies. It is the finishing step of the live route: it installs the reduction of **op-classical** to the *existence* of a positive approximate retract for a given almost-idempotent Q — stochastic A, M with $\|AM - Q\|_{\infty \rightarrow \infty} = O(\eta)$ and $\|MA - I\|_{\infty \rightarrow \infty} = O(\eta)$ — after which (5) delivers the exact idempotent at rate $\sqrt{\eta}$. Producing such a pair is what the COMP and H-CB tiers below are ultimately for, and that production is *not* established: **lem-prh** is a conditional finisher, and **op-classical** remains open. Downstream registry consumers are **lem-route-f-prh-finish** and **lem-route-f-k-ledger**, both non-rigorous.

4 Amplification naturality of the functional calculus

Lemma 3 (`lem-compcb-amplification-naturality`). *Let $\mathcal{B}$ be a unital Banach algebra and let $\iota_n : \mathcal{B} \rightarrow M_n(\mathcal{B})$, $X \mapsto 1_{M_n} \otimes X$, be the unital amplification, an isometric unital homomorphism. Let f be given on $\|X - x_0 I\| < r$ by the power series*

$$f(X) = a_0 I + \sum_{m \geq 1} a_m (X - x_0 I)^m. \quad (9)$$

Then for every $X \in \mathcal{B}$ with $\|X - x_0 I\| < r$ the element $f(\iota_n(X))$ is defined and

$$f(\iota_n(X)) = \iota_n(f(X)). \quad (10)$$

In particular, whenever $\|(2P - I)^2 - I\| < 1$,

$$\theta(\iota_n(2P - I)) = \iota_n(\theta(2P - I)), \quad \theta(X) = \frac{1}{2}(I + \operatorname{sgn} X).$$

Transcription note. θ and sgn are the power-series functional calculus of `def-theta-idempotent-approximation`; only its definitional displays are in scope here. The norm is the ambient Banach-algebra norm.

Registry contract (`verbatim`).

Amplification naturality of the power-series functional calculus: let B be a unital Banach algebra, ι_n : $B \rightarrow M_n(B)$ the unital amplification $X \mapsto 1_{\{M_n\}}$ tensor X (an isometric unital homomorphism), and f given on $\|X - x_0 I\| < r$ by the power series $f(X) = a_0 I + \sum_{m \geq 1} a_m (X - x_0 I)^m$; then for every X in B with $\|X - x_0 I\| < r$, $f(\iota_n(X))$ is defined and $f(\iota_n(X)) = \iota_n(f(X))$; in particular, whenever $\|(2P - I)^2 - I\| < 1$, $\theta(\iota_n(2P - I)) = \iota_n(\theta(2P - I))$ with $\theta(X) = (I + \operatorname{sgn}(X))/2$.

Proof (prose account of the af-validated tree). *The general statement.* Three structural facts about ι_n do all the work. Unitality gives $\iota_n(I_B) = I_{M_n(B)}$, hence $\iota_n(X) - x_0 I = \iota_n(X - x_0 I)$; isometry then transports the convergence hypothesis verbatim, $\|\iota_n(X) - x_0 I\| = \|X - x_0 I\| < r$, so the series (9) evaluated at $\iota_n(X)$ is inside its domain; and multiplicativity gives $(\iota_n(X) - x_0 I)^m = \iota_n((X - x_0 I)^m)$ for every $m \geq 0$. Consequently the N -th partial sums correspond exactly,

$$T_N = a_0 I + \sum_{m=1}^N a_m (\iota_n(X) - x_0 I)^m = \iota_n(S_N), \quad S_N = a_0 I + \sum_{m=1}^N a_m (X - x_0 I)^m.$$

Since $S_N \rightarrow f(X)$ in $\mathcal{B}$ and ι_n is isometric, hence continuous, $T_N \rightarrow \iota_n(f(X))$. The power-series definition identifies the limit of the same sequence T_N with $f(\iota_n(X))$, so uniqueness of limits yields both definedness and (10). No property of ι_n beyond continuity, unitality and multiplicativity is used.

The θ specialization. Here the functional calculus is not a single series but a composite built from the inverse square root, so the tree first pins that branch down. With $c_m = \binom{-1/2}{m}$ one has $c_0 = 1$ and $c_m/c_{m-1} = -(2m-1)/(2m)$, whose modulus tends to 1; the ratio test therefore gives radius exactly one for $h(w) = \sum_{m \geq 0} c_m w^m$. On $|w| < 1$ termwise differentiation is legitimate and the coefficient recurrence is precisely the differential equation $2(1+w)h'(w) + h(w) = 0$; hence $\frac{d}{dw}[(1+w)h(w)^2] = 0$, and $h(0) = 1$ gives $(1+w)h(w)^2 = 1$. So h is the analytic inverse-square-root branch normalised at 1.

Transporting this to a unital Banach algebra: for $\|Z - I\| < 1$ the series $H = \sum_{m \geq 0} c_m (Z - I)^m$ converges absolutely in norm, and absolute convergence licenses the same Cauchy products as in the scalar identity, so $ZH^2 = I$; the normalisation $H = I + (\text{terms of positive degree})$ identifies H with the branch $Z^{-1/2}$ used by the registered calculus. Applying the general statement to this series with $Z = Y^2$, $Y = 2P - I$, gives $(\iota_n(Y)^2)^{-1/2} = \iota_n((Y^2)^{-1/2})$ whenever $\|Y^2 - I\| < 1$; multiplicativity then propagates the identity through $\text{sgn}(Y) = Y(Y^2)^{-1/2}$ and finally through $\theta = \frac{1}{2}(I + \text{sgn})$.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-amplification-naturality: 12 nodes, all validated, root validated, taint clean.

Role in the chain. This lemma exists because of a stall, and the shape of the campaign record is worth stating: three separate blocking challenges in the first orchestration of `lem-compcb-amplified-compression` named this one identity, so it was factored out into its own shard and validated first, bottom-up. It has no dependencies. It is imported by `lem-compcb-amplified-compression` (§5) and by `lem-compcb-entrywise-compression-naturality` (§7), and it is the reason every compression identity in the COMP tier holds at *every* amplification level rather than only at level one.

5 The amplified compression identity

Lemma 4 (`lem-compcb-amplified-compression`). *There is a universal $e_{\text{cmp}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{cmp}}$, every pair of δ -projections P, Q in an extended ε - C^* -algebra and every $n \geq 1$ satisfy*

$$1_{M_n} \otimes \text{Co}_{P,Q} = \text{Co}_{I_n \otimes P, I_n \otimes Q} \quad (11)$$

and

$$M_n \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{I_n \otimes P, I_n \otimes Q}. \quad (12)$$

Transcription note. $\text{Co}_{P,Q}$ and $\mathcal{S}_{P,Q}$ are the compression map and compressed corner of **def-compressed-corner**; δ -projections are as in **def-delta-projection**, and “extended” is as in **def-extended-epsilon-cstar-algebra**, so that every $M_n \otimes \mathcal{A}$ is itself an ε - C^* -algebra. On the left of (11) $1_{M_n} \otimes (\cdot)$ denotes the entrywise amplification of an operator on $\mathcal{A}$; on the right $I_n \otimes P$ is the amplified element.

Registry contract (`verbatim`).

Amplified compression identity: there is a universal $e_{\text{cmp}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{cmp}}$, every pair of δ -projections P, Q in an extended ε - C^* -algebra and every $n \geq 1$ satisfy $1_{\{M_n\}} \text{ tensor } \text{Co}_{\{P,Q\}} = \text{Co}_{\{I_n \text{ tensor } P, I_n \text{ tensor } Q\}}$ and $M_n \text{ tensor } \mathcal{S}_{\{P,Q\}} = \mathcal{S}_{\{I_n \text{ tensor } P, I_n \text{ tensor } Q\}}$.

Proof (prose account of the af-validated tree). *A uniform threshold.* Write $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Because $\mathcal{A}$ is *extended*, each $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra with compatible matrix multiplication, involution, unit and amplification norm; since $P_n^\dagger = P_n$ and $\|P_n^2 - P_n\| = \|I_n \otimes (P^2 - P)\| = \|P^2 - P\| \leq \delta$, the amplified elements are δ -projections *with the same δ , uniformly in n* . Next, the compression is the value of θ at a fixed operator: writing $A_{P,Q} = \frac{1}{2}(L_P R_Q + R_Q L_P)$ and $T_{P,Q} = 2A_{P,Q} - I$, **def-compressed-corner** supplies a single universal C with $\|A_{P,Q}^2 - A_{P,Q}\| \leq Ce$ in *every* ε - C^* -algebra, and the exact operator identity

$$T_{P,Q}^2 - I = 4(A_{P,Q}^2 - A_{P,Q}) \quad (13)$$

converts this into $\|T_{P,Q}^2 - I\| \leq 4Ce$. Since the same C applies inside every $M_n \otimes \mathcal{A}$ to the pair (P_n, Q_n) , choosing e_{cmp} with $4Ce_{\text{cmp}} < 1$ puts $T_{P,Q}$ and every T_{P_n, Q_n} simultaneously inside the θ/sgn domain of **def-theta-idempotent-approximation**. Uniformity of the threshold in n is exactly what (13) buys.

The amplification is entrywise, not diagonal. Left and right multiplication by the block-diagonal P_n, Q_n act slotwise: for $X = [x_{ij}]$ one has $L_{P_n} R_{Q_n}(X) = [P(x_{ij}Q)]$ and $R_{Q_n} L_{P_n}(X) = [(Px_{ij})Q]$, whence $T_{P_n, Q_n} = 1_{M_n} \otimes T_{P,Q}$. Here the tree records a correction that is genuinely load-bearing. The relevant map is *not* the diagonal embedding of $B(\mathcal{A})$ into $M_n(B(\mathcal{A}))$ but the entrywise operator amplification

$$\Gamma_n : B(\mathcal{A}) \rightarrow B(M_n \otimes \mathcal{A}), \quad \Gamma_n(U)([x_{ij}]) = [U(x_{ij})],$$

and the tree proves exactly the properties Γ_n has and no more: coordinate compression gives $\|x_{ij}\| \leq \|X\|$ and $\|E_{ij} \otimes y\| = \|y\|$, so $\|\Gamma_n(U)\| \leq n^2\|U\|$ — bounded at each fixed n , with *no* claim of isometry and *no* uniform-in- n operator bound; and entrywise evaluation gives $\Gamma_n(I) = I$ and $\Gamma_n(UV) = \Gamma_n(U)\Gamma_n(V)$. Boundedness at fixed n is all that is needed, because the θ series is a norm limit for each fixed n separately.

Transporting θ . Put $T = T_{P,Q}$ and $R = T^2 - I$. On $\|R\| < 1$ the registered calculus writes $\theta(T) = \frac{1}{2}(I + T \sum_{k \geq 0} b_k R^k)$ with b_k the Taylor coefficients of $(1 + z)^{-1/2}$ — the branch identified in `lem-compcb-amplification-naturality`. Since Γ_n is a unital algebra homomorphism it commutes with every partial sum and satisfies $\Gamma_n(R) = \Gamma_n(T)^2 - I$; boundedness lets one pass to the norm limit; and the threshold above guarantees that the limit at $\Gamma_n(T) = T_{P_n, Q_n}$ is the value $\theta(T_{P_n, Q_n})$. Hence $\Gamma_n(\theta(T_{P,Q})) = \theta(T_{P_n, Q_n})$, which by $\text{Co}_{P,Q} = \theta(T_{P,Q})$ is precisely (11).

Ranges. For any linear $C : \mathcal{A} \rightarrow \mathcal{A}$ and finite n one has $\text{Im}(1_{M_n} \otimes C) = M_n \otimes \text{Im } C$: every output has entries $C(x_{ij})$, and conversely finitely many entrywise preimages assemble into a single matrix preimage. Since $\mathcal{S}_{P,Q} = \text{Im } \text{Co}_{P,Q}$ by `def-compressed-corner`, applying this to (11) gives (12).

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-amplified-compression: 13 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-amplification-naturality`. This is the structural key-stone of the COMP tier: every later result that says “uniformly in n ” does so because (11)–(12) let a level-one fact be read at level n with no loss. It is imported by `lem-compcb-amplified-compression-identities`, `lem-compcb-entrywise-compression-naturality`, `lem-compcb-rectangular-product`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-compcb-single-compression-transfer` and `lem-hcb2-product-defect`.

6 Idempotence and adjoint symmetry of amplified compressions

Lemma 5 (`lem-compcb-amplified-compression-identities`). *There is a universal $e_{\text{cmp}} > 0$ such that, whenever $\mathcal{A}$ is an extended ε - C^* -algebra, $e = \delta + \varepsilon \leq e_{\text{cmp}}$, P, Q are δ -projections in $\mathcal{A}$, $n \geq 1$, and $X \in M_n \otimes \mathcal{A}$, one has*

$$\text{Co}_{P_n, Q_n}^2 = \text{Co}_{P_n, Q_n}, \quad \text{Co}_{P_n, Q_n}(X)^\dagger = \text{Co}_{Q_n, P_n}(X^\dagger), \quad (14)$$

where $P_n = I_n \otimes P$ and $Q_n = I_n \otimes Q$.

Registry contract (verbatim).

Amplified compression identities: there is a universal $e_{\text{cmp}} > 0$ such that, whenever $\mathcal{A}$ is an extended ε - C^* -algebra, $e = \delta + \varepsilon \leq e_{\text{cmp}}$, P, Q are δ -projections in $\mathcal{A}$, $n \geq 1$, and X is in M_n tensor $\mathcal{A}$, one has $\text{Co}_{\{P_n, Q_n\}}^2 = \text{Co}_{\{P_n, Q_n\}}$ and $\text{Co}_{\{P_n, Q_n\}}(X)^\dagger = \text{Co}_{\{Q_n, P_n\}}(X^\dagger)$, where $P_n = I_n$ tensor P and $Q_n = I_n$ tensor Q .

Proof (prose account of the af-validated tree). The argument is short because the amplification lemma has already done the structural work; what remains is to check that two *exact* level-one identities survive the passage to matrices.

Identification. Take e_{cmp} to be the universal constant supplied by `lem-compcb-amplified-compression`. For $e \leq e_{\text{cmp}}$ that lemma gives $\text{Co}_{P_n, Q_n} = 1_{M_n} \otimes \text{Co}_{P, Q}$, and — applied a second time to the *ordered* pair (Q, P) , which is a distinct instance — also $\text{Co}_{Q_n, P_n} = 1_{M_n} \otimes \text{Co}_{Q, P}$. Both amplified compressions are therefore canonical amplifications of level-one maps.

The level-one identities. Writing $T = \text{Co}_{P, Q}$ and $U = \text{Co}_{Q, P}$, **def-compressed-corner** states exactly $T^2 = T$ and $T(x)^\dagger = U(x^\dagger)$ for all $x \in \mathcal{A}$. These are equalities, not estimates; no smallness is consumed by them, and the threshold above is needed only for the identification step.

Entrywise transfer. Let T, U be any linear maps on a self-adjoint operator space satisfying those two identities, and let $T_n = 1_{M_n} \otimes T$, $U_n = 1_{M_n} \otimes U$. For $X = [x_{ij}]$, idempotence is checked slotwise,

$$(T_n^2 X)_{ij} = T(T(x_{ij})) = (T^2)(x_{ij}) = T(x_{ij}) = (T_n X)_{ij},$$

so $T_n^2 = T_n$. Adjoint symmetry combines the matrix involution with the slotwise action: the (i, j) entry of $T_n(X)^\dagger$ is $T(x_{ji})^\dagger = U(x_{ji}^\dagger)$, while the (i, j) entry of $U_n(X^\dagger)$ is $U((X^\dagger)_{ij}) = U(x_{ji}^\dagger)$. The transposition of indices is what makes the *swap* $\text{Co}_{P_n, Q_n} \mapsto \text{Co}_{Q_n, P_n}$ in (14) the right statement. Combining the identification with the transfer gives the lemma.

A recorded contract amendment. The export’s final node is a contract-reconciliation step, and it documents why the statement is shaped as it is. The originally codified contract left the ambient algebra unbound: a challenge observed that the proof used, but the contract never asserted, that $\mathcal{A}$ is an extended ε - C^* -algebra with P, Q in $\mathcal{A}$. The contract was amended to the prover’s corrected root text — the version reproduced above — as a mechanical, verdict-driven typing repair. The added clauses are the intended and already-used hypotheses of the source argument, so nothing was strengthened or weakened.

Status. proved; af: validated (T0). Workspace `proofs/lem-compcb-amplified-compression-identities`: 7 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression`. These two identities are the algebraic skeleton on which every quantitative COMP estimate is hung: idempotence is what makes “ X lies in the corner” equivalent to “ X is fixed by its compression”, and the dagger swap

is what lets an estimate proved on one side of a rectangular corner be read on the other. It is imported by `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`, `lem-hcb2-amplified-adjointness` and `lem-hcb1-variational-`

7 Entrywise compression naturality at a matrix unit

Lemma 6 (`lem-compcb-entrywise-compression-naturality`). *There is a universal $e_{\text{nat}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{nat}}$, Q is a δ -projection in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and E_{11} is the $(1,1)$ matrix unit of M_n , every $Z \in \mathcal{A}$ satisfies*

$$\text{Co}_{I_n \otimes Q, I_n \otimes Q}(E_{11} \otimes Z) = E_{11} \otimes \text{Co}_Q(Z). \quad (15)$$

Transcription note. Co_Q is the $P = Q$ abbreviation $\text{Co}_{Q,Q}$ of **def-compressed-corner**. The point of the statement is that the amplified compression does not merely preserve the corner but acts in the single occupied slot, leaving the zero blocks zero.

Registry contract (verbatim).

Entrywise compression naturality: there is a universal $e_{\text{nat}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{nat}}$, Q is a δ -projection in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and E_{11} is the $(1,1)$ matrix unit of M_n , every Z in $\mathcal{A}$ satisfies $\text{Co}_{I_n \otimes Q, I_n \otimes Q}(E_{11} \otimes Z) = E_{11} \otimes \text{Co}_Q(Z)$.

Proof (prose account of the af-validated tree). Set $e_{\text{nat}} := e_{\text{cmp}}$, the universal threshold supplied by `lem-compcb-amplified-compression`. That lemma is an equality of *linear maps* on $M_n(\mathcal{A})$, and the proof consists of specializing it and then evaluating.

Specializing the imported identity to the pair $(P, Q) = (Q, Q)$ gives

$$\text{Co}_{I_n \otimes Q, I_n \otimes Q} = 1_{M_n} \otimes \text{Co}_{Q,Q} = 1_{M_n} \otimes \text{Co}_Q,$$

the last equality being nothing but the abbreviation $\text{Co}_{Q,Q} = \text{Co}_Q$ recorded in **def-compressed-corner**. This is where the whole quantitative content of the statement sits: the threshold, the θ -domain control and the amplification argument are all inherited from `lem-compcb-amplified-compression`, which in turn rests on `lem-compcb-amplification-naturality`.

What remains is a one-line evaluation. For any linear $T : \mathcal{A} \rightarrow \mathcal{A}$ the tensor amplification acts on elementary tensors by $(1_{M_n} \otimes T)(X \otimes Z) = X \otimes T(Z)$; taking $T = \text{Co}_Q$ and $X = E_{11}$ gives (15).

Status. `proved; af: validated (T0)`. Workspace `proofs/lem-compcb-entrywise-compression-naturality`: 4 nodes, all `validated`, root `validated`, taint `clean`.

Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplification-naturality`. Like §4, this shard exists because of a recorded stall: the column-Hilbert workspace (§16) reached its third consecutive stall on exactly the identity (15), whose statement was then extracted mechanically from the blocking challenge and validated on its own. It is a strictly sharper statement than the $1_{M_n} \otimes$ -clause it specializes — that clause identifies two maps, whereas (15) asserts what happens to a *single-slot* element — and that sharper form is what the column-Hilbert estimate consumes. It has one registry consumer, `lem-hcb-column-hilbert-squared`.

8 The uniform rectangular compressed-product estimate

Lemma 7 (`lem-compcb-rectangular-product`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular pair A, B satisfies*

$$\|A \cdot B - AB\| \leq C_{\text{co}} e \|A\| \|B\|. \quad (16)$$

Transcription note. The tree fixes “compatible amplified rectangular pair” as an explicit abbreviation, and the quantifier in (16) ranges over exactly these data and no others: an extended ε - C^ -algebra $\mathcal{A}$; an integer $n \geq 1$; δ -projections P, Q, R in $\mathcal{A}$; the amplifications $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$, $R_n = I_n \otimes R$; elements $A \in \mathcal{S}_{P_n, Q_n}$ and $B \in \mathcal{S}_{Q_n, R_n}$; and the compressed product $A \cdot B = \text{Co}_{P_n, R_n}(AB)$ formed as in **def-compressed-corner** inside $M_n \otimes \mathcal{A}$. In particular both factors live in one square amplification $M_n \otimes \mathcal{A}$ — a restriction that later forces a separate row/column statement (§13).*

Registry contract (`verbatim`).

Uniform rectangular compressed-product estimate: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular pair satisfies $\|A \cdot B - AB\| \leq C_{\text{co}} * e * \|A\| * \|B\|$.

Proof (prose account of the af-validated tree). The shape of the argument is: unpack a level-one universal estimate into named constants, verify that the amplified data satisfy its hypotheses, and apply it inside $M_n \otimes \mathcal{A}$. No new analysis is done at level n ; the whole point is that none is needed.

Unpacking the level-one estimate. **def-compressed-corner** states, for any triple U, V, W of δ -projections in an ε - C^* -algebra, that $X \cdot Y = \text{Co}_{U, W}(XY)$ on $\mathcal{S}_{U, V} \times \mathcal{S}_{V, W}$ and that $\|X \cdot Y - XY\| \leq O(\delta + \varepsilon) \|X\| \|Y\|$. The tree makes the big-O explicit: there are constants $C_{\text{corner}} < \infty$ and $e_{\text{corner}} > 0$, independent of the algebra, of the projections and of the elements, such that the bound reads $C_{\text{corner}} e \|X\| \|Y\|$ whenever $e \leq e_{\text{corner}}$. This independence is the reason the final constant is uniform in n : the level-one estimate is a statement about *every* ε - C^* -algebra, and $M_n \otimes \mathcal{A}$ is one of them.

The amplified data are admissible. Because $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra for every n ; and P_n is a δ -projection whenever P is, since $P_n^\dagger = P_n$, $P_n^2 - P_n = I_n \otimes (P^2 - P)$ and the block-diagonal norm satisfies $\|I_n \otimes Z\| = \|Z\|$ — likewise for Q_n, R_n . Applying **lem-compcb-amplified-compression** to the three pairs (P, Q) , (Q, R) , (P, R) identifies the amplified corners with $M_n \otimes \mathcal{S}_{P, Q}$, $M_n \otimes \mathcal{S}_{Q, R}$, $M_n \otimes \mathcal{S}_{P, R}$ and the amplified compressions with the entrywise amplifications; **lem-compcb-amplified-compression** supplies the idempotence and dagger identities for those maps. Hence a compatible amplified rectangular pair really is an ordinary rectangular pair in $M_n \otimes \mathcal{A}$, and its amplified compressed product is literally the ordinary compressed product there, $A \cdot B = \text{Co}_{P_n, R_n}(AB)$, which lands in $\mathcal{S}_{P_n, R_n}$.

Assembly. Take

$$C_{\text{co}} = C_{\text{corner}}, \quad e_{\text{co}} = \min(e_{\text{corner}}, e_{\text{cmp}}),$$

with e_{cmp} the threshold of the amplification dependencies; both are universal, finite and positive. For $e \leq e_{\text{co}}$ the level-one estimate applies inside $M_n \otimes \mathcal{A}$ with $U = P_n$, $V = Q_n$, $W = R_n$, $X = A$, $Y = B$, and gives (16) with constants independent of n and of all the data.

Status. `proved`; `af`: `validated (T0)`. Workspace `proofs/lem-compcb-rectangular-product`: 12 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-i`. This is the workhorse estimate of the COMP tier — the licence to replace a compressed product by an ambient product at cost $O(e)$ — and it is invoked by almost everything downstream: `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-hcb0-compressed-associator`, `lem-compcb-corner-algebra`, `lem-hcb-column-hilbert-squared`, `lem-compcb-single-compression-transfer`, and `lem-hcb2-product-defect`. Its square-amplification compatibility clause is a real restriction, not a formality: §13 exists precisely because a row/column pair fails it.

9 Amplified almost-containment of corners

Lemma 8 (`lem-compcb-amplified-almost-containment`). *There are universal $C_{ac} < \infty$ and $e_{ac} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ac}$, the elements P_1, P, Q_1, Q are δ -projections with*

$$\|P_1 P - P_1\| \leq \delta, \quad \|Q_1 Q - Q_1\| \leq \delta, \quad (17)$$

and $n \geq 1$, every $X \in M_n \otimes \mathcal{S}_{P_1, Q_1}$ satisfies

$$\|\text{Co}_{P_n, Q_n}(X) - X\| \leq C_{ac} e \|X\|, \quad (18)$$

with $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Transcription note. Hypothesis (17) is the approximate form of “ $P_1 \leq P$ and $Q_1 \leq Q$ ”; the conclusion says that the inner corner $\mathcal{S}_{P_1, Q_1}$ sits almost inside the outer one, uniformly at every amplification level.

Registry contract (`verbatim`).

Amplified almost-containment: there are universal $C_{ac} < \text{infinity}$ and $e_{ac} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ac}$, P_1, P, Q_1, Q are δ -projections with $\|P_1 P - P_1\|, \|Q_1 Q - Q_1\| \leq \delta$, every $n \geq 1$, and X in $M_n \text{ tensor } \mathcal{S}_{\{P_1, Q_1\}}$, one has $\|\text{Co}_{\{P_n, Q_n\}}(X) - X\| \leq C_{ac} * e * \|X\|$.

Proof (prose account of the af-validated tree). *A uniform ledger.* All four elements amplify to δ -projections in $M_n \otimes \mathcal{A}$, and the containment hypotheses amplify too, with the involution converting one side into the other: $P_n P_{1,n} - P_{1,n} = I_n \otimes (P P_1 - P_1) = I_n \otimes (P_1 P - P_1)^\dagger$ has norm at most δ , and $Q_{1,n} Q_n - Q_{1,n} = I_n \otimes (Q_1 Q - Q_1)$ likewise. A crude but uniform norm bound is also available: for $\varepsilon, \delta \leq \frac{1}{2}$ the ε - C^* axioms and `def-delta-projection` give $(1 - \varepsilon)\|R_n\|^2 \leq \|R_n^2\| \leq \|R_n\| + \delta$, hence $\|R_n\| \leq B := 3$ for each of the four. Finally `def-compressed-corner` supplies one universal k_{co} with

$$\|\text{Co}_{R,S}(Z) - R(ZS)\| \leq k_{co} e \|Z\|, \quad \|\text{Co}_{R,S}(Z) - (RZ)S\| \leq k_{co} e \|Z\|, \quad (19)$$

in every $M_n \otimes \mathcal{A}$. The threshold e_0 is the minimum of $\frac{1}{2}$, the threshold behind (19), and those of the two amplification dependencies.

Control in the source corner. By `lem-compcb-amplified-compression`, $M_n \otimes \mathcal{S}_{P_1, Q_1} = \mathcal{S}_{P_{1,n}, Q_{1,n}}$, so X lies in the image of $\text{Co}_{P_{1,n}, Q_{1,n}}$; idempotence (`lem-compcb-amplified-compression-identities`) then upgrades membership to the fixed-point identity $\text{Co}_{P_{1,n}, Q_{1,n}}(X) = X$. Feeding this into (19) with $R = P_{1,n}$, $S = Q_{1,n}$, $Z = X$ replaces the compression by X itself and yields the two bracketings

$$\|X - P_{1,n}(XQ_{1,n})\| \leq k_{co} e \|X\|, \quad \|X - (P_{1,n}X)Q_{1,n}\| \leq k_{co} e \|X\|. \quad (20)$$

Transfer to the outer projections. The claim is that X is almost invariant under P_n on the left and Q_n on the right, with a universal $L := k_{co}(2B + 1) + 2B(B^2 + 2)$. For the left estimate put $D = P_{1,n}(XQ_{1,n})$ and $Y = XQ_{1,n}$, and telescope

$$P_n X - X = P_n(X - D) + (P_n D - D) + (D - X).$$

The outer terms are controlled by (20) and submultiplicativity. The middle term is where the containment hypothesis and the *approximate* associativity of an ε - C^* -algebra both enter: $P_n D - D = P_n(P_{1,n}Y) - P_{1,n}Y$ is bounded by $[\varepsilon B^2 + (1 + \varepsilon)\delta]\|Y\|$, since re-associating costs $\varepsilon\|P_n\|\|P_{1,n}\|\|Y\|$

and $(P_n P_{1,n} - P_{1,n})Y$ costs $(1 + \varepsilon)\delta\|Y\|$. With $\|Y\| \leq 2B\|X\|$ and $\varepsilon, \delta \leq e$ the three terms sum to at most $Le\|X\|$. The right estimate is the mirror image, run with $D = (P_{1,n}X)Q_{1,n}$, $Y = P_{1,n}X$ and the hypothesis $\|Q_{1,n}Q_n - Q_{1,n}\| \leq \delta$.

Assembly. Insert $P_n(XQ_n)$ and P_nX :

$$\|\text{Co}_{P_n, Q_n}(X) - X\| \leq \|\text{Co}_{P_n, Q_n}(X) - P_n(XQ_n)\| + \|P_n(XQ_n - X)\| + \|P_nX - X\|.$$

The first term is $\leq k_{co}e\|X\|$ by (19); the second is $\leq (1 + \varepsilon)\|P_n\|Le\|X\| \leq 2BLE\|X\|$; the third is $\leq Le\|X\|$. Hence (18) holds with $C_{ac} = k_{co} + (2B + 1)L$ and $e_{ac} = e_0$, both finite, positive and universal.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-amplified-almost-containment: 13 nodes, all validated, root validated, taint clean.

Role in the chain. Imports lem-compcb-amplified-compression and lem-compcb-amplified-compression-i

It is the corner-inclusion primitive of the COMP tier: it says that once a corner is approximately contained in another, the outer compression is a near-identity on it, again with one constant at every level. Its registry consumer is lem-compcb-single-compression-transfer (§17).

10 The compressed-unit norm estimate

Lemma 9 (`lem-compcb-compressed-unit-norm`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$,*

$$\|u_T\| \leq 1 + C_{\text{co}} e \quad \text{for every } \delta\text{-projection } T, \quad (21)$$

and

$$|\|u_T\| - 1| \leq C_{\text{co}} e \quad \text{for every nonvanishing } T. \quad (22)$$

Transcription note. $u_T = \text{Co}_T(T)$ is the compressed unit of **def-compressed-corner**; “nonvanishing” is the second branch of the alternative in **def-delta-projection**. The asymmetry is deliberate and load-bearing: the two-sided bound is false in general, since a δ -projection may fall in the small branch $\|T\| = O(\delta)$, in which case u_T is small rather than close to a unit.

Registry contract (verbatim).

Compressed-unit norm estimate: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every δ -projection T satisfies $\|u_T\| \leq 1 + C_{\text{co}} * e$, and every nonvanishing T satisfies $\text{abs}(\|u_T\| - 1) \leq C_{\text{co}} * e$.

Proof (prose account of the af-validated tree). This shard’s tree is instructive twice over, because its first attempt was audited and found unsound, and the audit — not the estimate — is what is validated in the early nodes.

The blocked first attempt. The originally registered inputs (**def-extended-epsilon-cstar-algebra** together with **def-compressed-corner**) yield only the δ -projection dichotomy, the identity $u_T = \text{Co}_T(T)$, and the comparison $\|\text{Co}_T - L_T R_T\| \leq ke$. They do *not* yield the quantitative multiplication bound $\|XY\| \leq (1 + \varepsilon)\|X\|\|Y\|$ — the extended definition says only that each $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra, without supplying that term’s quantitative axioms — nor any unit-norm axiom that would give (22) from “ $\mathcal{S}_T$ is an $O(e)$ - C^* -algebra with unit u_T ”. A closure audit node records this verdict explicitly and refuses the root: without those axioms there is no sound bridge from the permitted premises to the contract.

The repair. The missing quantitative content was then registered in the workspace as byte-matched externals — the ε - C^* norm axioms (**epsilon-banach-cstar-norm-axioms**) and the nonvanishing alternative — and each $O(\cdot)$ in them unpacked into a named universal constant after shrinking the threshold. That gives universal $a, b, k < \infty$ and $e_0 > 0$ such that for $e \leq e_0$: (i) every δ -projection T satisfies $\|T\| \leq a\delta$ or is nonvanishing; (ii) $\|XY\| \leq (1 + \varepsilon)\|X\|\|Y\|$; (iii) $u_T = \text{Co}_T(T)$ with $\|\text{Co}_T - L_T R_T\| \leq ke$; and (iv) nonvanishing T obeys $|\|T\| - 1| \leq be$.

Small branch. Suppose $\|T\| \leq a\delta$ and $e \leq \min(e_0, 1)$. Since $(L_T R_T)(T) = T(T^2)$, two applications of (ii) give $\|(L_T R_T)(T)\| \leq (1 + \varepsilon)^2 \|T\|^3 \leq 4a^3 e^3 \leq 4a^3 e$, while (iii) gives $\|u_T - (L_T R_T)(T)\| \leq ke\|T\| \leq kae^2 \leq kae$. Hence $\|u_T\| \leq De$ with $D = 4a^3 + ka$, and in particular $\|u_T\| \leq 1 + De$. Note that here the compressed unit is *small*, which is why no two-sided statement can hold.

Nonvanishing branch. Shrink e_0 so that $e_0 \leq 1$ and $be_0 \leq 1$; then (iv) gives $\|T\| \leq 2$. The compressed unit is compared directly with T rather than through any unit axiom: bilinearity gives $T(T^2) - T = T(T^2 - T) + (T^2 - T)$, so by (iii), (ii) and $\|T^2 - T\| \leq \delta$,

$$\|u_T - T\| \leq \|u_T - (L_T R_T)(T)\| + \|T(T^2) - T\| \leq ke\|T\| + (1 + \varepsilon)\|T\|\delta + \delta \leq (2k + 5)e.$$

The reverse triangle inequality together with (iv) then gives $|\|u_T\| - 1| \leq (2k + 5 + b)e$, hence also the upper bound.

Assembly. With e_{co} the shrunk threshold and $C_{\text{co}} = \max(4a^3 + ka, 2k + 5 + b)$, the dichotomy routes every δ -projection to one of the two branches: the first gives (21), the second gives (21) and (22). The final assembly node states expressly that it uses neither the superseded audit node nor any compressed-unit norm axiom.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-compressed-unit-norm: 15 nodes (12 validated, 3 superseded nodes archived), root validated, taint clean.

Role in the chain. Imports lem-compcb-amplified-compression and lem-compcb-amplified-compression-i. It supplies the unit-size input to lem-compcb-corner-algebra (§15) and to lem-hcb-column-hilbert-squared (§16); in the latter it is applied to the one-dimensional projection Q of an H-CB datum, which is why the nonvanishing hypothesis of (22) has to be discharged there by a separate bridge rather than assumed.

11 Exact adjointness of the amplified Ha map

Lemma 10 (`lem-hcb2-amplified-adjointness`). *There is a universal $e_{\text{adj}} > 0$ such that every H-CB datum with $e \leq e_{\text{adj}}$, every $n \geq 1$ and every $Z \in M_n \otimes \mathcal{S}_{P,R}$ satisfy*

$$(\text{Ha}_{P,R}^Q)_n(Z)^\dagger = (\text{Ha}_{R,P}^Q)_n(Z^\dagger). \quad (23)$$

Transcription note. The datum is that of `def-hcb-datum` and $\text{Ha}_{P,R}^Q$ is the map of `def-ha-map`; $e = \delta + \varepsilon$. The two daggers in (23) are different operations: on the left it is the Hilbert-space adjoint of the operator $(\text{Ha}_{P,R}^Q)_n(Z)$ between amplified column spaces, on the right it is the involution of the ambient algebra applied to Z . The identity is exact — no $O(e)$ error — and the threshold is needed only so that the sesquilinear forms involved are genuine inner products.

Registry contract (`verbatim`).

Exact amplified Ha adjointness: there is a universal $e_{\text{adj}} > 0$ such that every H-CB datum with $e \leq e_{\text{adj}}$, every $n \geq 1$, and every Z in M_n tensor $\mathcal{S}_{\{P,R\}}$ satisfy $(\text{Ha}^Q_{P,R})_n(Z)^\dagger = (\text{Ha}^Q_{R,P})_n(Z^\dagger)$.

Proof (prose account of the af-validated tree). *Threshold.* Set $e_{\text{adj}} = \min(e_{\text{cmp}}, e_{\text{ip}})$, where e_{cmp} comes from `lem-compcb-amplified-compression-identities` and e_{ip} is a universal smallness threshold below which the Euclidean sesquilinear forms of `def-ha-map` and of the column-Hilbert displays are genuine inner products — such a threshold exists because those forms agree with the norms up to a $1 \pm O(e)$ factor. Fix an arbitrary datum, n and Z .

Corner symmetries. Applying `lem-compcb-amplified-compression-identities` at $n = 1$: if $a \in \mathcal{S}_{A,B} = \text{Im } \text{Co}_{A,B}$ then $a = \text{Co}_{A,B}(a)$, so $a^\dagger = \text{Co}_{B,A}(a^\dagger)$ and $a^\dagger \in \mathcal{S}_{B,A}$; and since $Q^\dagger = Q$, the compressed unit $\tilde{Q} = \text{Co}_Q(Q)$ is Hermitian.

Exact product reversal. The pivotal algebraic fact is

$$(a \cdot b)^\dagger = b^\dagger \cdot a^\dagger \quad (a \in \mathcal{S}_{A,B}, b \in \mathcal{S}_{B,C}), \quad (24)$$

and it is *exact*. The proof is three substitutions: the compressed-product display gives $a \cdot b = \text{Co}_{A,C}(ab)$; the imported identities at $n = 1$ give $\text{Co}_{A,C}(ab)^\dagger = \text{Co}_{C,A}((ab)^\dagger)$; the involution axiom gives $(ab)^\dagger = b^\dagger a^\dagger$; and a second use of the display, legitimate because the reversed corner memberships were just established, gives $\text{Co}_{C,A}(b^\dagger a^\dagger) = b^\dagger \cdot a^\dagger$. No associativity estimate and no approximation enter. The tree records a correction here worth one sentence: an earlier audit node concluded that (24) was *not* derivable, because the formula $a \cdot b = \text{Co}_{A,C}(ab)$ had not been registered as an allowed input in the workspace; after the compressed-product display was registered, that negative conclusion — and the associated demand for further definition provisioning — was explicitly withdrawn.

Level one. Fix $z \in \mathcal{S}_{P,R}$ and put $T = \text{Ha}_{P,R}^Q(z)$. The defining relation of `def-ha-map` reads, for all $x \in \mathcal{S}_{R,Q}$ and $y \in \mathcal{S}_{P,Q}$,

$$(y^\dagger \cdot z) \cdot x + y^\dagger \cdot (z \cdot x) = 2 \langle y, Tx \rangle \tilde{Q}. \quad (25)$$

Apply † to (25). On the left, (24) reverses each compressed product, turning the two summands into $x^\dagger \cdot (z^\dagger \cdot y)$ and $(x^\dagger \cdot z^\dagger) \cdot y$. On the right, $\tilde{Q}$ is Hermitian and the scalar is conjugated,

and conjugate symmetry together with the definition of the Hilbert-space adjoint gives $\overline{\langle y, Tx \rangle} = \langle Tx, y \rangle = \langle x, T^\dagger y \rangle$. Thus for all x, y ,

$$(x^\dagger \cdot z^\dagger) \cdot y + x^\dagger \cdot (z^\dagger \cdot y) = 2 \langle x, T^\dagger y \rangle \tilde{Q}.$$

This is precisely the defining relation of **def-ha-map** for the pair (R, P) , the symbol $z^\dagger$, the input y and the test vector x ; and that relation determines its solution uniquely. Hence $\text{Ha}_{R,P}^Q(z^\dagger)y = T^\dagger y$ for every y , i.e. $\text{Ha}_{R,P}^Q(z^\dagger) = \text{Ha}_{P,R}^Q(z)^\dagger$.

Amplification. Write $Z = [Z_{ij}]$. By the amplification convention of the contract, $(\text{Ha}_{P,R}^Q)_n(Z)$ is the block operator $[\text{Ha}_{P,R}^Q(Z_{ij})]$ from the n -fold Hilbert direct sum of $\mathcal{S}_{R,Q}$ to that of $\mathcal{S}_{P,Q}$; its Hilbert-space adjoint therefore has (i, j) block $\text{Ha}_{P,R}^Q(Z_{ji})^\dagger$. On the other side, $(Z^\dagger)_{ij} = Z_{ji}^\dagger$, which lies in $\mathcal{S}_{R,P}$ by the corner symmetry above, so the (i, j) block of $(\text{Ha}_{R,P}^Q)_n(Z^\dagger)$ is $\text{Ha}_{R,P}^Q(Z_{ji}^\dagger)$. The level-one identity equates the two blockwise, and equality of all blocks is (23). Since n and Z were arbitrary, the lemma follows.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb2-amplified-adjointness: 21 nodes, all validated, root validated, taint clean.

Role in the chain. Imports lem-compcb-amplified-compression-identities. Being exact rather than approximate, it is what allows a C^* -identity argument to be run on the amplified Ha map: it is used in lem-hcb3-diagonal-upper-norm (§22) to identify $(\text{Ha}_{P,P}^Q)_n(Z)^\dagger$ with $(\text{Ha}_{P,P}^Q)_n(Z^\dagger)$ inside $\|T\|^2 = \|T^\dagger T\|$, and the same role in lem-hcb3-diagonal-lower-modulus (§24). It is also a declared dependency of lem-hcb3-offdiagonal-inverse (§26) and lem-hcb4-canonical-close (§28), both now af: validated.

12 The uniform compressed-unit action

Lemma 11 (`lem-compcb-compressed-unit-action`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular corner satisfies*

$$\|u_T \cdot A - A\| \leq C_{\text{co}} e \|A\|, \quad \|A \cdot u_R - A\| \leq C_{\text{co}} e \|A\|. \quad (26)$$

Transcription note. As spelled out in the tree, the datum is an extended ε - C^* -algebra $\mathcal{A}$, an $n \geq 1$, δ -projections T, R with $T_n = I_n \otimes T$, $R_n = I_n \otimes R$, and an element $A \in \mathcal{S}_{T_n, R_n}$; the compressed units are $u_{T,n} = \text{Co}_{T_n, T_n}(T_n)$ and $u_{R,n} = \text{Co}_{R_n, R_n}(R_n)$. By `lem-compcb-amplified-compression` these equal $I_n \otimes \text{Co}_T(T)$ and $I_n \otimes \text{Co}_R(R)$, which is the notation u_T, u_R of the contract.

Registry contract (`verbatim`).

Uniform compressed-unit action: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular corner satisfies $\|u_T \cdot A - A\| \leq C_{\text{co}} e \|A\|$ and $\|A \cdot u_R - A\| \leq C_{\text{co}} e \|A\|$.

Proof (prose account of the af-validated tree). The estimate is assembled in three layers: replace the compressed units by the projections, act with the projections in the *ambient* product, then replace the ambient product by the compressed one.

Primitive bounds. Since $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra and T_n, R_n are δ -projections there (adjoints and products amplify entrywise and $\|I_n \otimes Z\| = \|Z\|$). Unpacking the $O(\cdot)$ terms of `def-delta-projection` and `def-compressed-corner` gives universal $a, k < \infty$ and $e_0 \leq 1$ such that every δ -projection S obeys $\|S\| \leq a\delta$ or $\|S\| \leq 1 + ae$ — so $\|T_n\|, \|R_n\| \leq M := \max(a, 1 + a)$ in either branch — and such that each compression is within ke of both $L \cdot R$ orderings. Idempotence of the amplified compression (`lem-compcb-amplified-compression-identities`) also gives the fixed-point identity $A = \text{Co}_{T_n, R_n}(A)$.

Compressed unit versus projection. For $S \in \{T_n, R_n\}$ and $u_S = \text{Co}_{S, S}(S)$ one has $(L_S R_S)(S) = S(S^2)$, and bilinearity gives $S(S^2) - S = S(S^2 - S) + (S^2 - S)$. With $\|S^2 - S\| \leq \delta$, the multiplication bound, and $\|\text{Co}_{S, S} - L_S R_S\| \leq ke$,

$$\|u_S - S\| \leq ke\|S\| + (1 + \varepsilon)\|S\|\delta + \delta \leq K_u e, \quad K_u := kM + 2M + 1, \quad (27)$$

and hence $\|u_S\| \leq M + K_u$, uniformly in n .

Ambient action of the projections. Put $D = T_n(AR_n)$. The fixed-point identity and the compression comparison give $\|A - D\| \leq ke\|A\|$, and one telescopes

$$T_n A - A = T_n(A - D) + (T_n D - D) + (D - A).$$

The outer terms cost $2Mke\|A\|$ and $ke\|A\|$. For the middle term write $Y = AR_n$, so that $T_n D - D = T_n(T_n Y) - T_n Y$; approximate associativity contributes $\varepsilon M^2 \|Y\|$ and $\|T_n^2 - T_n\| \leq \delta$ contributes $2\delta\|Y\|$, and $\|Y\| \leq 2M\|A\|$. Altogether $\|T_n A - A\| \leq K_T e\|A\|$ with $K_T = 2Mk + 2M(M^2 + 2) + k$; the right-hand estimate $\|AR_n - A\| \leq K_R e\|A\|$ is the mirror argument with $D = (T_n A)R_n$ and the same constant. Substituting the units via (27) — $\|(u_{T,n} - T_n)A\| \leq 2K_u e\|A\|$ and symmetrically — gives the ambient form

$$\|u_{T,n} A - A\| \leq K_a e\|A\|, \quad \|A u_{R,n} - A\| \leq K_a e\|A\|, \quad K_a := \max(K_T, K_R) + 2K_u. \quad (28)$$

From ambient to compressed. The pairs $(u_{T,n}, A)$ and $(A, u_{R,n})$ are compatible amplified rectangular pairs in the exact sense required by `lem-compcb-rectangular-product`, which therefore gives universal C_p with $\|u_{T,n} \cdot A - u_{T,n}A\| \leq C_p e \|u_{T,n}\| \|A\|$ and its mirror. Combining with (28) and $\|u_S\| \leq M + K_u$ yields (26) with

$$C_{\text{co}} = K_a + C_p (M + K_u), \quad e_{\text{co}} = \min(e_0, e_p, e_{\text{cmp}}, e_{\text{id}}, 1),$$

all universal and independent of $n, \mathcal{A}, T, R, A$.

A bookkeeping remark: the later nodes of the export re-derive (27)–(28) in “dependency-closed” form, so that the final assembly rests only on validated premises and not on any node still pending at the time. The mathematics is unchanged; the restructuring is what let the tree close bottom-up.

Status. `proved`; `af`: `validated (T0)`. Workspace `proofs/lem-compcb-compressed-unit-action`: 20 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-iden` and `lem-compcb-rectangular-product`. It supplies the unit axiom needed by `lem-compcb-corner-algebra` (§15) — a corner cannot be an approximate C^* -algebra without an approximate unit — and it is the input that drives the diagonal unit estimate `lem-hcb3-diagonal-unit` (§21).

13 The row-column compressed-product estimate

Lemma 12 (`lem-compcb-row-column-product`). *There are universal $C_{\text{rc}} < \infty$ and $e_{\text{rc}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{rc}}$, P, Q are δ -projections in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and $X, Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$, one has*

$$\|\text{Co}_Q(Y^\dagger X) - Y^\dagger X\| \leq C_{\text{rc}} e \|Y\| \|X\|, \quad (29)$$

where $Y^\dagger X$ is the ambient product of the $1 \times n$ row $Y^\dagger$ with the $n \times 1$ column X . Transcription note. The product $Y^\dagger X$ is a single element of $\mathcal{A}$, not a matrix: the row and column contract. This is exactly why `lem-compcb-rectangular-product` does not apply directly — its compatibility clause requires both factors inside one square amplification.

Registry contract (`verbatim`).

Row-column compressed-product estimate: there are universal $C_{\text{rc}} < \infty$ and $e_{\text{rc}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{rc}}$, P, Q are δ -projections in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and X, Y are in $M_{\{n,1\}} \otimes \mathcal{S}_{\{P,Q\}}$, one has $\|\text{Co}_Q(Y^\dagger X) - Y^\dagger X\| \leq C_{\text{rc}} e \|Y\| \|X\|$, where $Y^\dagger X$ is the ambient product of the 1-by- n row $Y^\dagger$ with the n -by-1 column X .

Proof (prose account of the af-validated tree). Set $C_{\text{rc}} = C_{\text{co}}$ and $e_{\text{rc}} = \min(e_{\text{co}}, e_{\text{cmp}})$ with the constants of `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`, and fix arbitrary admissible data. The whole proof is a change of ambient size: embed the row/column pair as off-diagonal blocks of one square amplification, apply the validated square estimate there, and read the result off the single occupied entry.

The embedding. Use the decomposition $\mathbb{C}^{n+1} = \mathbb{C}^n \oplus \mathbb{C}$ and let

$$C_X = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \quad R_Y = \begin{pmatrix} 0 & 0 \\ Y^\dagger & 0 \end{pmatrix}$$

in $M_{n+1} \otimes \mathcal{A}$. Writing $X = (X_i)$ and $Y = (Y_i)$ with entries in $\mathcal{S}_{P,Q}$, the dagger identity of `def-compressed-corner` gives $Y_i^\dagger \in \mathcal{S}_{Q,P}$, so X has entries in $\mathcal{S}_{P,Q}$ and $Y^\dagger$ has entries in $\mathcal{S}_{Q,P}$ with the indicated orientations. Since $e \leq e_{\text{cmp}}$, `lem-compcb-amplified-compression` at level $n+1$ identifies $M_{n+1} \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{I_{n+1} \otimes P, I_{n+1} \otimes Q}$ and the transposed statement, so zero-padding puts C_X and R_Y in exactly those compatible corners. The padding is isometric: in the operator-space matrix norms, insertion by a scalar coordinate isometry and extraction by its adjoint are both contractive, hence equalities; and involution is isometric because $\mathcal{A}$ is self-adjoint. Therefore

$$\|C_X\| = \|X\|, \quad \|R_Y\| = \|Y^\dagger\| = \|Y\|, \quad (30)$$

and (R_Y, C_X) is a compatible *square* amplified rectangular pair — the right projection of the first factor and the left projection of the second are both $I_{n+1} \otimes P$.

One occupied entry. Ordinary block multiplication gives

$$R_Y C_X = \text{diag}(0_{n \times n}, Y^\dagger X),$$

every other block containing a zero factor. On the compressed side, `def-compressed-corner` applied to the triple $(I_{n+1} \otimes Q, I_{n+1} \otimes P, I_{n+1} \otimes Q)$ gives $R_Y \cdot C_X = \text{Co}_{I_{n+1} \otimes Q}(R_Y C_X)$, using the

abbreviation $\text{Co}_{S,S} = \text{Co}_S$; and `lem-compcb-amplified-compression` applied to the pair (Q, Q) at level $n + 1$ says $\text{Co}_{I_{n+1} \otimes Q} = 1_{M_{n+1}} \otimes \text{Co}_Q$, an entrywise map. Applying it to the diagonal block matrix above,

$$R_Y \cdot C_X = \text{diag}(0_{n \times n}, \text{Co}_Q(Y^\dagger X)). \quad (31)$$

Conclusion. Since $e \leq e_{\text{co}}$ and (R_Y, C_X) is compatible, `lem-compcb-rectangular-product` yields $\|R_Y \cdot C_X - R_Y C_X\| \leq C_{\text{co}} e \|R_Y\| \|C_X\|$. By (31) the left-hand difference is the zero-padded element with single entry $\text{Co}_Q(Y^\dagger X) - Y^\dagger X$, whose norm equals the norm of that entry by the same corner isometry as in (30); and (30) converts the right-hand side. This is (29) with $C_{\text{rc}} = C_{\text{co}}$. Both thresholds are positive universal constants, so $e_{\text{rc}} > 0$, and universal generalisation over the arbitrary data completes the proof.

Status. `proved`; `af`: `validated` (T0). Workspace `proofs/lem-compcb-row-column-product`: 20 nodes (18 `validated`, 2 superseded nodes archived), root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. This shard was factored out of the column-Hilbert workspace at a recorded third stall: the blocking challenge observed that the validated rectangular-product contract quantifies only over square amplifications, so the pair $(Y^\dagger, X)$ — a $1 \times n$ row against an $n \times 1$ column — simply does not satisfy its hypotheses, and the gap had to be closed by a statement of its own rather than by an instantiation. It has one registry consumer, `lem-hcb-column-hilbert-squared` (§16), where $Y = X$ and the estimate becomes the bridge between the column inner product and the ambient norm.

14 The uniform compressed associator estimate

Lemma 13 (lem-hcb0-compressed-associator). *There are universal $C_{\text{as}} < \infty$ and $e_{\text{as}} > 0$ such that every H -CB datum with $e \leq e_{\text{as}}$ and all compatible amplified rectangular A, B, C satisfy*

$$\|(A \cdot B) \cdot C - A \cdot (B \cdot C)\| \leq C_{\text{as}} e \|A\| \|B\| \|C\|. \quad (32)$$

Transcription note. *The left-hand side is the compressed associator of **def-compressed-associator**; “compatible amplified rectangular” is the datum fixed in §8, here for a composable triple. Both the compressed product and the ambient product are only approximately associative — the ambient one by the ε -associator axiom of an ε - C^* -algebra — so the content is that the two failures add up to a single $O(e)$.*

Registry contract (verbatim).

Uniform compressed associator: there are universal $C_{\text{as}} < \text{infinity}$ and $e_{\text{as}} > 0$ such that every H -CB datum with $e \leq e_{\text{as}}$ and all compatible amplified rectangular A, B, C satisfy $\|(A \cdot B) \cdot C - A \cdot (B \cdot C)\| \leq C_{\text{as}} * e * \|A\| \|B\| \|C\|$.

Proof (prose account of the af-validated tree). Let $C_{\text{co}}, e_{\text{co}}$ be the universal witnesses of lem-compcb-rectangular-product and set

$$e_{\text{as}} = \min(e_{\text{co}}, 1), \quad C_{\text{as}} = 2C_{\text{co}}(2 + C_{\text{co}}) + 4C_{\text{co}} + 1.$$

A product-norm bound. For a compatible pair, the rectangular estimate plus the ambient bound $\|AB\| \leq (1 + \varepsilon)\|A\|\|B\|$ from the ε - C^* norm axioms give

$$\|A \cdot B\| \leq (1 + \varepsilon + C_{\text{co}}e)\|A\|\|B\|, \quad (33)$$

and likewise for $B \cdot C$. This is what keeps the outer terms of the telescope below from degrading.

The telescope. The compressed associator has the exact five-term expansion

$$\begin{aligned} (A \cdot B) \cdot C - A \cdot (B \cdot C) &= [(A \cdot B) \cdot C - (A \cdot B)C] + [(A \cdot B)C - (AB)C] \\ &\quad + [(AB)C - A(BC)] + [A(BC) - A(B \cdot C)] + [A(B \cdot C) - A \cdot (B \cdot C)], \end{aligned} \quad (34)$$

which routes both associations through the ambient product and isolates the ambient associator in the middle.

Write $N = \|A\|\|B\|\|C\|$. The first and fifth terms are rectangular-product errors for the compatible pairs $(A \cdot B, C)$ and $(A, B \cdot C)$, so by (33) each is at most $C_{\text{co}}e(1 + \varepsilon + C_{\text{co}}e)N$. The second and fourth are the rectangular-product errors for (A, B) and (B, C) multiplied by one ambient factor — bilinearity gives $(A \cdot B)C - (AB)C = (A \cdot B - AB)C$ and $A(BC) - A(B \cdot C) = A(BC - B \cdot C)$ — so each is at most $(1 + \varepsilon)C_{\text{co}}eN$. The third is precisely the ε -associator axiom, $\|(AB)C - A(BC)\| \leq \varepsilon N$.

Summation. Adding and using $0 \leq \varepsilon \leq e \leq 1$, so that $1 + \varepsilon + C_{\text{co}}e \leq 2 + C_{\text{co}}$ and $1 + \varepsilon \leq 2$, gives (32) with the stated C_{as} .

Two pieces of hygiene in the tree are worth recording, since they shape the constant. The summation is written so that the ambient-associator term is bounded directly by eN rather than as $(\varepsilon/e) \cdot eN$, because at the endpoint $e = 0$ one has $\varepsilon = 0$ and every error vanishes — the quotient ε/e is never formed. And the upstream witness is normalised to $C_{\text{co}} := \max\{C_{\text{co}}, 0\}$, which cannot

weaken the imported estimate since $e\|A\|\|B\| \geq 0$, but does make the sign-sensitive arithmetic above valid.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb0-compressed-associator: 20 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-rectangular-product`. It is the base of the H-CB tier: the defining relation of the Ha map (`def-ha-map`) pairs an element against precisely the difference of the two associations $(Y^\dagger \cdot Z) \cdot X$ and $Y^\dagger \cdot (Z \cdot X)$, so (32) is exactly the estimate that turns that relation into a quantitative bound. It feeds `lem-hcb1-column-action` (§19) and `lem-hcb2-product-defect` (§20).

15 The compressed corner as an extended approximate C^* -algebra

Lemma 14 (`lem-compcb-corner-algebra`). *There are universal $C_{ca} < \infty$ and $e_{ca} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ca}$ and P is a nonvanishing δ -projection, the corner $\mathcal{S}_P$ — equipped with the compressed product, the inherited involution and the compressed unit $u_P = \text{Co}_P(P)$ — is an extended $C_{ca}e$ - C^* -algebra. Transcription note. “Extended” is as in `def-extended-epsilon-cstar-algebra`: not merely that $\mathcal{S}_P$ satisfies the approximate C^* axioms, but that every $M_n \otimes \mathcal{S}_P$ does, with the same parameter $C_{ca}e$, independent of n . That uniformity is the whole content.*

Registry contract (`verbatim`).

Uniform compressed-corner algebra: there are universal $C_{ca} < \text{infinity}$ and $e_{ca} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ca}$ and P is a nonvanishing δ -projection, $\mathcal{S}_P$ with the compressed product, inherited involution, and compressed unit $u_P = \text{Co}_P(P)$ is an extended $C_{ca}e$ - C^* -algebra.

Proof (prose account of the af-validated tree). The tree is organised as one node per axiom family, over a shared constant ledger, with a preliminary step that moves everything to level n at once.

Constant ledger and amplified setup. Let C_r, e_r come from `lem-compcb-rectangular-product`; C_u, e_u from `lem-compcb-compressed-unit-action`; C_N, e_N from `lem-compcb-compressed-unit-norm`; and e_i from `lem-compcb-amplified-compression-identities`. Put

$$K_p = C_r + 1, \quad K_a = 2C_r(K_p + 3) + 1, \quad C_{ca} = \max\{K_p, K_a, C_u, C_N\}, \quad e_{ca} = \min\{e_r, e_u, e_N, e_i, 1\}.$$

With $P_n = I_n \otimes P$, the tree then *derives inside itself* — from `def-compressed-corner` and the registered θ calculus, rather than importing it — the naturality $\text{Co}_{P_n} = \text{id}_{M_n} \otimes \text{Co}_P$: writing $T_P = L_P R_P + R_P L_P - 1$, multiplication by the block-diagonal P_n acts entrywise so $T_{P_n} = \text{id}_{M_n} \otimes T_P$; powers of $T_{P_n} - x_0 I$ amplify by induction; so every partial sum of the θ series is the amplification of the corresponding level-one partial sum, and at fixed finite n the norm limits agree. Consequently $\mathcal{S}_{P_n} = M_n \otimes \mathcal{S}_P$, the P_n -compressed product of $[X_{ij}], [Y_{jk}]$ has entries $\text{Co}_P(\sum_j X_{ij} Y_{jk}) = \sum_j X_{ij} \cdot Y_{jk}$ — exactly the matrix amplification of the compressed product — and $\text{Co}_{P_n}(P_n) = I_n \otimes \text{Co}_P(P)$. Also P_n is a δ -projection, so all four imported estimates apply at every level.

Operator space and completeness. `lem-compcb-amplified-compression-identities` gives $\text{Co}_{P_n}^2 = \text{Co}_{P_n}$, so $\mathcal{S}_{P_n} = \text{Im } \text{Co}_{P_n}$ is a closed subspace of the complete space $M_n \otimes \mathcal{A}$ and hence complete in the inherited norm. With the inherited matrix norms $\mathcal{S}_P$ is therefore a complete operator space.

Multiplication and associator. By `lem-compcb-rectangular-product` and the ambient bound $\|AB\| \leq (1+\varepsilon)\|A\|\|B\|$, one gets $\|A \cdot B\| \leq (1+K_p e)\|A\|\|B\|$. For the associator, insert successively $(A \cdot B)C$, $(AB)C$, $A(BC)$ and $A(B \cdot C)$ between the two associations; the two outer rectangular errors cost $C_r e(1+K_p e)N$ each, the two errors obtained by multiplying $A \cdot B - AB$ or $BC - B \cdot C$ by an ambient factor cost $(1+\varepsilon)C_r e N$ each, and the ambient associator costs εN , where $N = \|A\|\|B\|\|C\|$. With $\varepsilon \leq e \leq 1$ the total is at most $K_a e N$.

Involution. If $A \in \mathcal{S}_{P_n}$ then $A = \text{Co}_{P_n}(A)$, so the imported dagger identity gives $A^\dagger = \text{Co}_{P_n}(A^\dagger)$ and the inherited isometric involution preserves the corner. Anti-multiplicativity is exact: $(A \cdot B)^\dagger = \text{Co}_{P_n}((AB)^\dagger) = \text{Co}_{P_n}(B^\dagger A^\dagger) = B^\dagger \cdot A^\dagger$. The unit is fixed, $u_{P_n}^\dagger = \text{Co}_{P_n}(P_n^\dagger) = u_{P_n}$, since $P_n^\dagger = P_n$. Finally the lower C^* -norm bound: the rectangular estimate applied to $A^\dagger$ and A , isometry of the involution, and the ambient lower estimate give

$$\|A^\dagger \cdot A\| \geq \|A^\dagger A\| - C_r e \|A\|^2 \geq (1 - \varepsilon - C_r e) \|A\|^2 \geq (1 - (C_r + 1)e) \|A\|^2.$$

Unit. `lem-compcb-compressed-unit-action` applied to the amplified square corner gives $\|u_{P,n} \cdot A - A\| \leq C_u e \|A\|$ and its right-hand mirror at every n . For the unit norm, each P_n is a δ -projection with $\|P_n\| = \|P\|$, so the *nonvanishing* alternative passes from P to P_n ; that is exactly the hypothesis under which `lem-compcb-compressed-unit-norm` gives the two-sided $|\|u_{P,n}\| - 1| \leq C_N e$, uniformly in n . This is where the nonvanishing hypothesis of the contract is used, and why it cannot be dropped.

Collecting the four families under the ledger constants gives the extended $C_{\text{ca}}e$ - C^* structure.

Status. `proved`; `af`: `validated` (T0). Workspace `proofs/lem-compcb-corner-algebra`: 14 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression-identities`, `lem-compcb-rectangular-pro`, `lem-compcb-compressed-unit-action` and `lem-compcb-compressed-unit-norm`. It is the output of the COMP tier: it converts a bundle of separate estimates into a single *structure* that later tiers may use as a black box. Its registry consumers are `lem-compcb-single-compression-transfer` (§17), `lem-hcb3-diagonal-upper-norm` (§22) — where the matrix-uniform product axiom is precisely the input that a level-one statement could not supply — `lem-hcb3-uniform-square-lower` (§23), which uses the matching lower axiom, and `lem-hcb3-diagonal-lower-modulus` (§24).

16 The corrected amplified column-Hilbert estimate

Lemma 15 (`lem-hcb-column-hilbert-squared`). *There are universal $C_{\text{col}} < \infty$ and $e_{\text{col}} > 0$ such that every H-CB datum with $e \leq e_{\text{col}}$, every $n \geq 1$ and every $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$\left| \langle X, X \rangle_n - \|X\|_{n,1}^2 \right| \leq C_{\text{col}} e \|X\|_{n,1}^2. \quad (35)$$

Transcription note. $\langle \cdot, \cdot \rangle_n$ is the amplified column inner product of **def-column-hilbert-corner**, defined by $Y^\dagger \cdot X = \langle Y, X \rangle_n u_Q$ with $u_Q = \text{Co}_Q(Q)$, and $\|\cdot\|_{n,1}$ is the ambient norm on the column space. The word “corrected” in the contract refers to the fact that this is the squared comparison: the unsquared display at the corresponding place in the pinned source is known to be false and is deliberately excluded from **def-column-hilbert-corner**.

Registry contract (`verbatim`).

Corrected amplified column-Hilbert estimate: there are universal $C_{\text{col}} < \text{infinity}$ and $e_{\text{col}} > 0$ such that every H-CB datum with $e \leq e_{\text{col}}$, every $n \geq 1$, and every X in $M_{\{n,1\}} \text{ tensor } S_{\{P,Q\}}$ satisfy $\text{abs}(\langle X, X \rangle_n - \|X\|_{\{n,1\}}^2) \leq C_{\text{col}} * e * \|X\|_{\{n,1\}}^2$.

Proof (**prose account of the af-validated tree**). *Scalarisation.* Fix an admissible datum, n and X , and abbreviate

$$r = \|X\|_{n,1}, \quad h = \langle X, X \rangle_n, \quad a = \|u_Q\|, \quad b = \|X^\dagger X\|.$$

The column space is a Hilbert space for its defined inner product, so $h \geq 0$ is real, and the defining amplified display of **def-column-hilbert-corner** at $Y = X$ reads

$$X^\dagger \cdot X = h u_Q. \quad (36)$$

The proof is then the interaction of three scalar estimates for h, a, b, r .

(i) *The compressed-versus-ambient comparison.* Applying `lem-compcb-row-column-product` with $Y = X$ gives $\|\text{Co}_Q(X^\dagger X) - X^\dagger X\| \leq C_{\text{rc}} e r^2$; and $\text{Co}_Q(X^\dagger X) = X^\dagger \cdot X = h u_Q$ by (36). Absolute homogeneity and the reverse triangle inequality therefore give

$$|ha - b| = \left| \|h u_Q\| - \|X^\dagger X\| \right| \leq \|h u_Q - X^\dagger X\| \leq C_{\text{rc}} e r^2. \quad (37)$$

(ii) *The C^* comparison.* The ε - C^* lower norm axiom at the amplified rectangular level gives $b \geq (1 - \varepsilon)r^2$, and submultiplicativity together with isometry of the involution gives $b \leq (1 + \varepsilon)\|X^\dagger\| r = (1 + \varepsilon)r^2$. Hence

$$|b - r^2| \leq \varepsilon r^2 \leq e r^2. \quad (38)$$

(iii) *The compressed unit is close to a unit.* Applying `lem-compcb-compressed-unit-norm` with $T = Q$ gives $|a - 1| \leq Ce$, and with $Ce \leq \frac{1}{2}$ also $a \geq \frac{1}{2}$. This step needs Q *nonvanishing*, which the datum does not literally state — **def-hcb-datum** says only that Q is a *one-dimensional* δ -projection. The tree first records this as a genuine gap, then closes it by a registered bridge: one-dimensionality means $\dim \mathcal{S}_Q = 1$, so $\mathcal{S}_Q \neq 0$; and $\mathcal{S}_Q = 0$ characterises the first (vanishing) alternative of **def-delta-projection**; hence Q lies in the second alternative, i.e. is nonvanishing.

Synthesis. Enlarge the constant and shrink the threshold so that one pair $C \geq 1$, $e_0 \leq 1/(2C)$ serves all three estimates. Then (37) and (38) give $|ha - r^2| \leq (C + 1)er^2$. The exact scalar identity

$$a(h - r^2) = (ha - r^2) + r^2(1 - a) \quad (39)$$

combined with $|a - 1| \leq Ce$ then yields $a|h - r^2| \leq (2C + 1)er^2$, and dividing by $a \geq \frac{1}{2}$ gives (35) with $C_{\text{col}} = 4C + 2$ and $e_{\text{col}} = e_0$.

The withdrawn route. A substantial part of this tree is a *negative* audit, and it is the reason two of the four dependencies exist. The natural first attempt was to square-pad the column, $X_c = Xe_1^*$ and $X_r = X_c^\dagger$, and apply `lem-compcb-rectangular-product` to the pair (X_r, X_c) . The audit found that under this shard's allowed inputs none of the three needed identifications was available: the corner memberships $M_n \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{P_n, Q_n}$ and $(\mathcal{S}_{P,Q})^\dagger \subseteq \mathcal{S}_{Q,P}$, the compressed-product identity $X_r \cdot X_c = E_{11} \otimes (X^\dagger \cdot X)$, and the matrix-norm identifications $\|X_c\| = r$ and $\|E_{11} \otimes z\| = \|z\|$. Each affirmative claim was withdrawn rather than repaired in place; a further over-claim — that the failure of this one construction proved an *exhaustive* impossibility — was also withdrawn when challenged. The actual repair was structural: register `lem-compcb-row-column-product` and `lem-compcb-entrywise-compression-naturality` as registry dependencies, after which step (i) above applies directly, with no square embedding and no E_{11} identification needed at this level.

Status. proved; af: validated (T0). Workspace `proofs/lem-hcb-column-hilbert-squared`: 50 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-rectangular-product`, `lem-compcb-compressed-unit-norm`, `lem-compcb-entrywise-compression-naturality` and `lem-compcb-row-column-product`. It is the analytic hinge of the H-CB tier: it makes the column-Hilbert norm q_P and the ambient norm interchangeable up to $1 \pm O(e)$, which is what lets estimates proved in one be read in the other. It feeds `lem-hcb1-variational-identity` (§18), `lem-hcb1-column-action` (§19) and `lem-hcb4-canonical-closeness` (§28), now af: validated, which consumes it at $n = 1$ to normalise the compressed unit.

17 Single-compression transfer of an extended inclusion

Lemma 16 (`lem-compcb-single-compression-transfer`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that restricting an extended α -inclusion to one ideal and following it by one compatible amplified compression produces an extended $C_{\text{co}}(\alpha + \varepsilon)$ -inclusion whenever $\alpha + \varepsilon \leq e_{\text{co}}$.
Transcription note. The tree fixes the datum: $\mathcal{A}$ an extended ε - C^* -algebra, $\mathcal{B}$ an exact unital C^* -algebra, $\mathcal{J} \subseteq \mathcal{B}$ a unital direct-summand ideal with unit q , and $v : \mathcal{B} \rightarrow \mathcal{A}$ an extended α -inclusion (*def-extended-delta-inclusion*). The “compatible single compression” is*

$$P := v(q), \quad T := \text{Co}_P \circ v|_{\mathcal{J}},$$

and throughout $e = \alpha + \varepsilon$.

Registry contract (`verbatim`).

Single-compression transfer: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that restricting an extended α -inclusion to one ideal and following it by one compatible amplified compression produces an extended $C_{\text{co}}(\alpha + \text{epsilon})$ -inclusion whenever $\alpha + \text{epsilon} \leq e_{\text{co}}$.

Proof (prose account of the af-validated tree). *The projection and the ledger.* Since $q = q^\dagger = q^2$ with $\|q\| = 1$ in the exact C^* -algebra $\mathcal{B}$, the inclusion clauses give $P^\dagger = v(q^\dagger) = P$, $\|P^2 - P\| = \|v(q)v(q) - v(q^2)\| \leq \alpha$ and $1 - \alpha \leq \|P\| \leq 1 + \alpha$; so P is an α -projection and, for small e , a *nonvanishing* one, its norm lying in the second alternative of **def-delta-projection**. That is what licenses `lem-compcb-corner-algebra` to make $\mathcal{S}_P$ an extended $C_{\text{cae}}C^*$ -algebra — the target structure. The remaining ledger constants are k_{cmp} from the compression comparison $\|\text{Co}_{P_n}(Z) - (P_n Z)P_n\| \leq k_{\text{cmp}}e\|Z\|$, then $D = 2k_{\text{cmp}} + 5$ and $K_N = D + 1$, together with C_r, e_r from `lem-compcb-rectangular-product`.

The exact clauses. By tensor functoriality $T_n = \text{id}_{M_n} \otimes T = (\text{id}_{M_n} \otimes \text{Co}_P) \circ v_n$, and `lem-compcb-amplified-comp` rewrites this as $T_n = \text{Co}_{P_n} \circ v_n$ with range $\text{Im } \text{Co}_{P_n} = \mathcal{S}_{P_n} = M_n \otimes \mathcal{S}_P$; linearity is inherited from v_n and Co_{P_n} . Two clauses hold *exactly*, with no error term. The star clause of the extended inclusion gives $v_n(x^\dagger) = v_n(x)^\dagger$, and `lem-compcb-amplified-compression-identities` for the square pair (P_n, P_n) gives $\text{Co}_{P_n}(Z^\dagger) = \text{Co}_{P_n}(Z)^\dagger$, so $T_n(x^\dagger) = T_n(x)^\dagger$. And the unit of $M_n \otimes \mathcal{J}$ is $q_n = I_n \otimes q$ with $v_n(q_n) = P_n$, so

$$T_n(q_n) = \text{Co}_{P_n}(P_n) = I_n \otimes \text{Co}_P(P) = I_n \otimes u_P :$$

the unit defect is zero.

Compression closeness. Because q is the unit of the ideal, $q_n x = x q_n = x$ for $x \in M_n \otimes \mathcal{J}$, so with $Z = v_n(x)$ the multiplicative clause of the inclusion gives $\|P_n Z - Z\| \leq \alpha\|x\|$ and $\|Z P_n - Z\| \leq \alpha\|x\|$, while its norm clause gives $\|P_n\| \leq 1 + \alpha$ and $\|Z\| \leq (1 + \alpha)\|x\|$. Combining with the compression comparison and $e \leq \frac{1}{2}$,

$$\|T_n(x) - v_n(x)\| \leq \|\text{Co}_{P_n}(Z) - (P_n Z)P_n\| + \|(P_n Z)P_n - Z\| \leq D e \|x\|, \quad (40)$$

uniformly in n ; the second term is bounded by $[(1 + \varepsilon)(1 + \alpha) + 1]\alpha\|x\| \leq 5e\|x\|$.

Norm distortion. The inclusion gives $(1 - \alpha)\|x\| \leq \|v_n(x)\| \leq (1 + \alpha)\|x\|$; with (40) the triangle and reverse-triangle inequalities give

$$(1 - K_N e)\|x\| \leq \|T_n(x)\| \leq (1 + K_N e)\|x\|, \quad K_N = D + 1. \quad (41)$$

Multiplicativity. Insert $v_n(xy)$, $v_n(x)v_n(y)$ and $T_n(x)T_n(y)$ between $T_n(xy)$ and $T_n(x) \cdot T_n(y)$. The first term is $\leq De\|x\|\|y\|$ by (40), using that $M_n \otimes \mathcal{J}$ is an exact C^* -algebra — so its norm is genuinely submultiplicative, $\|xy\| \leq \|x\|\|y\|$. The second is $\leq \alpha\|x\|\|y\|$ by the multiplicative clause. The third, after inserting $T_n(x)v_n(y)$, is $\leq 2D(K_N + 3)e\|x\|\|y\|$ by (40), (41) and the ε -product bound. The fourth needs a naturality observation: for $A, B \in M_n \otimes \mathcal{S}_P$ the amplified compressed product has entries $\sum_k \text{Co}_P(a_{ik}b_{kj}) = \text{Co}_P(\sum_k a_{ik}b_{kj})$, i.e. it equals $\text{Co}_{P_n}(AB)$ — so it is the rectangular compressed product for the compatible triple (P_n, P_n, P_n) , not merely an element of the same space, and **lem-compcb-rectangular-product** applies, giving $\leq C_r(1 + K_N)^2 e\|x\|\|y\|$. Summing, $K_M := D + 1 + 2D(K_N + 3) + C_r(1 + K_N)^2$ works.

Assembly. With $C_{\text{co}} = \max\{1, C_{\text{ca}}, K_N, K_M\}$ and e_{co} the positive minimum of all thresholds used, **def-extended-delta-inclusion** makes each T_n a $C_{\text{co}}e$ -inclusion; since n was arbitrary and every constant is universal, T is an extended $C_{\text{co}}(\alpha + \varepsilon)$ -inclusion, with no dependence on the amplification level or the ideal's dimension.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-single-compression-transfer: 33 nodes (32 validated, 1 superseded node archived), root validated, taint clean.

Role in the chain. Imports **lem-compcb-amplified-compression**, **lem-compcb-amplified-compression-iden**, **lem-compcb-amplified-almost-containment**, **lem-compcb-corner-algebra** and **lem-compcb-rectangular-product**; it is the only consumer of the almost-containment lemma. Unlike the rest of the COMP tier it is not used by the H-CB tier: it is the transfer primitive aimed at the still-open EXT and Stage-1 assembly, where inclusions have to be pushed through compressions repeatedly. It has no validated registry consumers yet.

18 The amplified Ha variational identity

Lemma 17 (`lem-hcb1-variational-identity`). *There is a universal $e_{\text{var}} > 0$ such that every H-CB datum with $e \leq e_{\text{var}}$, every $n \geq 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$, $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ and $Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$2 \langle Y, (\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X \rangle_n \text{Co}_Q(Q) = (Y^\dagger \cdot Z) \cdot X - Y^\dagger \cdot (Z \cdot X). \quad (42)$$

*Transcription note. The identity is exact; the threshold only guarantees that the objects involved are defined. Its content is that the “Ha defect” — the difference between letting $\text{Ha}_{P,R}^Q(Z)$ act on a column and simply multiplying by Z — is, after pairing, literally the compressed associator of **def-compressed-associator** for the triple $(Y^\dagger, Z, X)$. Contract amendment. The codified contract originally wrote the normalising factor as u_Q ; the ratified **def-ha-map** identity and the column-Hilbert displays write it as $\tilde{Q} = \text{Co}_Q(Q)$. The contract now spells out $\text{Co}_Q(Q)$. These are definitionally the same element — the compressed unit is $u_Q = \text{Co}_Q(Q)$ — so this was pure notation typing, neither a strengthening nor a weakening.*

Registry contract (verbatim).

Amplified Ha variational identity: there is a universal $e_{\text{var}} > 0$ such that every H-CB datum with $e \leq e_{\text{var}}$, every $n \geq 1$, Z in M_n tensor $\mathcal{S}_{\{P,R\}}$, X in $M_{\{n,1\}}$ tensor $\mathcal{S}_{\{R,Q\}}$, and Y in $M_{\{n,1\}}$ tensor $\mathcal{S}_{\{P,Q\}}$ satisfy $2 \langle Y, (\text{Ha}^Q_{P,R})_n(Z)X - Z \cdot X \rangle_n \text{Co}_Q(Q) = (Y^\dagger \cdot Z) \cdot X - Y^\dagger \cdot (Z \cdot X)$.

Proof (prose account of the af-validated tree). *Threshold.* Take $e_{\text{var}} = \min(e_{\text{cmp}}, e_{\text{col}})$, the thresholds of `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-squared`, and fix an arbitrary datum and arbitrary n, Z, X, Y .

Entrywise dictionary. Write $Z = (Z_{ij})_{1 \leq i,j \leq n}$, $X = (X_j)_j$, $Y = (Y_i)_i$. Matrix amplification, the matrix compressed product and the amplified sesquilinear form unfold as

$$\begin{aligned} ((\text{Ha}_{P,R}^Q)_n(Z)X)_i &= \sum_j \text{Ha}_{P,R}^Q(Z_{ij})(X_j), & (Z \cdot X)_i &= \sum_j Z_{ij} \cdot X_j, \\ (Y^\dagger \cdot Z)_j &= \sum_i Y_i^\dagger \cdot Z_{ij}, & \langle Y, W \rangle_n &= \sum_i \langle Y_i, W_i \rangle. \end{aligned}$$

Summing the level-one relation. For each pair (i, j) , the defining relation of **def-ha-map**, applied to $Z_{ij} \in \mathcal{S}_{P,R}$, $X_j \in \mathcal{S}_{R,Q}$ and $Y_i \in \mathcal{S}_{P,Q}$, gives

$$2 \langle Y_i, \text{Ha}_{P,R}^Q(Z_{ij})(X_j) \rangle \text{Co}_Q(Q) = (Y_i^\dagger \cdot Z_{ij}) \cdot X_j + Y_i^\dagger \cdot (Z_{ij} \cdot X_j).$$

Summing over i, j and using the dictionary together with sesquilinearity,

$$2 \langle Y, (\text{Ha}_{P,R}^Q)_n(Z)X \rangle_n \text{Co}_Q(Q) = \sum_{i,j} \left[(Y_i^\dagger \cdot Z_{ij}) \cdot X_j + Y_i^\dagger \cdot (Z_{ij} \cdot X_j) \right]. \quad (43)$$

The column display. The registered amplified column-Hilbert display $Y^\dagger \cdot W = \langle Y, W \rangle_n \text{Co}_Q(Q)$, applied to $W = Z \cdot X$ and expanded entrywise, gives

$$\langle Y, Z \cdot X \rangle_n \text{Co}_Q(Q) = Y^\dagger \cdot (Z \cdot X) = \sum_{i,j} Y_i^\dagger \cdot (Z_{ij} \cdot X_j). \quad (44)$$

Cancellation. Associating only as displayed — this is the point where the two sides differ — ordinary matrix multiplication gives $(Y^\dagger \cdot Z) \cdot X = \sum_{i,j} (Y_i^\dagger \cdot Z_{ij}) \cdot X_j$, whereas $Y^\dagger \cdot (Z \cdot X) = \sum_{i,j} Y_i^\dagger \cdot (Z_{ij} \cdot X_j)$. Subtracting twice (44) from (43) and using sesquilinearity in the second variable, the doubled $Y_i^\dagger \cdot (Z_{ij} \cdot X_j)$ sum cancels once and reverses sign, leaving exactly (42). Since the datum and n, Z, X, Y were arbitrary and e_{var} is universal and positive, the lemma follows.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb1-variational-identity: 8 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-`

It is the conversion device of the H-CB tier: on its own it proves no estimate, but it turns every bound on the compressed associator into a bound on the Ha defect. Its single consumer is `lem-hcb1-column-action` (§19), where it is combined with `lem-hcb0-compressed-associator` and the nondegeneracy of u_Q .

19 The uniform Ha column action

Lemma 18 (`lem-hcb1-column-action`). *There are universal $C_{\text{act}} < \infty$ and $e_{\text{act}} > 0$ such that every H-CB datum with $e \leq e_{\text{act}}$, every $n \geq 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$ and every $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ satisfy*

$$q_P((\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X) \leq C_{\text{act}} e \|Z\| q_R(X). \quad (45)$$

Transcription note. $q_P(U) = \langle U, U \rangle_n^{1/2}$ is the column-Hilbert norm of `def-column-hilbert-corner` on $M_{n,1} \otimes \mathcal{S}_{P,Q}$, and likewise q_R ; $\|Z\|$ is the ambient norm.

Registry contract (`verbatim`).

Uniform Ha column action: there are universal $C_{\text{act}} < \text{infinity}$ and $e_{\text{act}} > 0$ such that every H-CB datum with $e \leq e_{\text{act}}$, every $n \geq 1$, Z in M_n tensor $\mathcal{S}_{\{P,R\}}$, and X in $M_{\{n,1\}}$ tensor $\mathcal{S}_{\{R,Q\}}$ satisfy $q_P((\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X) \leq C_{\text{act}} * e * \|Z\| * q_R(X)$.

Proof (prose account of the af-validated tree). Throughout put $D := (\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X$, the object to be bounded.

Nondegeneracy of the compressed unit. Everything is divided by $\|u_Q\|$ at the end, so a lower bound is needed. Two ingredients combine. First, $\|u_Q - Q\| \leq K_1 e$. This is proved *inside the associative operator algebra* $B(\mathcal{A})$, not in the ambient ε - C^* -algebra, which is only approximately associative — a distinction the tree makes explicitly. With $L = L_Q$, $R = R_Q$ and $A = LR + RL - I$ one has $\text{Co}_Q = \theta(A)$; the δ -projection dichotomy bounds $\|Q\|$, hence $\|L\|$, $\|R\|$, and the ε -associator and δ -projection bounds give $\|LR - RL\| = O(e)$, $\|L^2 - L\| = O(e)$, $\|R^2 - R\| = O(e)$; from these, $T = \frac{1}{2}(LR + RL)$ satisfies $\|T^2 - T\| \leq C_T e$ and so $\|A^2 - I\| = 4\|T^2 - T\| \leq 4C_T e$. Writing $r = Q^2 - Q$ and keeping the displayed parentheses, $(A - I)Q = 2r + Qr + rQ$, so $\|(A - I)Q\| \leq C_Q e$. Finally, below the threshold making $\|A^2 - I\| < \frac{1}{2}$, the series $B = (A^2)^{-1/2}$ converges and factors as $B - I = (A^2 - I)K(A^2)$; since A commutes with B and $K(A^2)$, one gets the clean factorisation $\theta(A) - I = (A - I)H(A)$ with $\|H(A)\| \leq C_H$, whence $\|u_Q - Q\| = \|(\theta(A) - I)Q\| \leq C_H C_Q e$. Second, $\|Q\| \geq 1 - K_0 e$, which holds *provided Q is nonvanishing*. The tree runs this branch strictly conditionally — it refuses to infer nonvanishing from one-dimensionality — and discharges the antecedent only later, from the registered bridge $\dim \mathcal{S}_Q = 1 \Rightarrow \mathcal{S}_Q \neq 0 \Rightarrow Q$ nonvanishing. Combining the two by the reverse triangle inequality and shrinking the threshold gives

$$\|u_Q\| \geq c_u := \frac{1}{2}. \quad (46)$$

Column-norm comparison. By `lem-hcb-column-hilbert-squared`, $|q_T(U)^2 - \|U\|_{n,1}^2| \leq C_{\text{col}} e \|U\|_{n,1}^2$ for $T = P$ and $T = R$; with $C_{\text{col}} \leq \frac{1}{2}$ all terms are nonnegative and taking square roots gives $(1 - C_{\text{col}} e)^{1/2} \|U\|_{n,1} \leq q_T(U) \leq (1 + C_{\text{col}} e)^{1/2} \|U\|_{n,1}$, in particular $\|U\|_{n,1} \leq \sqrt{2} q_T(U)$.

Defect pairing. By `lem-hcb1-variational-identity`, $2 \langle Y, D \rangle_n u_Q = (Y^\dagger \cdot Z) \cdot X - Y^\dagger \cdot (Z \cdot X)$ for every compatible column Y . The right-hand side is precisely the compressed associator of $A = Y^\dagger$, $B = Z$, $C = X$ (`def-compressed-associator`), so `lem-hcb0-compressed-associator` bounds it by $C_{\text{as}} e \|Y^\dagger\| \|Z\| \|X\|_{n,1}$, and the involution is isometric, so $\|Y^\dagger\| = \|Y\|_{n,1}$. Taking norms on the left and using absolute homogeneity,

$$2 |\langle Y, D \rangle_n| \|u_Q\| \leq C_{\text{as}} e \|Y\|_{n,1} \|Z\| \|X\|_{n,1}. \quad (47)$$

Assembly by self-pairing. Insert (46) and the two norm comparisons into (47): for every compatible Y ,

$$|\langle Y, D \rangle_n| \leq 2C_{\text{as}} e q_P(Y) \|Z\| q_R(X).$$

Now the finishing move avoids any duality argument: since both $(\text{Ha}_{P,R}^Q)_n(Z)X$ and $Z \cdot X$ lie in $M_{n,1} \otimes \mathcal{S}_{P,Q}$, so does D , and $Y = D$ is an admissible test vector. Substituting it gives $q_P(D)^2 = \langle D, D \rangle_n \leq 2C_{\text{as}} e \|Z\| q_R(X) q_P(D)$; if $q_P(D) = 0$ the conclusion is immediate, and otherwise one divides. Hence (45) with $C_{\text{act}} = 2C_{\text{as}}$.

Two recorded refinements. The tree first proved the terminal cancellation only *conditionally*, explicitly declining to divide by an unavailable c_u and carrying a node that says so; the unconditional root appears only after the nonvanishing bridge is registered, and the earlier negative conclusion is marked stale rather than deleted. Separately, the threshold is set as $e_{\text{act}} = t/2$ for $t = \min(e_u, e_{\text{col}}, e_{\text{var}}, e_{\text{as}}, 1/(2 \max(C_{\text{col}}, 1)))$ so that the strict inequalities required upstream hold even at the endpoint, and the degenerate case $C_{\text{col}} = 0$ is split off so that the reciprocal $1/(2C_{\text{col}})$ is never formed.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb1-column-action: 38 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-hcb-column-hilbert-squared`, `lem-hcb0-compressed-associator` and `lem-hcb1-variational-identity`. This is the first genuinely quantitative statement about the Ha map: it says that $(\text{Ha}_{P,R}^Q)_n(Z)$ is, to within $O(e)$, left multiplication by Z . Everything later in the tier is built on it — `lem-hcb2-product-defect` (§20), `lem-hcb3-diagonal-unit` (§21) — and it is a declared dependency of `lem-hcb4-canonical-closeness` (§28), now af: validated.

20 The uniform amplified Ha product defect

Lemma 19 (`lem-hcb2-product-defect`). *There are universal $C_{\text{prod}} < \infty$ and $e_{\text{prod}} > 0$ such that every H-CB datum with $e \leq e_{\text{prod}}$, every $n \geq 1$, every $Z \in M_n \otimes \mathcal{S}_{P,S}$ and every $W \in M_n \otimes \mathcal{S}_{S,R}$ satisfy*

$$\|(\text{Ha}_{P,R}^Q)_n(Z \cdot W) - (\text{Ha}_{P,S}^Q)_n(Z) (\text{Ha}_{S,R}^Q)_n(W)\| \leq C_{\text{prod}} e \|Z\| \|W\|. \quad (48)$$

Transcription note. *On the left the juxtaposition of the two amplified Ha maps is composition of operators between amplified column spaces, whereas $Z \cdot W$ is the compressed product in the ambient corner; the norm is the operator norm of **def-ha-map**. So the statement is that $\text{Ha}_{\cdot}^Q$ is multiplicative up to $O(e)$.*

Registry contract (verbatim).

Uniform amplified Ha product defect: there are universal $C_{\text{prod}} < \text{infinity}$ and $e_{\text{prod}} > 0$ such that every H-CB datum with $e \leq e_{\text{prod}}$, every $n \geq 1$, Z in M_n tensor $S_{\{P,S\}}$, and W in M_n tensor $S_{\{S,R\}}$ satisfy $\|(\text{Ha}^Q_{P,R})_n(Z \cdot W) - (\text{Ha}^Q_{P,S})_n(Z) (\text{Ha}^Q_{S,R})_n(W)\| \leq C_{\text{prod}} * e * \|Z\| \|W\|$.

Proof (prose account of the af-validated tree). *Calibration.* Three uniform facts are set up first. The whole-column identity $X^\dagger \cdot X = q_T(X)^2 u_Q$ of **def-column-hilbert-corner**, combined with **lem-compcb-rectangular-product** and the ε - C^* norm axioms, gives $|\|X^\dagger \cdot X\| - \|X\|^2| \leq (C_{\text{co}} + 1)e\|X\|^2$; since Q is one-dimensional it is nonvanishing by the registered bridge, so u_Q is the unit of an $O(e)$ - C^* corner and its norm is within $O(e)$ of 1. Shrinking the threshold, this calibrates the column norm against the ambient norm,

$$\frac{1}{2}\|X\| \leq q_T(X) \leq 2\|X\|. \quad (49)$$

Next, the rectangular estimate and $\|AB\| \leq (1 + \varepsilon)\|A\|\|B\|$ give $\|A \cdot B\| \leq K_0\|A\|\|B\|$ with $K_0 = 2 + C_{\text{co}}$; and for a square-by-column pair, (49) converts this into the multiplier bound $q_P(A \cdot X) \leq K_{\text{mul}}\|A\|q_R(X)$ with $K_{\text{mul}} = 4K_0$. Finally, **lem-hcb1-column-action** together with the multiplier bound gives a uniform operator bound for the Ha map itself,

$$\|(\text{Ha}_{A,B}^Q)_n(U)\| \leq K_H\|U\|, \quad K_H := K_{\text{mul}} + C_{\text{act}}. \quad (50)$$

The telescope. Abbreviate $H_{A,B}(U) := (\text{Ha}_{A,B}^Q)_n(U)$. For every compatible column X the following identity is exact, obtained by adding and subtracting compressed multiplications:

$$\begin{aligned} [H_{P,R}(Z \cdot W) - H_{P,S}(Z)H_{S,R}(W)]X &= [H_{P,R}(Z \cdot W)X - (Z \cdot W) \cdot X] \\ &\quad + [(Z \cdot W) \cdot X - Z \cdot (W \cdot X)] \\ &\quad + Z \cdot [W \cdot X - H_{S,R}(W)X] \\ &\quad + [Z \cdot H_{S,R}(W)X - H_{P,S}(Z)H_{S,R}(W)X]. \end{aligned} \quad (51)$$

Each bracket is one of the two estimates already available. The first is a column-action defect for the element $Z \cdot W$, so **lem-hcb1-column-action** and $\|Z \cdot W\| \leq K_0\|Z\|\|W\|$ bound it by $C_{\text{act}}K_0e\|Z\|\|W\|q_R(X)$. The second is the compressed associator, bounded by **lem-hcb0-compressed-associator** in the ambient norm by $C_{\text{as}}e\|Z\|\|W\|\|X\|$; converting both ends through (49) costs two factors of 2, giving $4C_{\text{as}}e\|Z\|\|W\|q_R(X)$. The third is the column-action defect for W , pushed through

the square-by-column multiplier bound for Z , giving $K_{\text{mul}}C_{\text{act}}e\|Z\|\|W\|q_R(X)$. The fourth is the column-action defect for Z evaluated on the column $H_{S,R}(W)X$, whose column norm is controlled by the operator bound (50), giving $C_{\text{act}}K_H e\|Z\|\|W\|q_R(X)$.

Conclusion. Applying the q_P triangle inequality to (51) and adding the four bounds gives the pointwise estimate with

$$C_* := C_{\text{act}}K_0 + 4C_{\text{as}} + C_{\text{act}}K_{\text{mul}} + C_{\text{act}}K_H.$$

Taking the supremum over columns with $q_R(X) = 1$ converts it into the operator-norm statement (48), with $C_{\text{prod}} = C_*$ and $e_{\text{prod}} = \min\{e_{\text{cmp}}, e_{\text{co}}, e_{\text{as}}, e_{\text{act}}, e_q, 1\}$; `lem-compcb-amplified-compression` is what guarantees that all the corners and compressed products appearing at level n are the compatible amplified objects the imported estimates quantify over.

Status. `proved`; `af`: `validated` (T0). Workspace `proofs/lem-hcb2-product-defect`: 12 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-hcb0-compressed-associator`, `lem-hcb1-column-action`, `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. Together with the exact adjointness of §11 it makes $\text{Ha}_{\cdot}^Q$ an approximate $*$ -homomorphism at every amplification level, which is exactly the input the norm estimates of the next tier consume: it feeds `lem-hcb3-diagonal-upper-norm` (§22) and `lem-hcb3-diagonal-lower-modulus` (§24), and it is a declared dependency of `lem-hcb3-offdiagonal-inverse` (§26), now `af`: `validated`.

21 The uniform diagonal Ha unit estimate

Lemma 20 (`lem-hcb3-diagonal-unit`). *There are universal $C_{\text{unit}} < \infty$ and $e_{\text{unit}} > 0$ such that every H-CB datum with $e \leq e_{\text{unit}}$ and every $n \geq 1$ satisfy*

$$\|(\text{Ha}_{P,P}^Q)_n(I_n \otimes u_P) - I\| \leq C_{\text{unit}} e. \quad (52)$$

Transcription note. $u_P = \text{Co}_P(P)$ is the compressed unit of the corner $\mathcal{S}_P$, $I_n \otimes u_P$ its diagonal amplification, and I the identity operator on the amplified column space $M_{n,1} \otimes \mathcal{S}_{P,Q}$; the norm is the operator norm.

Registry contract (verbatim).

Uniform diagonal Ha unit estimate: there are universal $C_{\text{unit}} < \text{infinity}$ and $e_{\text{unit}} > 0$ such that every H-CB datum with $e \leq e_{\text{unit}}$ and every $n \geq 1$ satisfy $\|(\text{Ha}^Q_{P,P})_n(I_n \text{ tensor } u_P) - I\| \leq C_{\text{unit}} * e$.

Proof (prose account of the af-validated tree). Write $U = I_n \otimes u_P$. The estimate is pointwise on columns, then a supremum.

A uniform bound on the amplified unit. The obvious route — bound $\|u_P\|$ at level one and appeal to isometry of the diagonal amplification — is *not* taken, and the tree says why: no registered input supplies that isometry. Instead $\|U\|$ is bounded directly at level n by a C^* -style argument. Put $t = \|U\|$. The ε - C^* norm axioms at level n give $\|U^\dagger U\| \geq (1 - \varepsilon)t^2 \geq (1 - e)t^2$ and $\|U^\dagger\| = t$; the compressed-product display in the amplified P, P corner gives $\|U^\dagger U - U^\dagger \cdot U\| \leq C_{\text{pr}} e t^2$; and `lem-compcb-compressed-unit-action`, in its right-unit form with $A = U^\dagger$, gives $\|U^\dagger \cdot U - U^\dagger\| \leq C_{\text{co}} e t$. Chaining,

$$(1 - e)t^2 \leq \|U^\dagger U\| \leq C_{\text{pr}} e t^2 + (1 + C_{\text{co}} e)t.$$

If $t = 0$ there is nothing to prove; otherwise divide by t to get $[1 - (1 + C_{\text{pr}})e]t \leq 1 + C_{\text{co}} e$, and choosing $e_u \leq 1/[2(1 + C_{\text{pr}})]$ makes the bracket at least $\frac{1}{2}$, so

$$\|U\| \leq K_u := 2(1 + C_{\text{co}} e_u), \quad (53)$$

uniformly in n .

Two defects. For a column $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ split

$$((\text{Ha}_{P,P}^Q)_n(U) - I)X = ((\text{Ha}_{P,P}^Q)_n(U)X - U \cdot X) + (U \cdot X - X). \quad (54)$$

The first bracket is a column-action defect: `lem-hcb1-column-action` with $R = P$ and $Z = U$ gives $q_P((\text{Ha}_{P,P}^Q)_n(U)X - U \cdot X) \leq C_{\text{act}} e \|U\| q_P(X)$, which (53) turns into $C_{\text{act}} K_u e q_P(X)$.

The second bracket says that the amplified compressed unit is an approximate identity on columns, and it is assembled coordinatewise. Multiplication by the block-diagonal $U = I_n \otimes u_P$ acts entrywise, so writing $X = (X_1, \dots, X_n)^\top$ and applying `lem-compcb-compressed-unit-action` in each level-one P, Q corner gives $\|u_P \cdot X_j - X_j\| \leq C_{\text{co}} e \|X_j\|$ for every j . The level-one norm/inner-product comparison registered with `def-ha-map` and the column-Hilbert displays supplies a universal κ with $q_P(Y) \leq \kappa \|Y\|$ and $\|Y\| \leq \kappa q_P(Y)$, so each coordinate estimate becomes $q_P(u_P \cdot X_j - X_j) \leq \kappa^2 C_{\text{co}} e q_P(X_j)$. Finally the coordinate-sum display $q_P((Y_j)_j)^2 = \sum_j q_P(Y_j)^2$ lets the squared coordinate estimates be summed, giving $q_P(U \cdot X - X) \leq \kappa^2 C_{\text{co}} e q_P(X)$ uniformly

in n ; the existential constant C_{co} is then simply replaced by the larger universal $\kappa^2 C_{\text{co}}$ and the notation retained.

Conclusion. Setting $e_0 = \frac{1}{2} \min(e_u, e_{\text{act}}, e_{\text{co}})$ and applying the q_P triangle inequality to (54),

$$q_P\left(\left((\text{Ha}_{P,P}^Q)_n(U) - I\right)X\right) \leq (C_{\text{act}}K_u + C_{\text{co}})e_{q_P}(X),$$

and passing to the operator norm gives (52) with $C_{\text{unit}} = C_{\text{act}}K_u + C_{\text{co}}$ and $e_{\text{unit}} = e_0$.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb3-diagonal-unit: 18 nodes, all validated, root validated, taint clean.

Role in the chain. Imports lem-hcb1-column-action and lem-compcb-compressed-unit-action.

It is one of the estimates the parent contract conj-hcb collects — the clause asserting that the amplified Ha map is approximately unital — and its registry consumer is the parent contract itself (§30), now proved with af: validated; neither it nor this lemma bears on the status of op-classical, which remains open.

22 The uniform diagonal Ha upper norm

Lemma 21 (`lem-hcb3-diagonal-upper-norm`). *There are universal $C_{\text{up}} < \infty$ and $e_{\text{up}} > 0$ such that every H -CB datum with $e \leq e_{\text{up}}$, every $n \geq 1$ and every $Z \in M_n \otimes \mathcal{S}_P$ satisfy*

$$\|(\text{Ha}_{P,P}^Q)_n(Z)\| \leq (1 + C_{\text{up}} e) \|Z\|. \quad (55)$$

Transcription note. *The left norm is the operator norm on maps of the amplified column space; $\|Z\|$ is the ambient norm of Z in $M_n \otimes \mathcal{S}_P$. The estimate is uniform in n , which is the whole difficulty: the level-one statement is a routine consequence of the corner structure, the matrix-uniform one is not.*

Registry contract (verbatim).

Uniform diagonal Ha upper norm: there are universal $C_{\text{up}} < \text{infinity}$ and $e_{\text{up}} > 0$ such that every H -CB datum with $e \leq e_{\text{up}}$, every $n \geq 1$, and every Z in M_n tensor $\mathcal{S}_P$ satisfy $\|(\text{Ha}^Q_{P,P})_n(Z)\| \leq (1 + C_{\text{up}} * e) \|Z\|$.

Proof (prose account of the af-validated tree). Write $h_n = (\text{Ha}_{P,P}^Q)_n$ and $b_n = \|h_n\|$ for its operator norm; the goal is $b_n \leq 1 + C_{\text{up}} e$ for every n .

The matrix-uniform square estimate. The first ingredient is

$$\|Z^\dagger \cdot Z\| \leq (1 + \kappa e) \|Z\|^2, \quad Z \in M_n \otimes \mathcal{S}_P, \quad (56)$$

with $\kappa = C_{\text{ca}}$ and threshold $e_{\text{sq}} = e_{\text{ca}}$. If $\mathcal{S}_P = 0$ then $M_n \otimes \mathcal{S}_P = 0$ for every n , so $Z = 0$ and (56) is trivial. Otherwise $\mathcal{S}_P \neq 0$ puts P in the nonvanishing branch of the δ -projection dichotomy, and `lem-compcb-corner-algebra` makes $\mathcal{S}_P$ an *extended* $C_{\text{ca}}e$ - C^* -algebra — so, by the meaning of “extended”, every $M_n \otimes \mathcal{S}_P$ is a $C_{\text{ca}}e$ - C^* -algebra with the *same* parameter, independent of n . Its product upper axiom together with isometry of the inherited involution then gives (56).

This is the shard’s instructive correction, and it explains the shape of the dependency list. An earlier audit of the tree found that the permitted inputs at that time — the compressed-product display, the level-one ε - C^* axioms, `def-hcb-datum` (data and notation only), and the two Ha lemmas — gave (56) at $n = 1$ and *nowhere else*: passing to arbitrary n was not an inference from any allowed premise but a genuinely missing amplified-corner theorem. Once `lem-compcb-corner-algebra` became an allowed dependency, the negative audit nodes became obsolete and are marked as such.

A quadratic recurrence for b_n . Finiteness of b_n is established without any finite-dimensionality assumption on $\mathcal{S}_P$. From the defining Ha relation and the level-one bounds — $\|A \cdot B\| \leq K_{\text{dot}} \|A\| \|B\|$, the two-sided comparison $c_q^{-1} \|X\| \leq q(X) \leq c_q \|X\|$, and $\|u_Q\| \geq c_u > 0$ (with Q nonvanishing because it is one-dimensional) — one gets $2|\langle Y, h(Z)X \rangle| \|u_Q\| \leq 2K_{\text{dot}}^2 \|Y\| \|Z\| \|X\|$, hence $|\langle Y, h(Z)X \rangle| \leq K_1 q(Y) \|Z\| q(X)$ with $K_1 = K_{\text{dot}}^2 c_q^2 / c_u$, and Hilbert-space duality in the q norm gives $\|h(Z)\| \leq K_1 \|Z\|$ at level one. For fixed finite n , $h_n(Z) = [h(Z_{ij})]$ on the Hilbert direct sum, so $\|h_n(Z)\| \leq (\sum_{i,j} \|h(Z_{ij})\|^2)^{1/2} \leq n K_1 \|Z\|$, using contractivity of matrix-coordinate compressions. Crude in n , but finite — which is all that is needed to *form* b_n and take a supremum.

Now let $e \leq \min(e_{\text{sq}}, e_{\text{prod}}, e_{\text{adj}})$ and $Z \in E_n := M_n \otimes \mathcal{S}_P$. Put $T = h_n(Z)$. The operator C^* -identity gives $\|T\|^2 = \|T^\dagger T\|$; `lem-hcb2-amplified-adjointness` identifies $T^\dagger = h_n(Z^\dagger)$ — exactly, which is what makes this move legitimate; and `lem-hcb2-product-defect` with all

three corner indices equal to P gives $\|h_n(Z^\dagger)h_n(Z) - h_n(Z^\dagger \cdot Z)\| \leq C_{\text{prod}}e\|Z\|^2$. Chaining with boundedness of h_n and (56),

$$\|h_n(Z)\|^2 \leq b_n\|Z^\dagger \cdot Z\| + C_{\text{prod}}e\|Z\|^2 \leq [(1 + \kappa e)b_n + C_{\text{prod}}e]\|Z\|^2.$$

If $E_n = \{0\}$ then $b_n = 0$ and the recurrence below holds trivially; otherwise the unit sphere of E_n is nonempty, b_n is a finite supremum over it, and $\sup\|h_n(Z)\|^2 = (\sup\|h_n(Z)\|)^2 = b_n^2$, so

$$b_n^2 \leq (1 + \kappa e)b_n + C_{\text{prod}}e. \quad (57)$$

The root argument. It remains to solve (57). For nonnegative b, e, κ, C with $b^2 \leq ab + Ce$ and $a = 1 + \kappa e$, set $x = 1 + (\kappa + C)e$ and $f(t) = t^2 - at - Ce$. Direct expansion gives $f(x) = C(\kappa + C)e^2 \geq 0$, and f is strictly increasing on $[x, \infty)$ because $2x - a = 1 + (\kappa + 2C)e > 0$. The hypothesis is $f(b) \leq 0$; if $b > x$ then monotonicity would give $f(b) > f(x) \geq 0$, a contradiction. Hence $b \leq x$.

Applying this with $b = b_n$ and $C = C_{\text{prod}}$ gives $b_n \leq 1 + (\kappa + C_{\text{prod}})e$ for every n , which is (55) with $C_{\text{up}} = \kappa + C_{\text{prod}}$ and $e_{\text{up}} = \min(e_{\text{sq}}, e_{\text{prod}}, e_{\text{adj}})$; both are universal.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb3-diagonal-upper-norm: 30 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and `lem-compcb-corner-algebra`. It is the upper half of the norm control that the parent contract `conj-hcb` requires; the matching lower half is `lem-hcb3-diagonal-lower-modulus` (§24), which is validated but only *conditionally* — it assumes a level-one lower modulus of at least $\frac{1}{4}$, a hypothesis nothing in this document supplies, and the diagonal inverse estimate `lem-hcb3-diagonal-inverse` (§25) inherits it. The off-diagonal inverse (§26), the canonical tier (§§27–29) and `conj-hcb` itself (§30) are now all af: validated; the conditional hypotheses named above are supplied only inside §36, and nowhere else here. Nothing here bears on the status of `op-classical`.

23 The uniform square lower estimate

Lemma 22 (`lem-hcb3-uniform-square-lower`). *There are universal $K_{\text{sq}} < \infty$ and $e_{\text{sq}} > 0$ such that every H -CB datum with $e \leq e_{\text{sq}}$, every $n \geq 1$ and every $Z \in M_n \otimes \mathcal{S}_P$ satisfy*

$$\|Z^\dagger \cdot Z\| \geq (1 - K_{\text{sq}} e) \|Z\|^2. \quad (58)$$

Transcription note. The datum is that of `def-hcb-datum` and $e = \delta + \varepsilon$; $\cdot$ is the compressed product on the corner $\mathcal{S}_P$ of `def-compressed-corner`, and † the inherited involution. No hypothesis is placed on P beyond its being one of the datum's δ -projections — in particular no nonvanishing assumption — which is what makes the statement usable as a black box downstream.

Registry contract (verbatim).

Uniform square lower estimate: there are universal $K_{\text{sq}} < \text{infinity}$ and $e_{\text{sq}} > 0$ such that every H -CB datum with $e \leq e_{\text{sq}}$, every $n \geq 1$, and every Z in M_n tensor S_P satisfy $\|Z^\dagger \cdot Z\| \geq (1 - K_{\text{sq}} e) \|Z\|^2$.

Proof (prose account of the af-validated tree). The tree splits on the δ -projection dichotomy of `def-delta-projection`. The nonvanishing branch is a citation; the vanishing branch is the work, and it is settled by showing that the corner is then *literally zero*.

The vanishing branch. Suppose P is not nonvanishing, so the registered dichotomy puts it in the small alternative $\|P\| \leq C_v \delta$ for a universal C_v and e below a universal threshold. Put $T = L_P R_P + R_P L_P$. The ε -Banach product bound gives

$$\|T\| \leq 2(1 + \varepsilon)^2 \|P\|^2 \leq 2(1 + e)^2 C_v^2 e^2,$$

so after a further universal shrink of the threshold one may assume $\|T\| < \frac{1}{3}$.

At that point the θ calculus of `def-theta-idempotent-approximation` evaluates in closed form. Write $U = I - T$ and $X = T - I = -U$. Since $\|T\| < \frac{1}{3}$ the Neumann series $\sum_{k \geq 0} T^k$ converges absolutely to U^{-1} ; and $X^2 - I = -2T + T^2$ has norm at most $2\|T\| + \|T\|^2 < \frac{7}{9} < 1$, so the registered binomial series defining $(X^2)^{-1/2}$ converges absolutely. The two absolutely convergent series are then identified through their scalar counterparts: on the disc $|z| < \frac{1}{3}$ one has $\text{Re}(1 - z) > 0$, so the branch of the square root normalised to 1 at $z = 0$ satisfies $((1 - z)^2)^{-1/2} = (1 - z)^{-1} = \sum_{k \geq 0} z^k$; the two sides therefore have identical Taylor coefficients at 0, and evaluating both at the single Banach-algebra element T gives $(X^2)^{-1/2} = U^{-1}$. Consequently

$$\text{sgn}(X) = X (X^2)^{-1/2} = (-U)U^{-1} = -I, \quad \theta(X) = \frac{1}{2}(I + \text{sgn } X) = 0. \quad (59)$$

Since the registered compression construction is $\text{Co}_P = \theta(L_P R_P + R_P L_P - I) = \theta(T - I)$, (59) gives $\text{Co}_P = 0$ and hence $\mathcal{S}_P = \text{Im } \text{Co}_P = \{0\}$.

The nonvanishing branch. If P is nonvanishing and $e \leq e_{\text{ca}}$, then `lem-compccb-corner-algebra` makes $\mathcal{S}_P$, with the compressed product, the inherited involution and the compressed unit, an *extended* $C_{\text{ca}}e$ - C^* -algebra. By the meaning of “extended” in `def-extended-epsilon-cstar-algebra`, every $M_n \otimes \mathcal{S}_P$ is then a $C_{\text{ca}}e$ - C^* -algebra with the *same* parameter, independently of n , and its lower C^* -axiom applied to Z reads exactly $\|Z^\dagger \cdot Z\| \geq (1 - C_{\text{ca}}e) \|Z\|^2$. This — and not any level-one estimate — is where the uniformity in n comes from.

Exhaustion. Set

$$K_{\text{sq}} = C_{\text{ca}}, \quad e_{\text{sq}} = \min\left\{e_{\text{ca}}, e_v, \frac{1}{2 \max\{1, C_{\text{ca}}\}}\right\} > 0.$$

The two branches exhaust the possibilities for P . In the nonvanishing branch the displayed axiom is (58); in the vanishing branch $M_n \otimes \mathcal{S}_P = \{0\}$, so the only admissible Z is 0 and (58) reads $0 \geq 0$. The tree also records an explicit endpoint check: since $e_{\text{sq}} \leq e_{\text{ca}}$, the nonvanishing branch applies even at $e = e_{\text{sq}} = e_{\text{ca}}$, so no boundary point is omitted.

Status. `proved`; `af`: `validated` (T0). Workspace `proofs/lem-hcb3-uniform-square-lower`: 18 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-corner-algebra`. This shard exists because of a recorded balloon abort: the first orchestration of `lem-hcb3-diagonal-lower-modulus` tripped the node cap on exactly this claim, so it was factored out of that tree — its statement extracted mechanically from the blocking challenges — and validated on its own, bottom-up. Its value to the parent is that it carries the vanishing case internally, so the downstream argument may use (58) without re-splitting on the dichotomy. Its single registry consumer is `lem-hcb3-diagonal-lower-modulus` (§24).

24 Diagonal Ha lower-modulus propagation

Lemma 23 (`lem-hcb3-diagonal-lower-modulus`). *There are universal $C_{\text{diag}} < \infty$ and $e_{\text{diag}} > 0$ such that, for every H-CB datum with $e \leq e_{\text{diag}}$, if the level-one lower modulus of $\text{Ha}_{P,P}^Q$ is at least $\frac{1}{4}$, then*

$$\|(\text{Ha}_{P,P}^Q)_n(Z)\| \geq (1 - C_{\text{diag}} e) \|Z\| \quad \text{for every } n \geq 1 \text{ and } Z \in M_n \otimes \mathcal{S}_P. \quad (60)$$

Transcription note. The hypothesis is load-bearing and is not discharged here: the conclusion is asserted only for those data whose level-one map already has lower modulus $\geq \frac{1}{4}$. (This conditional shape is the corrected one; the corresponding clause of the parent contract `conj-hcb` was made conditional after an unconditional inverse claim was refuted by an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample.) The “lower modulus” of a bounded linear map is the largest c with $\|\cdot\| \geq c\|\cdot\|$, i.e. the infimum of the norm ratios over nonzero arguments.

Registry contract (verbatim).

Diagonal Ha lower-modulus propagation: there are universal $C_{\text{diag}} < \text{infinity}$ and $e_{\text{diag}} > 0$ such that, for every H-CB datum with $e \leq e_{\text{diag}}$, if the level-one lower modulus of $\text{Ha}^Q_{P,P}$ is at least $1/4$, then $\|(\text{Ha}^Q_{P,P})_n(Z)\| \geq (1 - C_{\text{diag}} * e) \|Z\|$ for every $n \geq 1$ and Z in M_n tensor S_P .

Proof (prose account of the af-validated tree). Write $T_n = (\text{Ha}_{P,P}^Q)_n$ throughout.

The root gap. Let $C_{\text{prod}}, e_{\text{prod}}$ come from `lem-hcb2-product-defect` — normalised to $C_{\text{prod}} \geq 0$, which cannot weaken that estimate since $e\|Z\|\|W\| \geq 0$ — let $K_{\text{sq}}, e_{\text{sq}}$ come from `lem-hcb3-uniform-square-lower` and put

$$C_{\text{diag}} := K_{\text{sq}} + 4C_{\text{prod}}, \quad A := 1 - K_{\text{sq}}e, \quad B := C_{\text{prod}}e.$$

Choose $e_{\text{diag}} > 0$ below $e_{\text{prod}}, e_{\text{sq}}$ and e_{adj} (from `lem-hcb2-amplified-adjointness`) and small enough that $A \geq \frac{3}{4}$, $A^2 - 4B \geq 0$, $4B < \frac{1}{8}$ and $C_{\text{diag}}e \leq \frac{1}{2}$; each requirement is a continuity condition at $e = 0$, where $A = 1$ and $A^2 - 4B = 1$, and is vacuous if the relevant constant vanishes, so the intersection is a positive universal interval. Let $r_{\mp} = \frac{1}{2}(A \mp \sqrt{A^2 - 4B})$ be the roots of $x^2 - Ax + B$. Rationalising the smaller root — legitimate because the denominator is at least $A > 0$, including when $B = 0$ — gives

$$0 \leq r_- = \frac{2B}{A + \sqrt{A^2 - 4B}} \leq \frac{2B}{A} \leq \frac{8}{3}B \leq 4B < \frac{1}{8}, \quad r_+ = A - r_- \geq 1 - C_{\text{diag}}e \geq \frac{1}{2}. \quad (61)$$

The gap between $r_- < \frac{1}{8}$ and $r_+ \geq \frac{1}{2}$ is the whole mechanism: any quantity known to satisfy the quadratic inequality and to exceed $\frac{1}{8}$ is forced up to r_+ .

The zero-corner branch. If $\mathcal{S}_P = \{0\}$ then $M_m \otimes \mathcal{S}_P = \{0\}$ for every m , so the only admissible Z is 0; linearity gives $T_m(0) = 0$ and $(1 - C_{\text{diag}}e)\|Z\| = 0$, so (60) holds at every level. The tree keeps this branch strictly separate, and *no* lower modulus a_m is defined in it — the infimum below would be over an empty set of ratios.

The dichotomy on the nonzero branch. Assume $\mathcal{S}_P \neq \{0\}$ and pick $0 \neq x \in \mathcal{S}_P$. For each n the element $E_{11} \otimes x$ is a nonzero member of $M_n \otimes \mathcal{S}_P$ — the operator-space direct-sum axiom gives $\|\text{diag}(x, 0, \dots, 0)\| = \|x\| > 0$ — so the ratio set $R_n = \{\|T_n(Z)\|/\|Z\| : 0 \neq Z \in M_n \otimes \mathcal{S}_P\}$ is nonempty and $a_n := \inf R_n$ is a finite nonnegative real.

Fix Z . The C^* -identity in the bounded-operator target gives $\|T_n(Z)\|^2 = \|T_n(Z)^\dagger T_n(Z)\|$; `lem-hcb2-amplified-adjointness` identifies $T_n(Z)^\dagger = T_n(Z^\dagger)$ *exactly*; and `lem-hcb2-product-defect`, applied with all three corner indices equal to P and with the factors $Z^\dagger, Z$, gives

$$\|T_n(Z)\|^2 \geq \|T_n(Z^\dagger \cdot Z)\| - C_{\text{prod}} e \|Z\|^2, \quad (62)$$

the involution being isometric. Applying the defining property $\|T_n(X)\| \geq a_n \|X\|$ at $X = Z^\dagger \cdot Z$ and then `lem-hcb3-uniform-square-lower` — multiplication by $a_n \geq 0$ preserving the inequality — turns (62) into $\|T_n(Z)\|^2 \geq [A a_n - B] \|Z\|^2$. Dividing by $\|Z\|^2$ and taking the infimum over nonzero Z , using that $\inf\{r^2 : r \in R\} = (\inf R)^2$ for a nonempty $R \subseteq [0, \infty)$ with finite infimum, gives

$$a_n^2 \geq A a_n - B, \quad \text{i.e.} \quad (a_n - r_-)(a_n - r_+) \geq 0, \quad (63)$$

so $a_n \leq r_-$ or $a_n \geq r_+$: the level- n lower modulus cannot sit in the gap.

Propagation upward. Two operator-space facts move the moduli between levels. First a halving estimate: writing $Z \in M_{2n} \otimes \mathcal{S}_P$ as a 2×2 matrix of $n \times n$ blocks and factoring $Z = A \cdot \text{diag}(Z_{11}, Z_{12}, Z_{21}, Z_{22}) \cdot B$ with the explicit scalar matrices $A = \begin{bmatrix} I & I & 0 & 0 \\ 0 & 0 & I & I \end{bmatrix}$ and $B = \begin{bmatrix} I & 0 \\ 0 & I \\ I & 0 \\ 0 & I \end{bmatrix}$ of norm $\sqrt{2}$, the Ruan axioms give $\|Z\| \leq 2 \max_{i,j} \|Z_{ij}\|$; and since T_{2n} is the entrywise amplification of T_n , compression to the (i, j) target block gives $\|T_{2n}(Z)\| \geq \max_{i,j} \|T_n(Z_{ij})\|$. Hence $\|T_{2n}(Z)\| \geq a_n \max_{i,j} \|Z_{ij}\| \geq \frac{1}{2} a_n \|Z\|$, i.e. $a_{2n} \geq a_n/2$. Second a monotonicity: the zero-corner inclusion $Z \mapsto \text{diag}(Z, 0_k)$ is isometric by the direct-sum axiom and is carried by T_m to $\text{diag}(T_n(Z), 0_k)$ of the same norm, so $a_n \geq a_m$ whenever $n \leq m$.

Now the bootstrap. The hypothesis of the lemma is exactly $a_1 \geq \frac{1}{4}$, while (61) gives $r_- < \frac{1}{8}$; so $a_1 > r_-$ and (63) at $n = 1$ forces $a_1 \geq r_+$. Inductively, if $a_{2^k} \geq r_+$ then the halving estimate gives

$$a_{2^{k+1}} \geq \frac{1}{2} a_{2^k} \geq \frac{1}{2} r_+ \geq \frac{1}{2} (1 - C_{\text{diag}} e) \geq \frac{1}{4} > r_-,$$

using $C_{\text{diag}} e \leq \frac{1}{2}$; so the dichotomy at level 2^{k+1} again excludes the lower alternative and $a_{2^{k+1}} \geq r_+$. Ordinary induction gives $a_{2^k} \geq r_+$ for every $k \geq 0$. Note that the halving alone would degrade the bound geometrically; it is the *gap* that repairs it at each step.

Conclusion. For arbitrary n choose a power of two $m \geq n$. Monotonicity gives $a_n \geq a_m \geq r_+$, and (61) gives $r_+ \geq 1 - C_{\text{diag}} e$. Hence $\|T_n(Z)\| \geq a_n \|Z\| \geq (1 - C_{\text{diag}} e) \|Z\|$ for every Z , which together with the zero-corner branch is (60); the constants are universal and $e_{\text{diag}} > 0$.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb3-diagonal-lower-modulus: 29 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. Together with `lem-hcb3-diagonal-upper` (§22) it pins the amplified diagonal Ha map between $1 \pm O(e)$ — but only under the level-one hypothesis, which nothing here supplies. It is the input to `lem-hcb3-diagonal-inverse` (§25), which inherits that hypothesis, and to `lem-hcb3-offdiagonal-inverse` (§26), which consumes it as its diagonal anchor. The parent `conj-hcb` (§30) is now proved with af: validated, and `op-classical` remains open.

25 Diagonal Ha inverse propagation

Lemma 24 (`lem-hcb3-diagonal-inverse`). *There are universal $C_{\text{inv}} < \infty$ and $e_{\text{inv}} > 0$ such that, for every H-CB datum with $e \leq e_{\text{inv}}$, if $\text{Ha}_{P,P}^Q$ has level-one lower modulus at least $\frac{1}{4}$ and is bijective at level one, then every amplification is bijective and*

$$\|((\text{Ha}_{P,P}^Q)_n)^{-1}\| \leq 1 + C_{\text{inv}} e. \quad (64)$$

Transcription note. **Both** level-one hypotheses are load-bearing and neither is discharged here: the conclusion is asserted only for those data whose level-one map is already bijective and already has lower modulus at least $\frac{1}{4}$. The two are used at genuinely different points of the proof — bijectivity supplies the inverse, the lower modulus supplies its norm — and neither implies the other in this setting. The conditional shape is the corrected one, inherited from the same verdict that made the lower-modulus clause of `conj-hcb` conditional after an unconditional inverse claim was refuted by an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample.

Registry contract (`verbatim`).

Diagonal Ha inverse propagation: there are universal $C_{\text{inv}} < \text{infinity}$ and $e_{\text{inv}} > 0$ such that, for every H-CB datum with $e \leq e_{\text{inv}}$, if $\text{Ha}^Q_{P,P}$ has level-one lower modulus at least $1/4$ and is bijective at level one, then every amplification is bijective and $\|((\text{Ha}^Q_{P,P})_n)^{-1}\| \leq 1 + C_{\text{inv}} * e$.

Proof (prose account of the af-validated tree). The tree is short — five nodes — because the analytic work has already been done by `lem-hcb3-diagonal-lower-modulus` (§24); what remains is an exact algebraic step and one scalar inequality. Write $T := \text{Ha}_{P,P}^Q$ and $T_n := (\text{Ha}_{P,P}^Q)_n$.

Constants. Let $C_{\text{diag}}, e_{\text{diag}}$ be universal witnesses supplied by `lem-hcb3-diagonal-lower-modulus`. Enlarging C_{diag} only weakens that external conclusion, so one may take $C_{\text{diag}} > 0$ and set

$$e_{\text{inv}} := \min\left\{e_{\text{diag}}, \frac{1}{2C_{\text{diag}}}\right\}, \quad C_{\text{inv}} := 2C_{\text{diag}}.$$

Both are universal, $e_{\text{inv}} > 0$, and the choice is made precisely so that

$$0 \leq C_{\text{diag}} e \leq \frac{1}{2} \quad \text{whenever } 0 \leq e \leq e_{\text{inv}}. \quad (65)$$

Bijectivity amplifies exactly. Fix an admissible datum and an arbitrary $n \geq 1$. This step uses the *first* hypothesis and nothing else, and it involves no estimate at all. By the meaning of matrix amplification, $T_n([Z_{ij}]) = [T(Z_{ij})]$ acts slotwise. Surjectivity therefore passes upward: for any target $[Y_{ij}]$ the matrix $[T^{-1}(Y_{ij})]$ is a preimage. So does injectivity: $T_n([Z_{ij}]) = 0$ forces $T(Z_{ij}) = 0$ for every i, j , hence $Z_{ij} = 0$ for every i, j and $[Z_{ij}] = 0$. Thus T_n is bijective and, moreover,

$$(T_n)^{-1} \text{ is the entrywise amplification of } T^{-1}. \quad (66)$$

Note what is *not* claimed: nothing here bounds $\|(T_n)^{-1}\|$ — entrywise inversion alone gives no uniform norm control, which is why the second hypothesis is needed.

The imported lower bound. Since $e \leq e_{\text{inv}} \leq e_{\text{diag}}$ and the level-one lower modulus is at least $\frac{1}{4}$ by the *second* hypothesis, `lem-hcb3-diagonal-lower-modulus` applies `verbatim` and gives

$$\|T_n(Z)\| \geq (1 - C_{\text{diag}} e) \|Z\| \quad \text{for every } Z \in M_n \otimes \mathcal{S}_P. \quad (67)$$

This is the only place the $\frac{1}{4}$ hypothesis enters, and it enters through the imported lemma rather than being re-proved.

The reciprocal estimate. Combine the two. For an arbitrary target element Y , put $Z = (T_n)^{-1}Y$, which exists by (66); then (67) reads $\|Y\| = \|T_n(Z)\| \geq (1 - C_{\text{diag}}e)\|Z\|$. By (65) the quantity $x := C_{\text{diag}}e$ lies in $[0, \frac{1}{2}]$, and on that interval the elementary identity

$$\frac{1}{1-x} = 1 + \frac{x}{1-x} \leq 1 + 2x$$

holds because $1 - x \geq \frac{1}{2}$. Therefore

$$\|(T_n)^{-1}Y\| \leq (1 - C_{\text{diag}}e)^{-1}\|Y\| \leq (1 + 2C_{\text{diag}}e)\|Y\| = (1 + C_{\text{inv}}e)\|Y\|.$$

Since the datum and n were arbitrary, every amplification is bijective with the inverse bound (64), and the constants are universal. The restriction to $x \leq \frac{1}{2}$ is not cosmetic: it is what converts the reciprocal into a linear bound in e , and it is exactly what the choice of e_{inv} secures.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb3-diagonal-inverse: 5 nodes, all validated, root validated, taint clean; first pass, two rounds, no challenges surviving.

Role in the chain. Imports lem-hcb3-diagonal-lower-modulus, lem-hcb2-amplified-adjointness, lem-hcb2-product-defect, lem-compcb-corner-algebra and lem-hcb3-uniform-square-lower. It completes the diagonal block of the H-CB tier: with §22 and §24 it says that, *under the two level-one hypotheses*, the amplified diagonal Ha map is a bijection whose norm and inverse norm are both within $O(e)$ of 1. The matching off-diagonal statement lem-hcb3-offdiagonal-inverse (§26), the hcb4 canonical tier (§§27–29) and the parent conj-hcb (§30) are now all proved with af: validated. Nothing here bears on the status of op-classical, which is open.

26 Off-diagonal Ha inverse propagation

Lemma 25 (`lem-hcb3-offdiagonal-inverse`). *There are universal $C_{\text{rect}} < \infty$ and $e_{\text{rect}} > 0$ such that, for every H-CB datum with $e \leq e_{\text{rect}}$, if $\text{Ha}_{P,R}^Q$ is bijective at level one and $\text{Ha}_{R,R}^Q$ has level-one lower modulus at least $\frac{1}{4}$, then every amplification of $\text{Ha}_{P,R}^Q$ is bijective with*

$$\|((\text{Ha}_{P,R}^Q)_n)^{-1}\| \leq 1 + C_{\text{rect}} e. \quad (68)$$

Transcription note. The two level-one hypotheses are placed on different maps: bijectivity is assumed of the rectangular map $\text{Ha}_{P,R}^Q$ itself, the lower-modulus hypothesis of the diagonal map $\text{Ha}_{R,R}^Q$ at the target index. Neither is discharged here, and neither implies the other. This is the same conditional shape as §25, inherited from the verdict that made the corresponding clause of `conj-hcb` conditional.

Registry contract (`verbatim`).

Off-diagonal Ha inverse propagation: there are universal $C_{\text{rect}} < \text{infinity}$ and $e_{\text{rect}} > 0$ such that, for every H-CB datum with $e \leq e_{\text{rect}}$, if $\text{Ha}^Q_{P,R}$ is bijective at level one and $\text{Ha}^Q_{R,R}$ has level-one lower modulus at least $1/4$, then every amplification of $\text{Ha}^Q_{P,R}$ is bijective with inverse norm at most $1+C_{\text{rect}}*e$.

Proof (prose account of the af-validated tree). Write $T_n := (\text{Ha}_{P,R}^Q)_n$ and $D_n := (\text{Ha}_{R,R}^Q)_n$. The tree has four movements: a constant ledger, a rectangular square estimate, the passage to a lower modulus for T_n , and a reciprocal bound. Only the third movement uses the diagonal hypothesis, and it uses it through `lem-hcb3-diagonal-lower-modulus` rather than re-proving it.

Constants. Let $C_{\text{prod}}, e_{\text{prod}}$ come from `lem-hcb2-product-defect`, let $C_{\text{diag}}, e_{\text{diag}}$ come from `lem-hcb3-diagonal-lower-modulus`, and let e_{adj} be the threshold of `lem-hcb2-amplified-adjointness`. The universal $O(e)$ constant of the compressed-product display (a workspace external, not a registry lemma) is unpacked as a finite C_{cp} valid below a positive e_{cp} . Put

$$B := 1 + C_{\text{cp}}, \quad A := \max\{1, C_{\text{diag}} + B + C_{\text{prod}}\}, \quad C_{\text{rect}} := 2A,$$

$$e_{\text{rect}} := \min\left\{e_{\text{adj}}, e_{\text{prod}}, e_{\text{diag}}, e_{\text{cp}}, \frac{1}{2A}\right\} > 0.$$

The last entry is what makes $Ae \leq \frac{1}{2}$ throughout, and every subsequent coefficient nonnegative.

The rectangular square estimate. Fix an admissible datum, an arbitrary $n \geq 1$ and $Z \in M_n \otimes \mathcal{S}_{P,R}$. Left and right multiplication by $I_n \otimes P$ and $I_n \otimes R$ act entrywise, so the operator to which the θ calculus is applied amplifies as $1_{M_n} \otimes (\cdot)$ and the registered power series gives $\text{Co}_{I_n \otimes P, I_n \otimes R} = 1_{M_n} \otimes \text{Co}_{P,R}$; hence Z lies in the rectangular corner of the amplified algebra, $Z^\dagger$ in the reversed one, and $Z^\dagger \cdot Z$ is the compressed product formed *inside* $M_n \otimes \mathcal{A}$. Comparing it with the ambient product and using the amplified lower C^* axiom,

$$\|Z^\dagger \cdot Z\| \geq (1 - \varepsilon - C_{\text{cp}}e)\|Z\|^2 \geq (1 - Be)\|Z\|^2, \quad (69)$$

uniformly in n . Note that this is a *rectangular* statement and is derived here from the compressed-product display and the norm axioms; the diagonal statement `lem-hcb3-uniform-square-lower` is not the vehicle.

Lower modulus for the rectangular map. Instantiate `lem-hcb2-product-defect` at corner indices (R, P, R) with first factor $Z^\dagger$ and second factor Z ; by `lem-hcb2-amplified-adjointness`

the first image is $(\text{Ha}_{R,P}^Q)_n(Z^\dagger) = T_n(Z)^\dagger$, and the C^* identity for bounded operators between the amplified column spaces gives $\|T_n(Z)^\dagger T_n(Z)\| = \|T_n(Z)\|^2$. Hence

$$\|T_n(Z)\|^2 \geq \|D_n(Z^\dagger \cdot Z)\| - C_{\text{prod}}e\|Z\|^2. \quad (70)$$

The diagonal hypothesis now enters exactly once: applying `lem-hcb3-diagonal-lower-modulus` at the index R to $X = Z^\dagger \cdot Z$ gives $\|D_n(X)\| \geq (1 - C_{\text{diag}}e)\|X\|$, and $1 - C_{\text{diag}}e \geq 1 - Ae \geq \frac{1}{2} > 0$ licenses multiplying (69) by it. Substituting into (70) and discarding the nonnegative quadratic remainder,

$$\|T_n(Z)\|^2 \geq [(1 - C_{\text{diag}}e)(1 - Be) - C_{\text{prod}}e]\|Z\|^2 \geq [1 - (C_{\text{diag}} + B + C_{\text{prod}})e]\|Z\|^2 \geq (1 - Ae)\|Z\|^2.$$

Since $0 \leq Ae \leq \frac{1}{2}$ and $\sqrt{1-x} \geq 1-x$ on $[0, 1]$,

$$\|T_n(Z)\| \geq (1 - Ae)\|Z\|. \quad (71)$$

Bijectivity and the reciprocal bound. Bijectivity amplifies algebraically and with no estimate: $\text{Ha}_{P,R}^Q$ is linear with a two-sided linear inverse S by the first hypothesis, $T_n = 1_{M_n} \otimes \text{Ha}_{P,R}^Q$ acts slotwise, and $1_{M_n} \otimes S$ is a two-sided inverse for it. For a target Y put $Z = T_n^{-1}Y$; then (71) reads $\|Y\| \geq (1 - Ae)\|Z\|$, and $1/(1-x) \leq 1+2x$ for $x = Ae \in [0, \frac{1}{2}]$ gives $\|T_n^{-1}Y\| \leq (1 + 2Ae)\|Y\| = (1 + C_{\text{rect}}e)\|Y\|$. The datum and n were arbitrary and the constants are universal, which is (68).

Status. proved; af: validated (T0). Workspace `proofs/lem-hcb3-offdiagonal-inverse`: 16 nodes, all validated, root validated, taint clean.

Role in the chain. Registry imports: `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-hcb3-diagonal-lower-modulus`, `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and `lem-hcb3-diagonal-inverse`; the exported tree visibly consumes the first three, the remaining three being available but unused in the validated derivation. Together with §§21–25 this completes the third layer of the H-CB tier: the diagonal maps are controlled on both sides, and the rectangular maps inherit an inverse bound from the diagonal anchor at their target index. It is the last clause consumed by the parent `conj-hcb` (§30). Nothing here bears on `op-classical`, which is open.

27 The canonical corner Gram estimate

Lemma 26 (`lem-hcb4-canonical-gram`). *There are universal $C_J < \infty$ and $e_J > 0$ such that every H -CB datum with $e \leq e_J$, every $n \geq 1$ and every Z in either special P, Q corner satisfy*

$$(1 - C_J e) \|Z\| \leq \|J_n(Z)\| \leq (1 + C_J e) \|Z\|. \quad (72)$$

Transcription note. *The special corners are the two involving the one-dimensional projection of the datum: $Z \in M_n \otimes \mathcal{S}_{P,Q}$ with $J_n = J_{P,Q,n}$ the canonical column identification, or $Z \in M_n \otimes \mathcal{S}_{Q,P}$ with $J_n = J_{Q,P,n}$ the canonical row identification, both fixed by `def-canonical-corner-identifications`. The two sides of (72) carry different norms: $\|Z\|$ is the ambient operator-space norm of the amplified corner, while $\|J_n(Z)\|$ is the operator norm of the resulting map between amplified column spaces — the Euclidean norm of `def-ha-map`. The content of the lemma is therefore that the canonical identification, which is pure notation and carries no estimate of its own, is a $1 \pm O(e)$ isometry between those two norms, uniformly in n .*

Registry contract (verbatim).

Canonical corner Gram estimate: there are universal $C_J < \text{infinity}$ and $e_J > 0$ such that every H -CB datum with $e \leq e_J$, every $n \geq 1$, and every Z in either special P, Q corner satisfy $(1 - C_J * e) \|Z\| \leq \|J_n(Z)\| \leq (1 + C_J * e) \|Z\|$.

Proof (prose account of the af-validated tree). The tree proves the column case and transfers the row case to it. The column case is one exact identity fed by two uniform inputs, followed by a scalar computation; there is no analysis beyond the two imports.

Exact Gram scalarisation. Fix an admissible datum, $n \geq 1$ and $Z \in M_n \otimes \mathcal{S}_{P,Q}$, and put $G := J_{P,Q,n}(Z)^\dagger J_{P,Q,n}(Z) \in M_n$. Reading the defining formula of `def-canonical-corner-identifications` against the column-Hilbert inner-product displays of `def-column-hilbert-corner` and the compressed-product display, the compressed square of Z is computed entrywise:

$$[Z^\dagger \cdot Z]_{ij} = \sum_k \langle Z_{ki}, Z_{kj} \rangle u_Q = G_{ij} u_Q, \quad \text{that is} \quad Z^\dagger \cdot Z = G \otimes u_Q. \quad (73)$$

This is an identity, not an estimate. Two norm facts turn it into one. The Ruan scalar-tensor invariance among the operator-space matrix-norm axioms gives $\|G \otimes u_Q\| = \|G\| \|u_Q\|$, and the Hilbert-space Gram identity gives $\|G\| = \|J_{P,Q,n}(Z)\|^2$. Hence

$$\|Z^\dagger \cdot Z\| = \|J_{P,Q,n}(Z)\|^2 \|u_Q\|. \quad (74)$$

Everything after this point is bookkeeping on two scalars: the compressed square on the left and the compressed unit on the right.

The two uniform inputs. There are universal $A, B < \infty$, valid below a universal threshold, with

$$|\|Z^\dagger \cdot Z\| - \|Z\|^2| \leq A e \|Z\|^2, \quad |\|u_Q\| - 1| \leq B e. \quad (75)$$

For the first, the ε - C^* norm axioms of `def-extended-epsilon-cstar-algebra` place the *ambient* square $\|Z^\dagger Z\|$ between $(1 - \varepsilon)\|Z\|^2$ and $(1 + \varepsilon)\|Z\|^2$ at level n , while `lem-compcb-rectangular-product` (§8) applied to the compatible amplified pair $(Z^\dagger, Z)$ gives $\|Z^\dagger \cdot Z - Z^\dagger Z\| \leq C_r e \|Z\|^2$; since $\varepsilon \leq e$, the reverse triangle inequality yields the first bound with $A = C_r + 1$, uniformly in n . For the

second, `lem-compcb-compressed-unit-norm` (§10) gives $|\|u_Q\| - 1| \leq C_u e$ once Q is known to be *nonvanishing*; the datum states only that Q is one-dimensional, and the gap is closed as elsewhere in this tier by the registered bridge $\dim \mathcal{S}_Q = 1 \Rightarrow \mathcal{S}_Q \neq 0 \Rightarrow Q$ nonvanishing. So $B = C_u$.

The scalar passage. Assume $Z \neq 0$ (for $Z = 0$ both sides of (72) vanish) and put $x := \|J_{P,Q,n}(Z)\|/\|Z\|$. Dividing (74) by $\|Z\|^2$ gives $x^2 \|u_Q\| = \|Z^\dagger \cdot Z\|/\|Z\|^2$, so (75) confines

$$x^2 \in \left[\frac{1-Ae}{1+Be}, \frac{1+Ae}{1-Be} \right].$$

Take e_{col} no larger than the two external thresholds, $1/(2 \max\{B, 1\})$ and $1/\max\{A+B, 1\}$. Then the denominators are bounded away from 0 and the two ends simplify to $x^2 \geq 1 - (A+B)e$ and $x^2 \leq 1 + 2(A+B)e$. Finally $\sqrt{1-t} \geq 1-t$ for $t \in [0, 1]$ and $\sqrt{1+t} \leq 1+t/2$ for $t \geq 0$ convert these into

$$1 - (A+B)e \leq x \leq 1 + (A+B)e, \quad (76)$$

which is (72) for the column corner with $C_{\text{col}} = A+B$. The asymmetric factor 2 in the upper bound for x^2 is exactly what the sharper square-root inequality absorbs, so the same constant serves both sides.

Row transfer. For $Z \in M_n \otimes \mathcal{S}_{Q,P}$ put $W := Z^\dagger \in M_n \otimes \mathcal{S}_{P,Q}$. The self-adjoint operator-space axiom gives $\|W\| = \|Z\|$ at every level; `def-canonical-corner-identifications` defines the row map by $J_{Q,P,n}(Z) = J_{P,Q,n}(W)^\dagger$; and Hilbert-space adjunction preserves the operator norm. Applying the column case to W therefore returns both row inequalities with the same constants, so $C_J = C_{\text{col}}$ and $e_J = e_{\text{col}}$ serve the root.

Status. proved; af: validated (T0). Workspace `proofs/lem-hcb4-canonical-gram`: 9 nodes, all validated, root validated, taint clean.

Role in the chain. Registry imports: `lem-compcb-rectangular-product`, `lem-compcb-compressed-unit-norm`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first two, the remaining two being available but unused in the validated derivation. This is the base of the canonical layer of the H-CB tier. Canonical closeness (§28) uses it once, to convert $\|J_{P,Q,n}(Z)\|$ into $\|Z\|$; the canonical inverse estimate (§29) uses its lower half twice, once for the bound $\|J_n^{-1}\| \leq \frac{4}{3}$ and once for the norm equivalence that makes the Neumann target complete. Through those two it reaches the canonical-identity clause of the parent `conj-hcb` (§30), which lists it among its sixteen dependencies. Nothing here bears on `op-classical`, which is open.

28 Canonical Ha closeness

Lemma 27 (`lem-hcb4-canonical-closeness`). *There are universal $C_{\text{sp}} < \infty$ and $e_{\text{sp}} > 0$ such that every H-CB datum with $e \leq e_{\text{sp}}$ satisfies*

$$\max\{\|(\text{Ha}_{P,Q}^Q)_n - J_{P,Q,n}\|, \|(\text{Ha}_{Q,P}^Q)_n - J_{Q,P,n}\|\} \leq C_{\text{sp}} e \quad \text{for every } n \geq 1. \quad (77)$$

Transcription note. Both differences are taken between maps defined on the same amplified corner — $M_n \otimes \mathcal{S}_{P,Q}$ in the first case, $M_n \otimes \mathcal{S}_{Q,P}$ in the second — and the outer norm is the induced one: the supremum over Z of the operator norm of the difference, divided by $\|Z\|$. The estimate is unconditional: unlike the inverse and lower-modulus results of §§24–26 it assumes nothing about the level-one behaviour of any map.

Registry contract (verbatim).

Canonical Ha closeness: there are universal $C_{\text{sp}} < \text{infinity}$ and $e_{\text{sp}} > 0$ such that every H-CB datum with $e \leq e_{\text{sp}}$ satisfies $\max\{\|(\text{Ha}^Q_{P,Q})_n - J_{P,Q,n}\|, \|(\text{Ha}^Q_{Q,P})_n - J_{Q,P,n}\|\} \leq C_{\text{sp}} e$ for every $n \geq 1$.

Proof (prose account of the af-validated tree). The tree has three movements: a scalar normalisation, the column estimate, and the same adjoint transfer used in §27. The organising observation is that the *entire* discrepancy between the special Ha map and the canonical map is carried by one number — the amount by which the compressed unit u_Q fails to be a unit vector for the column-Hilbert norm q_Q .

The normalised compressed unit. There are universal $A < \infty$ and $e_0 > 0$ such that, for $e \leq e_0$, the scalar

$$\alpha := q_Q(u_Q) \quad (78)$$

is strictly positive, the normalised element $q_0 = \alpha^{-1}u_Q$ of `def-canonical-corner-identifications` satisfies $q_Q(q_0) = 1$ by absolute homogeneity, and $|\alpha - 1| \leq Ae$. Two estimates combine. First, `lem-compcb-corner-algebra` (§15) applied with $P = Q$ — legitimate because Q is nonvanishing, by the registered one-dimensional-to-nonvanishing bridge — makes $\mathcal{S}_Q$ an extended γ - C^* -algebra with $\gamma = C_{\text{ca}}e$ and unit u_Q , so the ε -Banach C^* norm axioms give $|\|u_Q\| - 1| \leq \gamma$; after imposing $e \leq \min\{e_{\text{ca}}, 1/(2\max\{C_{\text{ca}}, 1\})\}$ one has $\frac{1}{2} \leq \|u_Q\| \leq \frac{3}{2}$ and in particular $u_Q \neq 0$. Second, `lem-hcb-column-hilbert-squared` (§16) at $n = 1$, with its variable corner specialised to Q and $X = u_Q \in \mathcal{S}_Q$, compares the inner product with the norm: $\langle u_Q, u_Q \rangle = \alpha^2$, so $|\alpha^2 - \|u_Q\|^2| \leq C_{\text{col}}e\|u_Q\|^2$, and for $C_{\text{col}}e \leq \frac{1}{2}$ the inequality $|\sqrt{1+t} - 1| \leq |t|$ on $|t| \leq \frac{1}{2}$ gives $\alpha > 0$ and $|\alpha/\|u_Q\| - 1| \leq C_{\text{col}}e$. Adding the two through $\|u_Q\| \leq \frac{3}{2}$,

$$|\alpha - 1| \leq |\alpha - \|u_Q\|| + |\|u_Q\| - 1| \leq \left(\frac{3}{2}C_{\text{col}} + C_{\text{ca}}\right)e,$$

so $A = 2C_{\text{col}} + C_{\text{ca}}$ works, with e_0 the minimum of the thresholds just used.

The column estimate. Fix $n \geq 1$ and $Z \in M_n \otimes \mathcal{S}_{P,Q}$ and put $H := (\text{Ha}_{P,Q}^Q)_n(Z)$. The comparison is made *on the adjoint side*, where the column-action lemma is directly applicable. By `lem-hcb2-amplified-adjointness` (§11), $H^\dagger = (\text{Ha}_{Q,P}^Q)_n(Z^\dagger)$, and by `def-canonical-corner-identification` $J_{P,Q,n}(Z)^\dagger = J_{Q,P,n}(Z^\dagger)$; so it suffices to compare those two. Applying `lem-hcb1-column-action` (§19) with corner variables (Q, P) , matrix symbol $Z^\dagger$ and an arbitrary column X , and using $\|Z^\dagger\| = \|Z\|$,

$$q_Q(H^\dagger X - Z^\dagger \cdot X) \leq C_{\text{act}} e \|Z\| q_P(X). \quad (79)$$

The remaining term is exactly scalar. Put $d := J_{P,Q,n}(Z)^\dagger X \in \mathbb{C}^n$. Entrywise, the column-Hilbert displays and the defining formula for J give $[Z^\dagger \cdot X]_i = \sum_k Z_{ki}^\dagger \cdot X_k = d_i u_Q$; under the canonical identification by q_0 that vector is αd , whereas $J_{P,Q,n}(Z)^\dagger X$ is d . Hence

$$q_Q(Z^\dagger \cdot X - J_{P,Q,n}(Z)^\dagger X) = |\alpha - 1| \|d\|_2 \leq A e (1 + C_J e) \|Z\| q_P(X), \quad (80)$$

the last step bounding $\|d\|_2$ by $\|J_{P,Q,n}(Z)\| q_P(X)$ and then invoking `lem-hcb4-canonical-gram` (§27) — its only use in this proof. Shrinking e below e_0 , the two external thresholds, e_J and $1/\max\{C_J, 1\}$ makes the factor $(1 + C_J e)$ at most 2, so adding (79) and (80) by the triangle inequality and taking the supremum over nonzero X gives $\|H^\dagger - J_{P,Q,n}(Z)^\dagger\| \leq C_* e \|Z\|$ with $C_* = C_{\text{act}} + 2A$; Hilbert-space adjunction preserves the operator norm, so $\|H - J_{P,Q,n}(Z)\|$ obeys the same bound. No constant and no threshold depends on n .

Row transfer and completion. For $W \in M_n \otimes \mathcal{S}_{Q,P}$, adjointness gives $(\text{Ha}_{Q,P}^Q)_n(W) = ((\text{Ha}_{P,Q}^Q)_n(W^\dagger))^\dagger$ and the definition gives $J_{Q,P,n}(W) = J_{P,Q,n}(W^\dagger)^\dagger$; with $\|W^\dagger\| = \|W\|$ and adjoint norm invariance, the column estimate transfers verbatim. Taking suprema over nonzero Z and W turns both pointwise bounds into map-norm bounds, so (77) holds with $C_{\text{sp}} = C_*$ and e_{sp} the positive minimum of the thresholds collected above.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb4-canonical-closeness: 11 nodes, all validated, root validated, taint clean.

Role in the chain. Registry imports: `lem-hcb-column-hilbert-squared`, `lem-hcb1-column-action`, `lem-hcb2-amplified-adjointness`, `lem-hcb4-canonical-gram`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first five. It is the “canonical-identity closeness” clause of the parent `conj-hcb` (§30) — one of the eight suppliers that parent’s tree cites directly — and it supplies the hypothesis $\|A_n - J_n\| \leq C_{\text{spe}} e$ of the Neumann step in §29. Nothing here bears on `op-classical`, which is open.

29 The canonical Ha inverse estimate

Lemma 28 (`lem-hcb4-canonical-inverse`). *There are universal $C_{\text{sp,inv}} < \infty$ and $e_{\text{sp,inv}} > 0$ such that every H-CB datum with $e \leq e_{\text{sp,inv}}$ has the special maps $\text{Ha}_{P,Q}^Q$ and $\text{Ha}_{Q,P}^Q$ completely bijective, with their amplified inverses differing from the corresponding canonical inverses by at most*

$$\|((\text{Ha}_{P,Q}^Q)_n)^{-1} - J_{P,Q,n}^{-1}\| \leq C_{\text{sp,inv}} e, \quad (81)$$

and likewise for the row pair $((\text{Ha}_{Q,P}^Q)_n, J_{Q,P,n})$. Transcription note. “Completely bijective” means bijective at every amplification level $n \geq 1$, not merely at level one; and the estimate is uniform in n . Unlike the diagonal and off-diagonal inverse statements of §§25–26, this lemma is unconditional: no level-one hypothesis is assumed, because the comparison map J_n is bijective by construction and the smallness of e does the rest.

Registry contract (`verbatim`).

Canonical Ha inverse estimate: there are universal $C_{\text{sp,inv}} < \text{infinity}$ and $e_{\text{sp,inv}} > 0$ such that every H-CB datum with $e \leq e_{\text{sp,inv}}$ has the special maps $\text{Ha}^Q_{Q\{P,Q\}}$ and $\text{Ha}^Q_{Q\{Q,P\}}$ completely bijective, with their amplified inverses differing from the corresponding canonical inverses by at most $C_{\text{sp,inv}}e$.

Proof (prose account of the af-validated tree). The mathematics is a perturbation of an invertible map by a small one; the length of the tree is due almost entirely to justifying that the target space is Banach, which is where the Neumann series lives.

The ledger and the canonical inverse. Let C_J, e_J come from `lem-hcb4-canonical-gram` (§27) and $C_{\text{sp}}, e_{\text{sp}}$ from `lem-hcb4-canonical-closeness` (§28), and put

$$e_{\text{sp,inv}} := \min\left\{e_J, e_{\text{sp}}, \frac{1}{4(C_J + C_{\text{sp}} + 1)}\right\}, \quad C_{\text{sp,inv}} := 4(C_{\text{sp}} + 1). \quad (82)$$

The last entry of the threshold forces both $C_J e \leq \frac{1}{4}$ and $C_{\text{sp}} e \leq \frac{1}{4}$ simultaneously. For either special ordered corner and every n , the canonical map J_n is the amplified coefficient-space identification of `def-canonical-corner-identifications` — the column map, or the row map defined as its adjoint — hence an algebraic linear bijection onto the indicated operator corner, with the corresponding canonical inverse. Its inverse is bounded: the lower half of the Gram estimate reads $\|J_n Z\| \geq (1 - C_J e)\|Z\|$, and applying it to $Z = J_n^{-1} Y$ with $C_J e \leq \frac{1}{4}$ gives

$$\|J_n^{-1}\| \leq \frac{4}{3}. \quad (83)$$

The quantitative Neumann step. The abstract step used twice below is: if $J : X \rightarrow Y$ is a bounded bijection onto a Banach space Y with $\|J^{-1}\| \leq \frac{4}{3}$, and $A : X \rightarrow Y$ satisfies $\|A - J\| \leq C_{\text{sp}} e$ with $C_{\text{sp}} e \leq \frac{1}{4}$, then A is bijective and $\|A^{-1} - J^{-1}\| \leq \frac{8}{3} C_{\text{sp}} e$. Put $T := (A - J)J^{-1} : Y \rightarrow Y$, so that $A = (I_Y + T)J$ and $\|T\| \leq \|A - J\|\|J^{-1}\| \leq \frac{4}{3} C_{\text{sp}} e \leq \frac{1}{3}$. Since Y is Banach so is $B(Y)$, and the partial sums $S_N = \sum_{k=0}^N (-T)^k$ are Cauchy there by $\sum_{k \geq 0} \|T\|^k \leq (1 - \|T\|)^{-1}$; passing to the limit in the finite geometric identities $(I_Y + T)S_N = S_N(I_Y + T) = I_Y - (-T)^{N+1}$ and using $\|T\|^{N+1} \rightarrow 0$ identifies the limit as the two-sided inverse, with $\|(I_Y + T)^{-1}\| \leq \frac{3}{2}$. Then $A^{-1} = J^{-1}(I_Y + T)^{-1}$ and $(I_Y + T)^{-1} - I_Y = -(I_Y + T)^{-1}T$ give

$$\|A^{-1} - J^{-1}\| \leq \frac{4}{3} \cdot \frac{3}{2} \cdot \frac{4}{3} C_{\text{sp}} e = \frac{8}{3} C_{\text{sp}} e. \quad (84)$$

Why the targets are Banach. This is not a formality, and the tree does not treat it as one. In an H-CB datum the ambient $\mathcal{A}$ is complete by `def-extended-epsilon-cstar-algebra`, and `def-compressed-corner` states that $\text{Co}_{P,Q}$ is an exact bounded idempotent with $\mathcal{S}_{P,Q} = \text{Im } \text{Co}_{P,Q} = \text{Ker}(1 - \text{Co}_{P,Q})$; a kernel of a continuous map is closed, and a closed subspace of a Banach space is Banach, so $\mathcal{S}_{P,Q}$ is complete in the *inherited* norm. The norm actually carried by the target is the Euclidean one, and the two are equivalent by the Gram estimate at $n = 1$: there $J_{P,Q,1}(Z)c = Zc$, so $\|J_{P,Q,1}(Z)\| = \|Z\|_{\text{Euc}}$ and $(1 - C_{Je})\|Z\| \leq \|Z\|_{\text{Euc}} \leq (1 + C_{Je})\|Z\|$ with $C_{Je} \leq \frac{1}{4}$. Hence $(\mathcal{S}_{P,Q}, \|\cdot\|_{\text{Euc}})$ is complete, its finite Hilbertian sum $\mathbb{C}^n \otimes \mathcal{S}_{P,Q}$ is complete, and since $B(E, F)$ is Banach whenever F is, both actual targets $B(\mathbb{C}^n, \mathbb{C}^n \otimes \mathcal{S}_{P,Q})$ and $B(\mathbb{C}^n \otimes \mathcal{S}_{P,Q}, \mathbb{C}^n)$ — the column and row targets of `def-ha-map` and `def-canonical-corner-identifications` — are Banach.

Assembly. Fix $n \geq 1$ and either special corner, and let A_n be $(\text{Ha}_{P,Q}^Q)_n$ or $(\text{Ha}_{Q,P}^Q)_n$. Canonical closeness gives $\|A_n - J_n\| \leq C_{\text{sp}}e$, (83) gives the bijection J_n with $\|J_n^{-1}\| \leq \frac{4}{3}$, and (82) gives $C_{\text{sp}}e \leq \frac{1}{4}$; so the Neumann step applies and $\|A_n^{-1} - J_n^{-1}\| \leq \frac{8}{3}C_{\text{sp}}e \leq 4(C_{\text{sp}} + 1)e = C_{\text{sp,inv}}e$. The constants do not depend on n or on the corner, so both special maps are completely bijective with the uniform estimate (81).

Status. proved; af: validated (T0). Workspace `proofs/lem-hcb4-canonical-inverse`: 19 nodes, all validated, root validated, taint clean. Six of the nineteen exported nodes form the completeness branch.

Role in the chain. Registry imports: `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-closeness`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first two. Its `defs` line records `def-ha-map`, `def-hcb-datum`, `def-canonical-corner-identifications`, `def-compressed-corner` (the range/kernel identity) and `def-extended-epsilon-cstar-algebra` (completeness of $\mathcal{A}$); the last two are consumed essentially in the validated closed-corner completeness branch. Downstream, this lemma is one of the sixteen declared dependencies of `conj-hcb` (§30); the parent’s own exported tree cites eight suppliers directly and this is not among them, so the parent consumes it through the registry rather than in its validated derivation. Nothing here bears on `op-classical`, which is open.

30 The H-CB parent contract

Lemma 29 (`conj-hcb`). *There are universal $C_H < \infty$ and $e_H > 0$ such that, whenever $e = \delta + \varepsilon \leq e_H$, Q is a level-one one-dimensional δ -projection in an extended ε - C^* -algebra $\mathcal{A}$, and P, R, S are δ -projections, the maps $1_{M_n} \otimes \text{Ha}_{P,R}^Q$ — read, under the COL-HILB identification, as operators from $\mathbb{C}^n \otimes \mathcal{S}_{R,Q}$ to $\mathbb{C}^n \otimes \mathcal{S}_{P,Q}$ — satisfy for every n the adjoint equality, a product defect at most $C_H e \|Z\| \|W\|$, and the uniform unit, upper-norm, homomorphism and canonical-identity closeness estimates required by `lem_extension`; moreover*

1. *if the level-one lower modulus of $\text{Ha}_{P,P}^Q$ is at least $\frac{1}{4}$, then every amplification has lower modulus at least $1 - C_H e$;*
2. *if $\text{Ha}_{P,P}^Q$ is also bijective at level one, then every amplification is bijective with inverse norm at most $1 + C_H e$;*
3. *the analogous off-diagonal inverse bound for $\text{Ha}_{P,R}^Q$ is asserted only when $\text{Ha}_{P,R}^Q$ is bijective at level one and $\text{Ha}_{R,R}^Q$ satisfies the diagonal lower-modulus hypothesis;*

all constants independent of n , $\dim \mathcal{A}$, block count and block dimensions. Transcription note. `lem_extension` is a label in the pinned source, not a registry identifier: it names the consumer whose required estimate list this contract is calibrated to. The three clauses above are conditional as written; nothing in this document establishes their level-one hypotheses for any particular datum.

Registry contract (verbatim).

H-CB: there are universal $C_H < \text{infinity}$ and $e_H > 0$ such that, whenever $e = \delta + \varepsilon \leq e_H$, Q is a level-one one-dimensional δ -projection in an extended ε - C^* -algebra $\mathcal{A}$, and P, R, S are δ -projections, the maps $1_{M_n} \otimes \text{Ha}_{P,R}^Q$, under the COL-HILB identification with operators on $\mathbb{C}^n \otimes \mathcal{S}_{R,Q}$ and $\mathbb{C}^n \otimes \mathcal{S}_{P,Q}$, satisfy for every n the adjoint equality, product defect at most $C_H e \|Z\| \|W\|$, and the uniform unit, upper-norm, homomorphism, and canonical-identity closeness estimates required by `lem_extension`; moreover, if the level-one lower modulus of $\text{Ha}_{P,P}^Q$ is at least $1/4$, then every amplification has lower modulus at least $1 - C_H e$, and if $\text{Ha}_{P,P}^Q$ is also bijective at level one then every amplification is bijective with inverse norm at most $1 + C_H e$; the analogous off-diagonal inverse bound for $\text{Ha}_{P,R}^Q$ is asserted only when $\text{Ha}_{P,R}^Q$ is bijective at level one and $\text{Ha}_{R,R}^Q$ satisfies that diagonal lower-modulus hypothesis; all constants independent of n , $\dim \mathcal{A}$, block count, and block dimensions.

Proof (prose account of the af-validated tree). The tree is a *composition*: eleven nodes, one for the constant ledger, one per asserted clause, one for uniformity. No new analysis is done at this level, and that is the point — every estimate has already been proved matrix-uniformly in §§11–26 and in the canonical tier (§§27–29), so the parent’s only work is to exhibit one constant pair that serves them all simultaneously.

The ledger. Let the named coefficients and thresholds be those supplied by `lem-hcb2-amplified-adjointness` (e_{adj}), `lem-hcb2-product-defect` ($C_{\text{prod}}, e_{\text{prod}}$), `lem-hcb3-diagonal-unit` ($C_{\text{unit}}, e_{\text{unit}}$), `lem-hcb3-diagonal-up` ($C_{\text{up}}, e_{\text{up}}$), `lem-hcb3-diagonal-lower-modulus` ($C_{\text{diag}}, e_{\text{diag}}$), `lem-hcb3-diagonal-inverse` ($C_{\text{inv}}, e_{\text{inv}}$), `lem-hcb3-offdiagonal-inverse` ($C_{\text{rect}}, e_{\text{rect}}$) and `lem-hcb4-canonical-closeness` ($C_{\text{sp}}, e_{\text{sp}}$). Put

$$C_H := \max\{1, C_{\text{prod}}, C_{\text{unit}}, C_{\text{up}}, C_{\text{diag}}, C_{\text{inv}}, C_{\text{rect}}, C_{\text{sp}}\}, \quad e_H := \min\{1, e_{\text{adj}}, e_{\text{prod}}, e_{\text{unit}}, e_{\text{up}}, e_{\text{diag}}, e_{\text{inv}}, e_{\text{rect}}, e_{\text{sp}}\}. \quad (85)$$

A finite maximum of finite universal constants is finite and universal; a finite minimum of positive universal thresholds is positive and universal. Consequently $e \leq e_H$ meets *every* imported threshold at once, and each imported error coefficient is dominated by C_H — which is the only step where the clauses interact.

The clauses. Each is now the corresponding import, read at $e \leq e_H$ and with its coefficient enlarged to C_H . For $Z \in M_n \otimes \mathcal{S}_{P,R}$ the adjoint clause is the exact identity $(\text{Ha}_{P,R}^Q)_n(Z)^\dagger = (\text{Ha}_{R,P}^Q)_n(Z^\dagger)$; for $Z \in M_n \otimes \mathcal{S}_{P,S}$ and $W \in M_n \otimes \mathcal{S}_{S,R}$ the product-defect clause is

$$\|(\text{Ha}_{P,R}^Q)_n(Z \cdot W) - (\text{Ha}_{P,S}^Q)_n(Z) (\text{Ha}_{S,R}^Q)_n(W)\| \leq C_H e \|Z\| \|W\|, \quad (86)$$

which is simultaneously the asserted uniform amplified homomorphism estimate; the unit clause is $\|(\text{Ha}_{P,P}^Q)_n(I_n \otimes u_P) - I\| \leq C_H e$; the upper-norm clause is $\|(\text{Ha}_{P,P}^Q)_n(Z)\| \leq (1 + C_H e) \|Z\|$; and the canonical-identity clause is

$$\max\{\|(\text{Ha}_{P,Q}^Q)_n - J_{P,Q,n}\|, \|(\text{Ha}_{Q,P}^Q)_n - J_{Q,P,n}\|\} \leq C_H e,$$

with J the maps fixed by **def-canonical-corner-identifications**. The three conditional clauses are, in order, **lem-hcb3-diagonal-lower-modulus**, **lem-hcb3-diagonal-inverse** (§25) and **lem-hcb3-offdiagonal** (§26), each carrying its level-one hypotheses unchanged into the parent statement.

Uniformity. The witnesses (85) are finite maxima and minima of universal quantities only. No entry depends on n , on $\dim \mathcal{A}$, on the block count or on the block dimensions, and no clause involves an entrywise sum over the amplification index; the hostile verifier looked for an n -growth family and found none.

Status. proved; af: validated (T0). Workspace **proofs/conj-hcb**: 11 nodes, all validated, root validated, taint clean. The identifier keeps its historical **conj-** slug; registry ids are stable.

Contract amendment of record. The conjecture as originally transcribed asserted the inverse estimate *unconditionally*. That clause is false: an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample refutes it, and the hostile verifier confirmed the refutation as genuine. The contract above carries the verifier’s replacement clause verbatim. The verifier also recorded that this does not weaken what the EXT tier consumes, because at the point of use the relevant map h_{11} is first shown to be a level-one $O(e)$ -isomorphism, so both conditional hypotheses are met at a universal smallness threshold. Three further corrections were wording-level: the compressed-unit input splits into a general upper bound and a nonvanishing-only two-sided bound; the phrase “established separately” means a universal $1 - O(e)$ lower bound threshold-shrunk to $\frac{1}{4}$, not bare bijectivity; and bijectivity of the canonical corners comes from the Neumann condition, not from two-sided norm bounds alone.

Role in the chain. The registry shard lists sixteen dependencies — the whole COMP-fed H-CB tier including **lem-hcb0-compressed-associator**, **lem-hcb-column-hilbert-squared**, **lem-hcb1-variational**, **lem-hcb1-column-action**, **lem-compcb-corner-algebra**, **lem-hcb3-uniform-square-lower** and the canonical tier **lem-hcb4-canonical-gram** (§27), **lem-hcb4-canonical-closeness** (§28), **lem-hcb4-canonical** (§29); the exported tree cites eight of them directly, the others entering through those eight. All sixteen are proved with af: validated, and all sixteen are now reproduced in this document. Downstream, **conj-hcb** is the principal input to **conj-extcb** (§36), itself proved with af: validated and reproduced below; the conditional clauses above are consumed there, after their level-one hypotheses have been established. The registry shard additionally records the paper-proof’s explicit relative witnesses $C_H = 4000c$ and $e_H = 1/(10000c)$, with c the maximum of the sanctioned COMP-CB and corrected COL-HILB constants; the validated tree does not use them, deriving (85) instead. Nothing here bears on **op-classical**, which is open.

31 One-dimensional corner calculus

The EXT tier works with the corners of a *closed EXT-CB datum* (`def-extcb-datum`) and asks for one map on a matrix algebra of one larger size. The two results of this section are its arithmetic base: through a one-dimensional δ -projection the compressed product is multiplicative on norms up to $1 \pm O(e)$, and consequently a corner between two one-dimensional projections is at most a line. Both are level-one statements; the amplified content of the tier is carried elsewhere.

Lemma 30 (`lem-extcb-one-dimensional-product`). *There are universal $C_{PQR} < \infty$ and $e_{PQR} > 0$ such that, for $e = \delta + \varepsilon \leq e_{PQR}$, if Q is one-dimensional then*

$$| \|X \cdot Y\| - \|X\| \|Y\| | \leq C_{PQR} e \|X\| \|Y\| \quad \text{for } X \in \mathcal{S}_{P,Q}, Y \in \mathcal{S}_{Q,R}. \quad (87)$$

Transcription note. *One-dimensionality is required of the middle index Q — the index shared by the two corners — and of nothing else; P and R are arbitrary δ -projections. “One-dimensional” means $\dim \mathcal{S}_Q = 1$ (`def-one-dimensional-delta-projection`), and the statement is level-one.*

Registry contract (`verbatim`).

Level-one one-dimensional corner product: there are universal $C_{PQR} < \text{infinity}$ and $e_{PQR} > 0$ such that, for $e = \delta + \varepsilon \leq e_{PQR}$, if Q is one-dimensional then $\text{abs}(|\|X \cdot Y\| - \|X\| \|Y\||) \leq C_{PQR} * e * \|X\| \|Y\|$ for X in $\mathcal{S}_{\{P,Q\}}$ and Y in $\mathcal{S}_{\{Q,R\}}$.

Proof (prose account of the af-validated tree). The tree proves the *squared* comparison and takes a square root at the end; the squared form is what the algebra produces, because the only handle on X is $X^\dagger \cdot X$.

A uniform ledger. One-dimensionality gives $\mathcal{S}_Q \neq 0$, and vanishing of the corner characterises the small alternative of `def-delta-projection`; hence Q is nonvanishing and `lem-compcb-corner-algebra` makes $\mathcal{S}_Q$ an extended $C_{ca}e$ - C^* -algebra with unit $u_Q = \text{Co}_Q(Q)$ and $|\|u_Q\| - 1| \leq C_{ca}e$. The dichotomy bounds $\|P\|, \|Q\|, \|R\|$ by a universal B ; the compressed-product display supplies a universal C_{dot} with $\|A \cdot B - AB\| \leq C_{\text{dot}} e \|A\| \|B\|$ on compatible pairs. One consequence is used repeatedly: u_Q is a left unit on $\mathcal{S}_{Q,R}$ up to $O(e)$,

$$\|u_Q Y - Y\| \leq C_L e \|Y\|, \quad (88)$$

obtained by combining $\|u_Q - Q\| = O(e)$ (from $\text{Co}_Q(A) = Q(AQ) + O(e)\|A\|$ and $\|Q^2 - Q\| \leq \delta$) with $\|QY - Y\| = O(e)\|Y\|$ (from $Y = \text{Co}_{Q,R}(Y) = Q(YR) + O(e)\|Y\|$ and one associator move). Write C_0 for the maximum of these constants and shrink the threshold below $1/(2C_0)$, so that $\|u_Q\| \geq \frac{1}{2}$.

Scalarisation. Fix $X \in \mathcal{S}_{P,Q}$. The compression adjoint identity puts $X^\dagger$ in $\mathcal{S}_{Q,P}$, so $a := X^\dagger \cdot X$ lies in $\mathcal{S}_Q$; since $\dim \mathcal{S}_Q = 1$ and $u_Q \neq 0$, the unit spans, and $a = h u_Q$ for a unique scalar h , real because $a^\dagger = a$ and $u_Q^\dagger = u_Q$. Comparing a with the ambient square and using the ε - C^* axioms, $|\|a\| - \|X\|^2| \leq (C_0 + 1)e\|X\|^2$; and $\|a\| = |h| \|u_Q\|$ together with $|\|u_Q\| - 1| \leq C_0 e$ and $\|u_Q\| \geq \frac{1}{2}$ converts this into

$$|\|h\| - \|X\|^2| \leq C_1 e \|X\|^2, \quad |h| \leq C_h \|X\|^2. \quad (89)$$

The squared comparison. Put $r = \|X\|$, $s = \|Y\|$, $W = X \cdot Y$ and $Z = XY$ for the ambient product. Three estimates combine. First, $\|W - Z\| \leq C_0 e r s$ and $|\|Z^\dagger Z\| - \|Z\|^2| \leq \varepsilon \|Z\|^2$ give

$|||W||^2 - ||Z^\dagger Z|| \leq C_3 e r^2 s^2$. Second, exact anti-multiplicativity of the involution, two applications of the ε -associator axiom, the replacement of $X^\dagger X$ by $a = h u_Q$ and then (88) give

$$|(XY)^\dagger(XY) - h Y^\dagger Y| \leq C_4 e r^2 s^2. \quad (90)$$

Third, $|||Y^\dagger Y|| - s^2| \leq \varepsilon s^2$ and (89) give $|||h Y^\dagger Y|| - r^2 s^2| \leq C_5 e r^2 s^2$. The reverse triangle inequality then yields $|||W||^2 - r^2 s^2| \leq C_2 e r^2 s^2$ with $C_2 = C_3 + C_4 + C_5$.

Square root. If $rs = 0$ then $X = 0$ or $Y = 0$, and bilinearity of the compressed product makes $W = 0$, so (87) holds exactly. Otherwise $||W|| + rs \geq rs > 0$ and the factorisation $|\alpha - \beta| = |\alpha^2 - \beta^2|/(\alpha + \beta)$ for nonnegative α, β turns the squared comparison into (87) with $C_{PQR} = C_2$ and e_{PQR} the minimum of the finitely many universal thresholds.

Status. proved; af: validated (T0). Workspace proofs/lem-extcb-one-dimensional-product: 25 nodes, all validated, root validated, taint clean.

Lemma 31 (lem-extcb-one-dimensional-corner-dimension). *For sufficiently small universal $\delta + \varepsilon$, if P and Q are one-dimensional δ -projections then $\dim \mathcal{S}_{P,Q} \leq 1$.*

Registry contract (verbatim).

Level-one one-dimensional corner dimension: for sufficiently small universal delta+epsilon, if P and Q are one-dimensional delta-projections then dim S_{P,Q} <= 1.

Proof (prose account of the af-validated tree). Set $e_0 = \min\{e_{PQR}, 1/(2(C_{PQR} + 1))\}$ and let $e \leq e_0$. For nonzero $X, Y \in \mathcal{S}_{P,Q}$ the compression adjoint identity puts $X^\dagger$ in $\mathcal{S}_{Q,P}$ with $||X^\dagger|| = ||X||$, and $X^\dagger \cdot Y$ lies in $\mathcal{S}_Q$. Apply Lemma 30 to the triple (Q, P, Q) — whose middle index is P , one-dimensional by hypothesis — to get

$$||X^\dagger \cdot Y|| \geq (1 - C_{PQR} e) ||X|| ||Y|| \geq \frac{1}{2} ||X|| ||Y|| > 0. \quad (91)$$

This is the whole analytic content; the rest is linear algebra. If $\mathcal{S}_{P,Q} = 0$ there is nothing to prove. Otherwise fix a nonzero X and let $T_X : \mathcal{S}_{P,Q} \rightarrow \mathcal{S}_Q$ be $T_X(Y) = X^\dagger \cdot Y$, linear because the compressed product is bilinear. By (91), $T_X(Y) \neq 0$ whenever $Y \neq 0$, so $\text{Ker } T_X = 0$; an injective linear map into $\mathcal{S}_Q$, which is one-dimensional by hypothesis, forces $\dim \mathcal{S}_{P,Q} \leq \dim \mathcal{S}_Q = 1$.

Status. proved; af: validated (T0). Workspace proofs/lem-extcb-one-dimensional-corner-dimension: 8 nodes, all validated, root validated, taint clean.

Role in the chain. Both lemmas import `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the second imports the first. Together they are the approximate- C^* substitute for the exact fact that a corner through a rank-one projection is a line: the product statement is the result the pinned source invokes for transitivity of equivalence of one-dimensional δ -projections (`def-one-dimensional-delta-projection`), and the dimension statement is consumed by `lem-extcb1-cross-co` (§33), which is where that transitivity is actually used.

32 Corner-dimension additivity

Lemma 32 (`lem-extcb-corner-dimension-additivity`). *For two finite-dimensional commutative C^* -algebras with projection bases and non-unital sufficiently accurate inclusions v, w , the compressed corner $\mathcal{S}_{v(I), w(I)}$ is linearly bijective to*

$$\bigoplus_{j,k} \mathcal{S}_{v(\Pi_j), w(\Sigma_k)}. \quad (92)$$

Transcription note. “Projection basis” is **def-projection-basis** ($\Pi_j^\dagger = \Pi_j$, $\Pi_j \Pi_k = \delta_{jk} \Pi_k$, $\sum_j \Pi_j = I$, and likewise $\{\Sigma_k\}$); “inclusion” is **def-extended-delta-inclusion**, and non-unital means the unit of the source need not be carried to the unit of the ambient algebra — $v(I)$ is only an approximate projection there. “Sufficiently accurate” is a universal ceiling on the inclusion defect plus the ambient ε ; the tree fixes it as the minimum of finitely many universal thresholds, independent of the two basis cardinalities. The conclusion is a linear bijection, bounded with bounded inverse; no isometry and no bound uniform in the number of blocks is asserted, and none is needed downstream, where only the dimension count is consumed.

Registry contract (verbatim).

Level-one corner-dimension additivity: for two finite-dimensional commutative C^* -algebras with projection bases and non-unital sufficiently accurate inclusions v, w , the compressed corner $\mathcal{S}_{\{v(I), w(I)\}}$ is linearly bijective to the direct sum over j, k of $\mathcal{S}_{\{v(\Pi_j), w(\Sigma_k)\}}$.

Proof (prose account of the af-validated tree). The tree proves one binary splitting lemma, transports it across the second index by the involution, and then runs two finite inductions. Write $P_j = v(\Pi_j)$, $Q_k = w(\Sigma_k)$, $P_{[1,j]} = \sum_{r \leq j} P_r$ and $Q_{[1,k]} = \sum_{s \leq k} Q_s$.

Images of a projection basis. Each Π_j and each partial sum $\sum_{r < j} \Pi_r$ is an exact Hermitian projection of norm one in the source, and distinct basis projections annihilate each other. Since v is linear, star-preserving, δ -multiplicative and two-sidedly norm-bounded, every image is Hermitian with norm at most $1 + \delta$ and $\|P_j^2 - P_j\| \leq \delta$; the partial-sum identities $P_{[1,j]} = P_{[1,j-1]} + P_j$ and $P_{[1,p]} = v(I)$ are *exact*, by linearity alone, while orthogonality survives only approximately,

$$\|P_{[1,j-1]} P_j\| \leq \delta, \quad \|P_j P_{[1,j-1]}\| \leq \delta, \quad (93)$$

the second from the first by star-preservation. The same holds for w and the Q_k . This mixture — exact additivity, approximate orthogonality — is what the splitting lemma is built to consume.

Binary splitting. Let $R = R_0 + R_1$ and T be uniformly bounded approximate projections with R_0, R_1 approximately orthogonal, and let e dominate the ambient associativity and product errors, all projection defects and both orthogonality defects. Define

$$A(X_0, X_1) = \text{Co}_{R,T}(X_0 + X_1), \quad B(X) = (\text{Co}_{R_0,T} X, \text{Co}_{R_1,T} X),$$

bounded and linear because every compression is a bounded idempotent whose range is the displayed corner. The calculus of **def-compressed-corner** gives, uniformly, $X = U(XV) + O(e)\|X\|$ for $X \in \mathcal{S}_{U,V}$. Hence for $X_i \in \mathcal{S}_{R_i,T}$, expanding $R = R_0 + R_1$ reproduces X_i from the matching summand and leaves $O(e)\|X_i\|$ from the other — one reassociation plus $R_{1-i}R_i = O(e)$ — so $\text{Co}_{R,T} X_i = X_i + O(e)\|X_i\|$; applying $\text{Co}_{R_a,T}$ and using $R_a R_i = \delta_{ai} R_i + O(e)$ gives $\text{Co}_{R_a,T} \text{Co}_{R,T} X_i =$

$\delta_{ai}X_i + O(e)\|X_i\|$, i.e. $\|BA - I\| \leq Ce$ for the max norm on the sum. In the other order, for $X \in \mathcal{S}_{R,T}$ the same replacement gives $X = R(XT) + O(e)\|X\|$ and $\text{Co}_{R_i,T}X = R_i(XT) + O(e)\|X\|$; adding the two and using $R_0 + R_1 = R$ *exactly* gives $\text{Co}_{R_0,T}X + \text{Co}_{R_1,T}X = X + O(e)\|X\|$, and applying the bounded $\text{Co}_{R,T}$, which fixes X , gives $\|AB - I\| \leq Ce$. Shrink the universal threshold so that $Ce < 1$: the Neumann series invert BA and AB , so $(BA)^{-1}B$ is a left inverse of A and $B(AB)^{-1}$ a right inverse; a map with both is a bounded linear bijection.

The second index. By the compression adjoint identity and isometry of the involution, $X \mapsto X^\dagger$ is a conjugate-linear isometric bijection $\mathcal{S}_{U,V} \rightarrow \mathcal{S}_{V,U}$. Conjugating the first-index assembly for a split $T = T_0 + T_1$ by this map on domain and codomain gives a *complex*-linear bounded bijection — two conjugate-linear maps around a complex-linear one — and the adjoint identity identifies it with the canonical second-index assembly $\mathcal{S}_{R,T_0} \oplus \mathcal{S}_{R,T_1} \rightarrow \mathcal{S}_{R,T}$. No second calculation is needed.

Two inductions. Fix j . The base $k = 1$ is the identity, since $Q_{[1,1]} = Q_1$ exactly; for $k \geq 2$ the second-index splitting applies to $Q_{[1,k]} = Q_{[1,k-1]} + Q_k$, whose hypotheses are supplied uniformly by (93). Finite induction over the q basis projections, composing each step with the preceding bijection and using $Q_{[1,q]} = w(I)$, produces a bounded linear bijection $\mathcal{S}_{P_j,w(I)} \rightarrow \bigoplus_k \mathcal{S}_{P_j,Q_k}$. The first-index splitting run along $P_{[1,j]} = P_{[1,j-1]} + P_j$ produces, in the same way, $\mathcal{S}_{v(I),w(I)} \rightarrow \bigoplus_j \mathcal{S}_{P_j,w(I)}$. Composing the second with the direct sum of the first family and reassociating the finite double sum — an isometric relabelling — gives (92). Each direct sum of finitely many bounded bijections is bounded with bounded inverse, the bound being a maximum over a finite index set.

Status. proved; af: validated (T0). Workspace proofs/lem-extcb-corner-dimension-additivity: 39 nodes, all validated, root validated, taint clean. Several of those nodes are *re-derivations*: where a child had been written to consume a still-pending sibling, the tree carries a sharper child that discharges the same step from validated material only. That is a validation-ordering artifact of the bottom-up protocol, not additional mathematics; the mathematical content is the four movements above.

Role in the chain. Imports `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. It is the counting engine of the EXT tier: a corner over a matrix-unit basis decomposes into its blocks, so a dimension over a composite projection is the sum of the dimensions over its parts. Its single registry consumer is `lem-extcb1-cross-corner-dimension` (§33), which applies it to the diagonal subalgebra of M_r and to the scalars, and reads off $\dim \mathcal{S}_{v(I_r),Q} = r d$.

33 Dimension selection in an EXT-CB datum

A closed EXT-CB datum (`def-extcb-datum`) carries δ -projections P, Q with $\|P + Q - I\| \leq \delta$, an extended δ -isomorphism $v : M_r \rightarrow \mathcal{S}_P$, $\dim \mathcal{S}_Q = 1$ and $\mathcal{S}_{P,Q} \neq 0$. Nothing in the datum says how large the cross corner is. The two lemmas of this section settle it: the corner has dimension exactly r , and the diagonal Q -corner exactly 1 — the shape needed to build one map on M_{r+1} . The first lemma is the technical enabler; the second is the count.

Lemma 33 (`lem-extcb1-close-corner-dimension`). *There is a universal $e_{\text{close}} > 0$ such that, in an EXT-CB datum with $e \leq e_{\text{close}}$, the compression ranges $\mathcal{S}_{v(I_r),Q}$ and $\mathcal{S}_{P,Q}$ have the same dimension. Transcription note. $v(I_r)$ is the image of the source unit; it is an approximate projection close to P , not equal to it, and the two corners it and P cut out are literally different subspaces. Only their dimensions are claimed equal.*

Registry contract (`verbatim`).

Close-compression range invariance: there is a universal $e_{\text{close}} > 0$ such that, in an EXT-CB datum with $e \leq e_{\text{close}}$, the compression ranges $\mathcal{S}_{\{v(I_r),Q\}}$ and $\mathcal{S}_{\{P,Q\}}$ have the same dimension.

Proof (prose account of the af-validated tree). Write $R = v(I_r)$, $E = \text{Co}_{R,Q}$ and $F = \text{Co}_{P,Q}$.

R is close to P , and is itself an approximate projection. Bijectivity of v onto $\mathcal{S}_P$ makes that corner nonzero, so P is nonvanishing; $\dim \mathcal{S}_Q = 1$ makes Q nonvanishing; the nonvanishing alternative of `def-delta-projection` then bounds $\|P\|, \|Q\|$ by a universal M_1 . By `lem-compcb-corner-algebra` the corner $\mathcal{S}_P$ is an extended $C_{\text{ca}}e$ - C^* -algebra with unit $u_P = \text{Co}_P(P)$, and the unit clause for the extended δ -isomorphism v gives $\|R - u_P\| \leq \delta$; combining with $\|u_P - P(PP)\| = O(e)\|P\|$ and $\|P(PP) - P\| \leq ((1 + \varepsilon)\|P\| + 1)\delta$ yields

$$\|R - P\| \leq C_1 e. \quad (94)$$

Star-preservation gives $R^\dagger = R$, δ -multiplicativity gives $\|R \cdot R - R\| \leq \delta$ inside $\mathcal{S}_P$, and the compressed-versus-ambient comparison converts this into $\|R^2 - R\| \leq C_2 e$ in $\mathcal{A}$: R is a (C_{Re}) -projection there.

Two close bounded idempotents. With the common adjusted defect $\delta' = C_{Re}$, the pairs (R, Q) and (P, Q) both satisfy the hypotheses of `lem-compcb-amplified-compression-identities` at $n = 1$, so $E^2 = E$ and $F^2 = F$. Each compression is $O(e)$ -close to the corresponding left-times-right multiplication, and $(L_R R_Q - L_P R_Q)(X) = (R - P)(XQ)$ has norm at most $(1 + \varepsilon)^2 \|R - P\| \|Q\| \|X\|$; with (94) this gives universal C_0, M_0 with

$$\|E - F\| \leq C_0 e, \quad \|E\| \leq M_0. \quad (95)$$

Range transport. If E, F are bounded idempotents on a Banach space with $\|E\| \leq M$ and $\|E - F\|(2M + 1) < 1$, put $W = FE + (I - F)(I - E)$. Idempotence gives the two exact identities

$$W - I = (F - E)(2E - I), \quad WE = FW = FE, \quad (96)$$

so $\|W - I\| \leq \|F - E\|(2\|E\| + 1) < 1$ and W is invertible by its Neumann series. The relation $WE = FW$ sends $\text{Im } E$ into $\text{Im } F$ and, read as $EW^{-1} = W^{-1}F$, sends $\text{Im } F$ back; hence W restricts to a linear bijection $\text{Im } E \rightarrow \text{Im } F$ and the two ranges have equal dimension. Taking e_{close}

below the thresholds of the two preceding steps and below $1/[2C_0(2M_0 + 1)]$ makes (95) satisfy that hypothesis, and $\text{Im Co}_{R,Q} = \mathcal{S}_{R,Q}$, $\text{Im Co}_{P,Q} = \mathcal{S}_{P,Q}$ finish it.

Status. proved; af: validated (T0). Workspace proofs/lem-extcb1-close-corner-dimension: 16 nodes, all validated, root validated, taint clean. This lemma exists because a hostile verdict on the EXT-CB transcription demanded that the level-one close-idempotent normalisation be made explicit, and that the selection threshold cover it.

Lemma 34 (lem-extcb1-cross-corner-dimension). *There is a universal $e_{\text{sel}} > 0$ such that every EXT-CB datum with $e \leq e_{\text{sel}}$ satisfies*

$$(\dim \mathcal{S}_{P,Q}, \dim \mathcal{S}_{Q,Q}) = (r, 1). \quad (97)$$

Registry contract (verbatim).

EXT-CB cross-corner dimension: there is a universal $e_{\text{sel}} > 0$ such that every EXT-CB datum with $e \leq e_{\text{sel}}$ satisfies $(\dim S_{\{P,Q\}}, \dim S_{\{Q,Q\}}) = (r, 1)$.

Proof (prose account of the af-validated tree). The count is produced by transporting the diagonal matrix units through v , showing that each transported unit is a one-dimensional projection equivalent to all the others, and then summing.

Two inclusions. Let $D_r \subseteq M_r$ be the diagonal subalgebra with projection basis $\Pi_a = E_{aa}$, let v_D be the restriction of v to D_r , put $R_a = v(E_{aa})$, and let $w : \mathbb{C} \rightarrow \mathcal{A}$ be $w(\lambda) = \lambda Q$, whose source has the one-element projection basis $\{1\}$. Restriction preserves linearity, star-preservation, δ -multiplicativity and the two-sided norm bounds, and the compressed product on $\mathcal{S}_P$ differs from the ambient one by $O(e)$ on bounded elements, so v_D is a non-unital $O(e)$ -inclusion into $\mathcal{A}$ with $v_D(I) = v(I_r)$, uniformly in r . For w : nonvanishing of Q gives $|||Q|| - 1| \leq C_{\text{nv}}e$, the operator-space axioms give the exact cross-norm identity $\|X \otimes Q\| = \|X\| \|Q\|$ at every level, and $\|(XY) \otimes Q - (X \otimes Q)(Y \otimes Q)\| = \|(XY) \otimes (Q - Q^2)\| \leq \delta \|X\| \|Y\|$; so every amplification of w is an $O(e)$ -inclusion as well.

The transported matrix units. For $a, b \leq r$ let $c_{ab}(X) = E_{aa} X E_{bb}$ on M_r and put $T_{ab} = v c_{ab} v^{-1} \text{Co}_P$. Since Co_P is an idempotent with range $\mathcal{S}_P = \text{Im } v$, one has $\text{Co}_P v = v$ and $v^{-1} \text{Co}_P v = \text{id}$; as $c_{ab}^2 = c_{ab}$ with one-dimensional range, T_{ab} is an *exact* bounded idempotent of rank one, with range $\mathbb{C} v(E_{ab})$. The tree then compares it with the corresponding compression: uniformly in a, b and r ,

$$\|\text{Co}_{R_a, R_b} - T_{ab}\| \leq C_m e. \quad (98)$$

Three fixed-length estimates give this — each R_a is a Hermitian $C_1 e$ -projection (from $\|E_{aa}\| = 1$, δ -multiplicativity and the compressed-versus-ambient comparison); the subordination bound $\|R_a(XR_b) - R_a((\text{Co}_P X)R_b)\| \leq C_s e \|X\|$, a finite telescope using $R_i = \text{Co}_P(R_i)$ and $\text{Co}_P(Y) = P(YP) + O(e)\|Y\|$; and the transported-source comparison $\|T_{ab}X - R_a((\text{Co}_P X)R_b)\| \leq C_3 e \|X\|$, from two applications of δ -multiplicativity in M_r and the corner-algebra bounds — together with $\|\text{Co}_{R_a, R_b} X - R_a(XR_b)\| \leq C_4 e \|X\|$. Since all source matrix factors have norm one, no constant depends on r .

Rank invariance. If E, F are bounded idempotents with $\|E - F\| < 1$ then $x \in \text{Im } E$ with $Fx = 0$ gives $x = Ex = (E - F)x$, hence $x = 0$; interchanging E and F gives the reverse, so the two ranges have equal dimension when either is finite-dimensional. Shrinking e so that $C_m e < 1$ and applying this to (98) gives $\dim \mathcal{S}_{R_a, R_b} = \text{rank } T_{ab} = 1$ for all a, b : taking $a = b$, each

R_a is a one-dimensional projection; taking $a \neq b$, the R_a are pairwise equivalent in the sense of `def-one-dimensional-delta-projection`.

Dichotomy, additivity, selection. Put $d_a = \dim \mathcal{S}_{R_a, Q}$. By `lem-extcb-one-dimensional-corner-dimension` (§31), applied to the one-dimensional projections R_a and Q with the enlarged common accuracy, $d_a \leq 1$, so $d_a \in \{0, 1\}$. The values coincide: if some $d_b = 1$ then R_b is equivalent to Q , every R_a is equivalent to R_b , and transitivity of equivalence — the registered property whose source proof is the one-dimensional product lemma of §31 — gives $d_a = 1$ for all a ; otherwise all $d_a = 0$. Write d for the common value. Now `lem-extcb-corner-dimension-additivity` (§32), applied to D_r and $\mathbb{C}$ with the inclusions v_D, w , gives a linear bijection $\mathcal{S}_{v(I_r), Q} \rightarrow \bigoplus_{a=1}^r \mathcal{S}_{R_a, Q}$ and hence $\dim \mathcal{S}_{v(I_r), Q} = \sum_a d_a = r d$; Lemma 33 turns this into $\dim \mathcal{S}_{P, Q} = r d$. The datum assumes $\mathcal{S}_{P, Q} \neq 0$ and $r \geq 1$, so $d \neq 0$, i.e. $d = 1$ and $\dim \mathcal{S}_{P, Q} = r$. Finally $\mathcal{S}_{Q, Q} = \mathcal{S}_Q$ has dimension 1 by hypothesis, and e_{sel} is the minimum of the finitely many universal thresholds used, giving (97).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb1-cross-corner-dimension`: 30 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. The close-corner lemma imports `lem-compcb-amplified-compression-identities`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the cross-corner lemma imports both of the preceding sections and the close-corner lemma. The pair is the *selection* step of the EXT tier: it is what licenses treating $\mathcal{S}_{P, Q}$ as an r -dimensional column and $\mathcal{S}_Q$ as a line, i.e. the block shape that the four-corner merge (§35) assembles into a single map on M_{r+1} . Its consumer is `conj-extcb` (§36), now `proved` with `af: validated`; the close-corner lemma is there because a hostile verdict on that parent demanded it.

34 The four-corner assembled norm estimate

Lemma 35 (`lem-extcb-four-corner-norm`). *There are universal $K_{\text{norm}} < \infty$ and $e_{\text{norm}} > 0$ such that, for every four-corner merging datum with $e = \rho + \varepsilon \leq e_{\text{norm}}$, every $n \geq 1$ and every $X \in M_n \otimes \mathcal{B}$, the assembled map $\gamma_n = \alpha_n \Gamma_n \mu_n$, where*

$$\mu_n(X) = ((I_n \otimes \Pi_j) X (I_n \otimes \Pi_k))_{j,k}, \quad \Gamma_n = \bigoplus_{j,k} \gamma_{jk,n}, \quad \alpha_n((Y_{jk})) = \sum_{j,k} Y_{jk},$$

satisfies

$$(1 - K_{\text{norm}} e) \|X\| \leq \|\gamma_n(X)\| \leq (1 + K_{\text{norm}} e) \|X\|. \quad (99)$$

Transcription note. ρ is the datum's common defect bound of `def-four-corner-merging-datum` — involution compatibility, compressed-product compatibility, diagonal-unit approximation, two-sided norm control, and the quantitative complementarity $\|P_1 + P_2 - I\| \leq \rho$ of the target projections — and not the signed-picture scale of the same name in `CONVENTIONS.md`. The four maps γ_{jk} are fixed at level one and reused at every amplification: γ_n is not a separately chosen family.

Registry contract (`verbatim`).

Four-corner assembled norm estimate: there are universal $K_{\text{norm}} < \text{infinity}$ and $e_{\text{norm}} > 0$ such that, for every four-corner merging datum with $e = \text{rho} + \text{epsilon} \leq e_{\text{norm}}$, every $n \geq 1$, and every X in M_n tensor $\mathcal{B}$, the assembled map $\text{gamma}_n = \alpha_n \Gamma_n \mu_n$ (where $\mu_n(X) = ((I_n \text{ tensor } \Pi_j) X (I_n \text{ tensor } \Pi_k))_{j,k}$, Γ_n is the direct sum of the four fixed amplifications $\text{gamma}_{jk,n}$, and $\alpha_n((Y_{jk})) = \sum_{j,k} Y_{jk}$) satisfies $(1 - K_{\text{norm}} * e) \|X\| \leq \|\text{gamma}_n(X)\| \leq (1 + K_{\text{norm}} * e) \|X\|$.

Proof (prose account of the af-validated tree). The estimate is proved in three stages, and the order matters: crude two-sided bounds first, an approximate multiplicativity of the *square* second, and only then a scalar bootstrap that turns the crude constants into $1 \pm O(e)$. A direct triangle-inequality attack on (99) does not work, because the four blocks can cancel.

Stage 1: block conditioning. Put $\pi_j = I_n \otimes \Pi_j$ and $X_{jk} = \pi_j X \pi_k$. The source projections are exactly complementary, so $X = \sum_{j,k} X_{jk}$, $\|X_{jk}\| \leq \|X\|$ and

$$\max_{j,k} \|X_{jk}\| \leq \|X\| \leq \sum_{j,k} \|X_{jk}\| \leq 4 \max_{j,k} \|X_{jk}\|. \quad (100)$$

On the target side put $p_j = I_n \otimes P_j$ and $C_{jk} = 1_{M_n} \otimes \text{Co}_{P_j, P_k}$. The compression construction commutes with amplification — the operator $L_{P_j} R_{P_k} + R_{P_k} L_{P_j} - 1$ amplifies entrywise, every partial sum of the defining power series satisfies $f_N(T_n) = 1_{M_n} \otimes f_N(T)$, and matrix entries are bounded coordinate maps — so $C_{jk} = \text{Co}_{p_j, p_k}$ is genuinely the compression built inside $M_n \otimes \mathcal{A}$ and $\text{Im } C_{jk} = M_n \otimes \mathcal{S}_{P_j, P_k}$. Quantitative complementarity gives $\|p_j^2 - p_j\| \leq \rho$ and $\|p_1 + p_2 - I\| \leq \rho$, whence $\|p_j\| \leq 2$ and, for $j \neq l$, $\|p_j p_l\| \leq 4e$. Consequently C_{jk} fixes its own corner and almost annihilates the other three: $\|C_{jk}(Y_{lm})\| \leq L_c e \|Y_{lm}\|$ for $(j, k) \neq (l, m)$, with universal L_c . With $\beta(Y) = (C_{jk} Y)_{j,k}$ and the max norm on the fourfold sum this reads $\|\beta \alpha - I\| \leq 3L_c e$ and $\|\beta\| \leq M_c$. Feeding (100) and the datum's two-sided norm clause through this extraction gives universal $c_{\text{blk}} > 0$ and $B_{\text{blk}} < \infty$ — concretely $c_{\text{blk}} = 1/(16M_c)$ and $B_{\text{blk}} = 5$ — with

$$c_{\text{blk}} \|X\| \leq \|\gamma_n(X)\| \leq B_{\text{blk}} \|X\|, \quad \text{uniformly in } n. \quad (101)$$

Stage 2: the square defect. For U in the jk target corner and V in the lm one, two cases occur. If $k = l$ the compressed-product estimate applies at matrix level: $\|UV - U \cdot V\| \leq c_c e \|U\| \|V\|$. If $k \neq l$ the product is *small*: replacing U, V by their compressed representatives $p_j(U p_k)$ and $p_l(V p_m)$ and moving the middle factors together by exactly three ε -associator steps exposes the factor $p_k p_l$, so $\|UV\| \leq K_{\text{inc}} e \|U\| \|V\|$. Expanding $\gamma_n(X^\dagger X)$ and $\gamma_n(X)^\dagger \gamma_n(X)$ over the blocks produces sixteen terms, eight compatible and eight incompatible; the compatible ones are matched using the datum's involution and compressed-product clauses, the incompatible ones are simply small. Hence

$$\|\gamma_n(X^\dagger X) - \gamma_n(X)^\dagger \gamma_n(X)\| \leq D_{\text{sq}} e \|X\|^2, \quad D_{\text{sq}} = 8(4 + 2c_c + 2K_{\text{inc}}). \quad (102)$$

Stage 3: the scalar bootstrap. Fix n and let a_n and b_n be the infimum and supremum of $\|\gamma_n(X)\|/\|X\|$ over $X \neq 0$; by (101), $0 < c_{\text{blk}} \leq a_n \leq b_n \leq 5$. The source is an exact C^* -algebra, so $\|X^\dagger X\| = \|X\|^2$. For the upper bound, the target lower C^* inequality and (102) give $(1 - \varepsilon)\|\gamma_n(X)\|^2 \leq \|\gamma_n(X^\dagger X)\| + D_{\text{sq}} e \|X\|^2 \leq (b_n + D_{\text{sq}} e)\|X\|^2$, so a maximising sequence yields $(1 - \varepsilon)b_n^2 \leq b_n + D_{\text{sq}} e$; if $b_n > 1$ then $b_n - 1 \leq b_n(b_n - 1) \leq \varepsilon b_n^2 + D_{\text{sq}} e \leq (25 + D_{\text{sq}})e$. For the lower bound, the same identity read in the other direction gives $a_n \leq (1 + \varepsilon)a_n^2 + D_{\text{sq}} e$, and if $a_n < 1$ then $c_{\text{blk}}(1 - a_n) \leq a_n(1 - a_n) \leq (25 + D_{\text{sq}})e$. Therefore

$$b_n \leq 1 + (25 + D_{\text{sq}})e, \quad a_n \geq 1 - \frac{25 + D_{\text{sq}}}{c_{\text{blk}}} e,$$

which is (99) with $K_{\text{norm}} = \max\{25 + D_{\text{sq}}, (25 + D_{\text{sq}})/c_{\text{blk}}\}$ and e_{norm} the minimum of the universal ceilings of Stages 1 and 2 (and, harmlessly, $1/(2K_{\text{norm}})$). The crude constant c_{blk} survives only in K_{norm} : it is what prevents the bootstrap from being vacuous, not what the final estimate looks like.

Status. proved; af: validated (T0). Workspace `proofs/lem-extcb-four-corner-norm`: 18 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. This shard exists because of a recorded *third-stall* classification: three successive orchestrations of `lem-extcb-four-corner-merge` stalled on exactly this claim, so it was factored out of that tree — its statement extracted mechanically from the blocking challenges, with the assembly maps written out from an already-validated node — and validated on its own. Its single registry consumer is `lem-extcb-four-corner-merge` (§35), which uses it as the entire norm-control half of the merge.

35 The complete four-corner merge

Lemma 36 (`lem-extcb-four-corner-merge`). *There are universal $C_{\text{merge}} < \infty$ and $a_{\text{merge}} > 0$ such that four fixed bijective level-one corner maps satisfying `def-four-corner-merging-datum` with common defect ρ and $\rho + \varepsilon \leq a_{\text{merge}}$ combine into one extended $C_{\text{merge}}(\rho + \varepsilon)$ -isomorphism. Transcription note. The smallness hypothesis constrains the total defect $\rho + \varepsilon$, not ρ alone; that is a verdict-driven amendment of the transcribed contract, and the record of why it cannot be dropped is reproduced below. “Extended isomorphism” is `def-extended-delta-inclusion`: every amplification is a δ -homomorphism with two-sided $(1 \pm \delta)$ norm bounds, and the level-one map is bijective.*

Registry contract (`verbatim`).

Complete four-corner merge: there are universal $C_{\text{merge}} < \text{infinity}$ and $a_{\text{merge}} > 0$ such that four fixed bijective level-one corner maps satisfying `def-four-corner-merging-datum` with common defect ρ and $\rho + \varepsilon \leq a_{\text{merge}}$ combine into one extended $C_{\text{merge}}(\rho + \varepsilon)$ -isomorphism.

Proof (prose account of the af-validated tree). Write $\pi_j = I_n \otimes \Pi_j$, $X_{jk} = \pi_j X \pi_k$, and let $\mu_n, \Gamma_n, \alpha_n$ and $\gamma_n = \alpha_n \Gamma_n \mu_n$ be the assembly maps of §34. Set $e = \rho + \varepsilon$.

One map, not a family. Exact source complementarity gives $\pi_j \pi_k = \delta_{jk} \pi_j$, $\pi_1 + \pi_2 = I$ and $X = \sum_{jk} X_{jk}$; since the four $\gamma_{jk,n}$ are the amplifications of the same four level-one maps,

$$\gamma_n = 1_{M_n} \otimes \gamma_1, \quad \gamma_1(X) = \sum_{j,k} \gamma_{jk}(\Pi_j X \Pi_k). \quad (103)$$

This is the structural point of the lemma: the conclusion is about *one* linear map and its amplifications, so nothing may be chosen level by level. Two clauses follow at once. Adjoints of source corners transpose indices, $(\pi_j X \pi_k)^\dagger = \pi_k X^\dagger \pi_j$, and the datum’s involution clause matches them, so $\gamma_n(X^\dagger) = \gamma_n(X)^\dagger$ *exactly*. The source unit has only diagonal blocks, so $\gamma_n(I) - I$ splits into the two diagonal-unit errors and $1_{M_n} \otimes (P_1 + P_2 - I)$, whose norm equals its level-one value by the direct-sum norm axiom; hence

$$\|\gamma_n(I) - I\| \leq 3\rho, \quad \text{uniformly in } n. \quad (104)$$

Corner separation and the multiplicative defect. As in §34, compression commutes with amplification, so all matrix-level corner estimates are available with level-one constants; and from $p_k p_l = p_k(s - I) + (p_k I - p_k) - (p_k^2 - p_k)$ with $s = p_1 + p_2$ one gets $\|p_k p_l\| \leq 5e$ for $k \neq l$ once $e \leq \frac{1}{4}$. Hence a product of corner elements is compressed-product-close when the middle indices match ($\|UV - U \cdot V\| \leq c_{\text{prod}} e \|U\| \|V\|$) and is itself small when they do not ($\|UV\| \leq c_{\text{cross}} e \|U\| \|V\|$, by three associator moves that expose $p_k p_l$). Expanding $\gamma_n(XY)$ and $\gamma_n(X)\gamma_n(Y)$ over the blocks gives sixteen terms — eight with matching middle index, compared through the datum’s compressed-product clause, and eight without, which are small — so

$$\|\gamma_n(XY) - \gamma_n(X)\gamma_n(Y)\| \leq D_{\text{mult}} e \|X\| \|Y\|, \quad D_{\text{mult}} = 8[1 + 2c_{\text{prod}} + 2c_{\text{cross}}], \quad (105)$$

independently of n and of all dimensions.

Norm control. The datum and the assembly (103) are exactly the hypotheses of `lem-extcb-four-corner-norm` (§34), which supplies the two-sided bound $(1 - K_{\text{norm}} e) \|X\| \leq \|\gamma_n(X)\| \leq (1 + K_{\text{norm}} e) \|X\|$ at every level simultaneously.

Coverage, and bijectivity at level one. Injectivity of γ_1 is immediate from the lower bound once $K_{\text{norm}}e \leq \frac{1}{2}$. Surjectivity is the delicate half, and it is obtained without assuming that the target corner sum is direct: with $\beta(Y) = (\text{Co}_{P_j, P_k} Y)_{jk}$, each compression is $O(e)$ -close to $Y \mapsto P_j(Y P_k)$, so $\alpha_1 \beta$ is $O(e)$ -close to $Y \mapsto s(Y s)$ with $s = P_1 + P_2$ — an identity of bilinearity alone, no associativity used — and $\|s - I\| \leq \rho$ together with the product and approximate-unit axioms gives

$$\|\alpha_1 \beta - I\| \leq C_{\text{cov}} e. \quad (106)$$

Choosing e with $C_{\text{cov}} e < 1$ makes $\alpha_1 \beta$ invertible by its Neumann series, so $Y = \alpha_1 (\beta ((\alpha_1 \beta)^{-1} Y))$ for every Y and α_1 is onto. Now μ_1 is a bijection by exact source complementarity, Γ_1 is a bijection because all four fixed level-one maps are bijective by hypothesis, and α_1 is onto; therefore $\gamma_1 = \alpha_1 \Gamma_1 \mu_1$ is onto, hence bijective, and every $1_{M_n} \otimes \gamma_1$ is algebraically bijective with inverse $1_{M_n} \otimes \gamma_1^{-1}$.

Conclusion. Let a_{merge} be the minimum of the thresholds above — e_{corner} , e_{norm} , e_{cov} , $1/(2K_{\text{norm}})$ and the universal compression radius — and put $C_{\text{merge}} = \max\{3, D_{\text{mult}}, K_{\text{norm}}\}$. Then (104) and (105) make every amplification a $C_{\text{merge}} e$ -homomorphism, the norm lemma supplies the two-sided $(1 \pm C_{\text{merge}} e)$ bounds, and level-one bijectivity supplies the isomorphism clause; **def-extended-delta-inclusion** then returns exactly the asserted extended $C_{\text{merge}}(\rho + \varepsilon)$ -isomorphism.

Why the hypothesis is on the total defect. The transcribed contract bounded only ρ , leaving ε free. That version is *false*, and the tree records an exact counterexample rather than a repair. Take $\mathcal{A} = \mathbb{C}^3$ with coordinatewise involution, the multiplication $(a, b, c)(a', b', c') = (aa', bb', 0)$ and designated unit $I = (1, 1, 0)$: at every matrix level this is an extended 1 - C^* -algebra, the lower C^* inequality being vacuous at $\varepsilon = 1$. The elements $P_1 = (1, 0, 0)$ and $P_2 = (0, 1, 0)$ are exact projections with $P_1 + P_2 = I$, so $\rho = 0$; the registered θ formula gives $\text{Co}_{P_j} = E_j$ and $\text{Co}_{P_j, P_k} = 0$ for $j \neq k$, so the four target corners are $\mathbb{C}P_1, 0, 0, \mathbb{C}P_2$. With $\mathcal{B} = \mathbb{C}^2$ and its exact structure, the two diagonal identities and the two zero maps form a datum with common defect $\rho = 0$ whose four maps are bijective complete isometries; yet the assembled map is $\gamma(a, b) = (a, b, 0)$, which is not surjective, and no linear bijection $\mathbb{C}^2 \rightarrow \mathbb{C}^3$ exists at all. So ρ may be arbitrarily small while the conclusion fails, and only a ceiling on $\rho + \varepsilon$ repairs it. The prover amended the root accordingly and the fresh verifiers validated the amended statement; the registry contract carries that amended root verbatim, and downstream consumption supplies a small ambient ε , so the conditional form is the consumable one.

Status. proved; af: validated (T0). Workspace proofs/lem-extcb-four-corner-merge: 22 nodes validated plus 4 archived, root validated, taint clean. The archived nodes are dependency-gated duplicates superseded by children that discharge the same step from validated material only.

Role in the chain. Imports `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and `lem-extcb-four-corner-norm` (§34). It is the assembly primitive of the EXT tier: given the block shape selected in §33 and four corner maps matched to it, the merge produces the single map that the tier's conclusion is about. Its consumer is `conj-extcb` (§36), now proved with af: validated: this merge is the last step of that proof. Nothing here bears on `op-classical`, which is open.

36 The EXT-CB parent contract

Lemma 37 (conj-extcb). *There are universal $C_{\text{ext}} < \infty$ and $e_{\text{ext}} > 0$ such that if $e = \delta + \varepsilon \leq e_{\text{ext}}$, P, Q are δ -projections in an extended ε - C^* -algebra $\mathcal{A}$ with $\|P + Q - I\| \leq \delta$, $v : M_r \rightarrow \mathcal{S}_P$ is an extended δ -isomorphism, $\dim \mathcal{S}_Q = 1$ at level one, and $\mathcal{S}_{P,Q} \neq 0$, then there is one map $v_+ : M_{r+1} \rightarrow \mathcal{A}$ whose every amplification is a $C_{\text{ext}}e$ -isomorphism; the same level-one unitary and the same four corner maps carry all amplification levels, with constants independent of r, n and $\dim \mathcal{A}$. Transcription note. The hypothesis list is exactly **def-extcb-datum**; the conclusion's “isomorphism” is **def-extended-delta-inclusion** read at accuracy $C_{\text{ext}}e$. The clause $\dim \mathcal{S}_Q = 1$ is used only at level one: no amplification of Q is anywhere called one-dimensional. The final clause is not decoration — it is the content of “CB”: one unitary and four fixed level-one maps serve every n , so nothing is chosen level by level.*

Registry contract (verbatim).

EXT-CB: there are universal $C_{\text{ext}} < \text{infinity}$ and $e_{\text{ext}} > 0$ such that if $e = \delta + \varepsilon \leq e_{\text{ext}}$, P, Q are δ -projections in an extended ε - C^* -algebra $\mathcal{A}$ with $\|P + Q - I\| \leq \delta$, $v : M_r \rightarrow \mathcal{S}_P$ is an extended δ -isomorphism, $\dim \mathcal{S}_Q = 1$ at level one, and $\mathcal{S}_{\{P, Q\}}$ is nonzero, then there is one map $v_+ : M_{\{r+1\}} \rightarrow \mathcal{A}$ whose every amplification is a $C_{\text{ext}}e$ -isomorphism; the same level-one unitary and the same four corner maps carry all amplification levels, with constants independent of r, n , and $\dim \mathcal{A}$.

Proof (prose account of the af-validated tree). Put $P_1 = P$, $P_2 = Q$, $H_j = \mathcal{S}_{P_j, Q}$, $K_1 = \mathbb{C}^r$, $K_2 = \mathbb{C}$ and $h_{jk} = \text{Ha}_{P_j, P_k}^Q$. The hypotheses are a closed EXT-CB datum, so for e below the universal e_{sel} of **lem-extcb1-cross-corner-dimension** (§33) the cross corners have dimensions $\dim H_1 = r$ and $\dim H_2 = 1$. Fix once and for all a unitary $U_2 : K_2 \rightarrow H_2$.

A dimension-free exact-target correction. The tree proves in-workspace, from no external theorem, the following. *There are universal $a_{\text{corr}} > 0$ and $C_{\text{corr}} < \infty$ such that if $\mathcal{B}$ is a finite-dimensional C^* -algebra, H a finite-dimensional Hilbert space, and $T : \mathcal{B} \rightarrow B(H)$ is linear, $\dagger$ -preserving and satisfies $\|T_n(XY) - T_n(X)T_n(Y)\| \leq a\|X\|\|Y\|$ and $\|T_n(I) - I\| \leq a$ at every n , with $0 \leq a \leq a_{\text{corr}}$, then one unital $\dagger$ -homomorphism $\mu : \mathcal{B} \rightarrow B(H)$ satisfies $\|\mu_n - T_n\| \leq C_{\text{corr}}a$ at every n — the same level-one μ at every amplification.* It is a Newton iteration on the multiplicative defect.

At $\|X\| = 1$, $\dagger$ -preservation and the defect estimate give $\|T_n(X)\|^2 \leq \|T_n(X^\dagger X)\| + a$, so $M_n = \|T_n\|$ obeys $M_n^2 \leq M_n + a$ and $\|T\|_{\text{cb}} \leq 2$ for $a \leq 1$. Fixing a state ϕ on $\mathcal{B}$, $S_0(x) = T(x) + \phi(x)(I - T(I))$ is $\dagger$ -preserving and *exactly* unital, with $\|S_0 - T\|_{\text{cb}} \leq a$, $\|S_0\|_{\text{cb}} \leq 3$ and multiplicative defect $b_0 \leq c_u a$, $c_u = 7$. Normalised Haar measure on the compact unitary group of $\mathcal{B}$ gives $D = \int U^\dagger \otimes U dU$, in finite dimension a finite convex combination $\sum_s p_s U_s^\dagger \otimes U_s$ of projective norm one, with $XD = DX$ and $\text{mult}(D) = I$ by invariance. For a defect G of a map S , the correction $w'(x) = \sum_s S(p_s U_s^\dagger)G(U_s, x)$ then obeys $\|w'\|_{\text{cb}} \leq \|S\|_{\text{cb}}\|G\|$, and at level n the same formula is $1_{M_n} \otimes w'$ entrywise, because the diagonal identity moves arbitrary matrix entries. This is the step that makes every correction *one level-one map* with n -free, dimension-free bounds.

Now the Newton step. Let S be exactly unital, $\dagger$ -preserving, $\|S\|_{\text{cb}} \leq 4$, with defect G of uniform amplified bilinear norm b ; put $w''(x) = w'(x^\dagger)^\dagger$, $w = (w' + w'')/2$ and $S^+ = S + w$. Since $G(y, I) = G(I, y) = 0$ one has $w(I) = 0$, so S^+ is still exactly unital and $\dagger$ -preserving, with $\|w\|_{\text{cb}} \leq 4b$. Exact associativity in $B(H)$ gives $S(x)G(y, z) - G(xy, z) + G(x, yz) - G(x, y)S(z) = 0$, and expanding $S(x)w(y) - w(xy) + w(x)S(y)$ with $XD = DX$, $\text{mult}(D) = I$ and $S(I) = I$ cancels

G exactly to first order, every surviving term carrying two defect factors; a triangle estimate gives $\|G_{S^+}\| \leq K_N b^2$ at every amplification, for one numerical universal K_N , while $\|S^+ - S\|_{\text{cb}} \leq 4b$. Choosing a_{corr} with $K_N c_u a_{\text{corr}} \leq \frac{1}{2}$ and $8c_u a_{\text{corr}} < 1$ and iterating from S_0 , the recursion $b_{k+1} \leq K_N b_k^2$ gives $b_k \leq b_0 2^{-k}$, hence $\|S_k - S_0\|_{\text{cb}} \leq \sum_{j < k} 4b_j \leq 8b_0 \leq 8c_u a < 1$ and $\|S_k\|_{\text{cb}} < 4$, so every iteration is licensed — including the degenerate case $b_0 = 0$, where every increment vanishes. The sequence converges in cb norm to one unital $\dagger$ -preserving μ ; continuity of multiplication and $b_k \rightarrow 0$ give $\mu(xy) = \mu(x)\mu(y)$, and cb convergence means every μ_n is the amplification of that same level-one μ . Finally $\|\mu - T\|_{\text{cb}} \leq a + 8c_u a$, i.e. $C_{\text{corr}} = 1 + 8c_u = 57$.

One spatial four-corner system. Let $T = h_{11} \circ v$. For $X, Y \in M_n \otimes M_r$, **def-extended-delta-inclusion** gives $\|v_n(XY) - v_n(X) \cdot v_n(Y)\| \leq \delta \|X\| \|Y\|$, $\|v_n(X)\| \leq (1 + \delta) \|X\|$, exact $\dagger$ -preservation and the compressed unit estimate in $\mathcal{S}_P$; feeding these through the upper-norm, product, unit and adjoint clauses of **conj-hcb** (§30) yields

$$\|T_n(XY) - T_n(X)T_n(Y)\| \leq [(1 + C_{He})\delta + C_{He}(1 + \delta)^2] \|X\| \|Y\|, \quad \|T_n(I) - I\| \leq A_0 e,$$

so below one universal ceiling both defects are at most $A_0 e$ with $A_0 = 4(C_H + 1)$, uniformly in n , r and $\dim \mathcal{A}$. The correction now applies, in two cases that exhaust $e \geq 0$: if $e = 0$ then $\delta = \varepsilon = 0$, both defects vanish, T is itself an exact unital $\dagger$ -homomorphism and one takes $\mu_{11} := T$; if $e > 0$, shrink the threshold so that $a = A_0 e < a_{\text{corr}}$ and correct. Either way there is one level-one unital $\dagger$ -homomorphism $\mu_{11} : M_r \rightarrow B(H_1)$ with

$$\|\mu_{11,n} - h_{11,n} v_n\| \leq \kappa e \quad \text{for every } n, \quad \kappa = C_{\text{corr}} A_0. \quad (107)$$

It is unital and nonzero, so its kernel is a two-sided ideal in the simple algebra M_r and vanishes; domain and codomain both have dimension r^2 by the level-one count, so it is onto. A unital $\dagger$ -isomorphism between full matrix algebras preserves matrix units, hence is spatial: there is one unitary $U_1 : \mathbb{C}^r \rightarrow H_1$ with $\mu_{11}(A) = U_1 A U_1^\dagger$. With the already fixed U_2 , set $\mu_{jk}(A) = U_j A U_k^\dagger$. These four fixed level-one maps are exact, completely isometric, adjoint-compatible and multiplicative at every amplification via $1_{M_n} \otimes U_j$, with no level-dependent choice.

The conditional H-CB clauses, in the required order. Shrink the threshold so that $\kappa e < 1$ and $C_{He} < \frac{1}{4}$. Since μ_{11} is invertible and $\|T - \mu_{11}\| < 1$, Neumann inversion makes T bijective at level one; v is bijective, so h_{11} is, and for $Z = v(A)$ one has $\|h_{11}(Z)\|/\|Z\| \geq (1 - \kappa e)/(1 + \delta) > \frac{1}{4}$. The canonical-identity closeness clause of **conj-hcb** makes each of h_{12}, h_{21}, h_{22} a strict-norm perturbation of its canonical level-one bijection, hence level-one bijective by Neumann inversion, and gives h_{22} level-one lower modulus above $\frac{1}{4}$. *Only now* are the conditional clauses invoked: the diagonal clause for h_{11} and h_{22} , then the off-diagonal clause for h_{12} anchored at h_{22} and for h_{21} anchored at h_{11} . Every $h_{jk,n}$ is then bijective with

$$(1 - C_{He})\|Z\| \leq \|h_{jk,n}(Z)\| \leq (1 + C_{He})\|Z\|, \quad \|h_{jk,n}^{-1}\| \leq 1 + C_{He}, \quad (108)$$

and $h_{jk,n}^{-1} = 1_{M_n} \otimes h_{jk}^{-1}$ by uniqueness of inverses, so no inverse is chosen afresh at level n . The order is the delicate point, and the hostile verifier checked it: the parent's inverse clauses are conditional (§30), and their level-one hypotheses are established *before* use, never assumed.

Four fixed corner maps form a merging datum. Put $\gamma_{11} = v$ and $\gamma_{jk} = h_{jk}^{-1} \circ \mu_{jk}$ for $(j, k) \neq (1, 1)$, and $d_{jk,n} = h_{jk,n} \gamma_{jk,n} - \mu_{jk,n}$. Off $(1, 1)$ the composition is exact, so $d_{jk,n} = 0$; at $(1, 1)$, (107) gives $\|d_{11,n}\| \leq \kappa e$. *Involution* is exact: off $(1, 1)$, exact H-CB adjointness and spatial adjoint compatibility give $h_{kj,n}(\gamma_{jk,n}(X)^\dagger) = h_{kj,n} \gamma_{kj,n}(X^\dagger)$ and injectivity of $h_{kj,n}$ cancels it, while $\gamma_{11,n} = v_n$ preserves $\dagger$ exactly. For the *product*, (108) and isometry of μ give $\|\gamma_{jk,n}(X)\| \leq$

$(1 + C_H e)(1 + \kappa e)\|X\| \leq \frac{25}{16}\|X\| \leq 2\|X\|$; with $D = \gamma_{jl,n}(XY) - \gamma_{jk,n}(X) \cdot \gamma_{kl,n}(Y)$, the product-defect clause of **conj-hcb** supplies E with $\|E\| \leq 4C_H e\|X\|\|Y\|$ and the exact spatial multiplication $\mu_{jl,n}(XY) = \mu_{jk,n}(X)\mu_{kl,n}(Y)$ turns the comparison into

$$h_{jl,n}(D) = d_{jl,n}(XY) - \mu_{jk,n}(X)d_{kl,n}(Y) - d_{jk,n}(X)\mu_{kl,n}(Y) - d_{jk,n}(X)d_{kl,n}(Y) - E,$$

whence $\|h_{jl,n}(D)\| \leq [3\kappa + \kappa^2 e + 4C_H]e\|X\|\|Y\| \leq 4(C_H + \kappa)e\|X\|\|Y\|$ and, applying $\|h_{jl,n}^{-1}\| \leq \frac{5}{4}$, $\|D\| \leq 5(C_H + \kappa)e\|X\|\|Y\|$. For the *unit and norm* clauses, spatiality gives $\mu_{jj,n}(I) = I$, so the HCB diagonal-unit estimate, $\|u_{P_j} - P_j\| \leq C_H e$ and (108) give $\|\gamma_{jj,n}(I) - I_n \otimes P_j\| \leq 3(C_H + \kappa)e$, while isometry of μ and the d -bound give $(1 - (C_H + \kappa)e)\|X\| \leq \|\gamma_{jk,n}(X)\| \leq (1 + 2(C_H + \kappa)e)\|X\|$. Finally $\gamma_{11} = v$ is bijective by hypothesis and every other γ_{jk} composes two bijections. Since the source block projections in $M_{r+1} = B(K_1 \oplus K_2)$ are exactly complementary and $\|P_1 + P_2 - I\| \leq \delta \leq e$, every clause of **def-four-corner-merging-datum** holds with the one common defect $\rho = D_0 e$, $D_0 = 5(C_H + \kappa)$ enlarged so that $D_0 \geq 1$, for the same four fixed level-one maps at every amplification.

Merge, and the root. Let e_{ext} be the positive minimum of the thresholds above, of $a_{\text{merge}}/(D_0 + 1)$, and of the finitely many Neumann and $\frac{1}{4}$ ceilings. Then $\rho + \varepsilon \leq (D_0 + 1)e \leq a_{\text{merge}}$, so **lem-extcb-four-corner-merge** (§35) — whose norm half is **lem-extcb-four-corner-norm** (§34) — applies and returns the single sum map $v_+ : M_{r+1} \rightarrow \mathcal{A}$ as an extended $C_{\text{merge}}(\rho + \varepsilon)$ -isomorphism, hence an extended $C_{\text{ext}}e$ -isomorphism with $C_{\text{ext}} = C_{\text{merge}}(D_0 + 1)$. Its n -th map is $1_{M_n} \otimes v_+$, assembled from $1_{M_n} \otimes U_1$, $1_{M_n} \otimes U_2$ and the same four γ_{jk} ; every constant used is universal, so none depends on r , on n , on $\dim \mathcal{A}$ or on the block data. That is the root.

Status. proved; af: validated (T0). Workspace **proofs/conj-extcb**: 40 nodes validated plus 6 archived, root validated, taint clean. The archived nodes are dependency-gated duplicates superseded by siblings that discharge the same step from validated material only: they are not part of the validated spine, and nothing above rests on them. The identifier keeps its historical **conj-slug**; registry ids are stable.

Correction of record (proof-level, not contract-level). The registry shard records one verifier-mandated correction: the level-one close-idempotent normalisation in the selection step must be explicit — the added-dimension corner is stated for $\hat{P} = v(I_r)$, with $\|\hat{P} - P\| = O(e)$ and range identification by close idempotents — and the selection threshold must cover it, at a universal threshold with no r - or level-dependence. That is what **lem-extcb1-close-corner-dimension** (§33) supplies; the contract itself was not amended.

Role in the chain. The registry shard lists ten dependencies: **conj-hcb** (§30), the EXT front end (§§31–35), **lem-compcb-corner-algebra** (§15) and **lem-hcb3-uniform-square-lower** (§23); all ten are proved with af: validated and all ten are reproduced in this document. The exported tree cites four directly — **conj-hcb**, **lem-extcb1-cross-corner-dimension**, **lem-extcb-four-corner-merge**, **lem-extcb-four-corner-norm** — the rest entering through the selection lemma. The registry body additionally records the paper-proof’s witness in the expanded form $C_{\text{ext}} = C_{\text{merge}}[1 + 5C_H + 20C_{\text{app}}(C_H + 1)]$; the validated tree does not use it, deriving $C_{\text{merge}}(D_0 + 1)$ instead. Downstream, **conj-extcb** with **conj-hcb** discharges both gap nodes of the extended assembly, whose registry carrier **lem-thmainext-conditional** is proved-mod-audit with af: none — **open** here and **not** reproduced — and the unconditional end-to-end constant ledger for that route remains separate open work. Nothing here bears on **op-classical**, which is open.

37 Status and outlook

What is validated here

The preceding sections reproduce thirty-seven registry results, each proved with `af: validated`: the signed-stochastic bridge `lem-classical-equiv` (§2); positive-retract hardening `lem-prh` (§3); the COMP tier — `lem-compcb-amplification-naturality`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-entrywise-compression-naturality`, `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`; the H-CB tier — `lem-hcb0-compressed-associator`, `lem-hcb-column-hilbert-squared`, `lem-hcb1-variational-identity`, `lem-hcb1-column-action`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-hcb3-diagonal-unit`, `lem-hcb3-diagonal-lower-modulus`, `lem-hcb3-diagonal-inverse`, `lem-hcb3-offdiagonal-inverse` — its canonical layer — `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-close-corner-dimension`, `lem-hcb4-canonical-inverse` — together with the tier’s parent contract `conj-hcb` (§30); and the front end of the EXT tier — `lem-extcb-one-dimensional-product`, `lem-extcb-one-dimensional-corner-dimension`, `lem-extcb-corner-dimension-additivity`, `lem-extcb1-close-corner-dimension`, `lem-extcb1-cross-corner-dimension`, `lem-extcb-four-corner-norm`, `lem-extcb-four-corner-merge`; and, closing that tier, its parent contract `conj-extcb` (§36). The supplied generated registry index contains **70** results with that status pair. The remaining thirty-three such rows are listed in the table below.

The headline also needs qualifications. The inverse and lower-modulus results — `lem-hcb3-diagonal-lower-modulus`, `lem-hcb3-diagonal-inverse`, `lem-hcb3-offdiagonal-inverse` and the corresponding clauses of `conj-hcb` — are validated as *conditional* statements. Their level-one hypotheses are established in exactly one place in this document and for exactly one datum: §36 establishes them for the EXT-CB datum, in the order the tree fixes, before invoking the clauses. Nothing else here supplies them for any datum. Their conditional shape is itself a correction of record: the parent’s inverse clause was unconditional until an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample refuted it. `lem-extcb-four-corner-merge` (§35) carries a smallness hypothesis on the *total* defect $\rho + \varepsilon$; the transcribed version that constrained ρ alone is false, and the refuting example is reproduced there. Both corrections were produced by hostile verification, and both are reasons to distrust unconditional restatements of this material.

What is next, and what it is not

With §30 the H-CB tier is closed, and with §36 so is the EXT tier: `conj-extcb` is proved with `af: validated` and is reproduced above as Lemma 37. What it delivers is one extension step — from M_r to M_{r+1} , at one universal accuracy, with one unitary and four fixed corner maps serving every amplification level. It is not an induction, and nothing here iterates it.

Beyond the EXT tier lies the assembly, and that is where the live route now stops. Its registry carrier `lem-thmainext-conditional` is `proved-mod-audit` with `af: none` — **open** at this document’s rigour bar, and **not** reproduced here. The MAIN-CB assembly rows and the ledger rows are explicitly *not* in the pipeline and must not be treated as pending detail. They are **quarantined** pending a focused repair and a fresh hostile review, along with the named gap interfaces (exact-target approximation, the polar contract, the MAIN structure, the ledger domains, and the final bridge); and several external topology sources are not yet in `refs/`, which is a provisioning risk, not a proof step. The unconditional end-to-end constant ledger for Route F remains separate open work.

The honest summary is therefore: the base, the middle and both tier parents of the live route have been made rigorous here, and the assembly above them is quarantined. `op-classical` is

open.

Validated registry results not reproduced here

The table is the exact set difference between the supplied index rows carrying **proved/validated** and the reproduced ids listed above. Its ordering follows the generated index.

Registry id	Status	Note
lem-always-tight-dual-support	proved, validated	af: off the live route; registry record retained
lem-collateral-import	proved, validated	af: off the live route; registry record retained
lem-cross-pivot-cancellation	proved, validated	af: off the live route; registry record retained
lem-cs-low-slab-pincer	proved, validated	af: off the live route; registry record retained
lem-depth-d-halo-collapse	proved, validated	af: off the live route; registry record retained
lem-factorization	proved, validated	af: off the live route; registry record retained
lem-fan-payment	proved, validated	af: off the live route; registry record retained
lem-fan-payment-restricted	proved, validated	af: off the live route; registry record retained
lem-genuine-disintegration	proved, validated	af: off the live route; registry record retained
lem-halo-collapse	proved, validated	af: off the live route; registry record retained
lem-harmonic-affine-bridge	proved, validated	af: off the live route; registry record retained
lem-hiddenness-depth-markov	proved, validated	af: off the live route; registry record retained
lem-hiddenness-dual-witness	proved, validated	af: off the live route; registry record retained
lem-hx-financing-floor	proved, validated	af: off the live route; registry record retained
lem-hx-forced-exterior-coupling	proved, validated	af: off the live route; registry record retained
lem-hx-robust-scalar-starvation	proved, validated	af: off the live route; registry record retained
lem-hx-signed-variation-ledger	proved, validated	af: off the live route; registry record retained
lem-hx-transverse-moment-identity	proved, validated	af: off the live route; registry record retained
lem-import-reduction	proved, validated	af: off the live route; registry record retained
lem-mass-split	proved, validated	af: off the live route; registry record retained

Registry id	Status	Note
lem-negpart-subadditive	proved, validated	af: off the live route; registry record retained
lem-parametric-halo-collapse	proved, validated	af: off the live route; registry record retained
lem-pivot-removing-move	proved, validated	af: off the live route; registry record retained
lem-residual-lower	proved, validated	af: off the live route; registry record retained
lem-residual-upper	proved, validated	af: off the live route; registry record retained
lem-row-far-dual-certificate	proved, validated	af: off the live route; registry record retained
lem-row-zero-capacity	proved, validated	af: off the live route; registry record retained
lem-starvation-completion-obstruction	proved, validated	af: off the live route; registry record retained
lem-top-concentration	proved, validated	af: off the live route; registry record retained
lem-top-slab-companion	proved, validated	af: off the live route; registry record retained
lem-weighted-min	proved, validated	af: off the live route; registry record retained
lem-zerosum-triangle	proved, validated	af: off the live route; registry record retained
obs-height-collapse	proved, validated	af: off the live route; registry record retained

Each id and status pair in the table is transcribed from `repo-inputs/argument-INDEX.md`. In particular, `conj-extcb` is absent from the table because it is *reproduced* above (§36), not because of its status: the index records it as `proved` with `af: validated`.